A Complexity-Theoretic Foundation for Metrology

Book III: Finite Tests, Fact Decomposition-Independence, and Physics

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Abstract

This volume develops the operational and computational backbone for single-run verification in measurement science. Building on the Simple Test formalism and the ABET scaffolding from earlier books, we prove the *Fundamental Test Equivalence* (FTE)—showing that tail facts are certifiable *iff* they admit a finite guarded decomposition—and the *Fact Decomposition-Independence* (FDI) theorem—characterizing when no such decomposition can exist and why the corresponding certification language necessarily jumps to at least Turing degree $0'$.

We isolate precisely which axioms (Compact Continuity, Traceability, Time-Irreversibility, Effectivity EM1–EM3) are responsible for *hypercomputational* behavior and exhibit clean embeddings of HALT (and, in globalized settings, TOT) that attach hardness either to run time, compile time, or both. We then audit collapse and interpretation frameworks in quantum measurement, distinguishing compact laboratory slices (where ABET/FTE/FDI apply) from non-compact, global ontologies (out of scope). Along the way, we use Weihrauch reductions (closed choice) to formalize the intrinsic non-computability of single-outcome selection and identify bundles of assumptions that force Π^0_2-hardness. The goal is a neutral, model-independent yardstick for *epistemic certifiability* in physics.

Foreword and Guide to This Collection

This book continues the same editorial approach as Books I and II: a guided tour through a research program as it crystallized. Chapters are written to be self-contained while sustaining a cumulative argument. The first part fixes the operational language and the temporal discipline (fairness, ranking). The second part proves the central equivalence and its sharp converse under axiom failures. The third part turns to physics: quantum measurement, laboratory vs. global scope, and an *epistemic-complexity* audit of leading theories.

Readers familiar with Book I (motivation, metrology and complexity) and Book II (Simple Tests, ABET, certification standards) can enter here directly. Newcomers may wish to

skim Book I's primer on computational classes and Book II's definitions of basins, shells, and the minimal certification standard $\mathcal{S}$.

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Part I
Operational Foundations

This part fixes the process language and temporal discipline for single-run tests. We formalize guarded programs, traceability, and certification shells, then require time to flow irreversibly via weak-fairness and ranking functions.

Chapter 1: Foundational Process Structure

- The Simple Test tuple $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ and tail facts.
- Guarded program primitives: Sequential, Conditional, Loop; PL vs. UI traceability.
- Axioms (F) invariant basin, (S) certification standard, (C) compact continuity; shells and the Minimal Standard.
- What "finite decomposition" means: guards as effectively testable open cells; robustness under $\mathcal{S}$.
- *Editor's Note:* This chapter sets the operational vocabulary that all later equivalences will compile into.

Chapter 2: Temporal Fairness and Ranking Functions

- Axiom (I): time-irreversibility and weak-fairness; progress via ranking functions.
- Halting of guarded programs under (I); unrolling loops without Zeno paths.
- Reduction lemmas linking NP-style witnesses to finite guarded branches.
- Separation of compile time vs. run time; local vs. uniform traceability.
- **Key Result:** Under (I) and (F,S,C), fair executions admit finite witnessing structure for tail facts.

Part II
Equivalences and Failure Modes

Here we prove that finite guarded decompositions and NP-verification coincide (FTE), and that when no such decomposition exists (FDI) the certification language must already be hypercomputational—pinpointing exactly which axiom(s) break.

Chapter 3: Finite-Test Equivalence (FTE)

- **Main Theorem:** A tail fact Φ admits a finite guarded decomposition $\iff$ Φ is verifiable by a polytime checker (NP/$\mathcal{V}$).
- ATRT (Adaptive Traceability Resolution Theorem): synthesizing open guards under PL-traceability.
- Effectivity EM1–EM3 (representation, modulus, guard tests) $\Rightarrow$ polynomial verifiers.
- Consequences: extraction of decision trees; robustness to shell tolerances; uniformity in instance size.
- *Editor's Note:* FTE is the "if" direction that explains why proofs look like small finite programs.

Chapter 4: Fact Decomposition-Independence Theorem (FDI)

- **FDI Theorem:** If no finite guarded decomposition exists (with (F) and (S) in scope), then $\deg_T(L_\Phi) \geq 0'$.
- Clean embeddings for each single-axiom failure:
  - (I) Zeno subroutines $\Rightarrow$ HALT at run time.
  - (C) non-compactness $\Rightarrow$ ABET uniformization Σ^0_1-hard.
  - (T) non-open preimages $\Rightarrow$ compile-time stabilization Σ^0_1-hard.
  - (EM1) real-constant encoding; (EM2) super-polynomial modulus; (EM3) r.e.-open guards.
- Hierarchy-collapse corollary: with all axioms, languages lie in NP; otherwise $\geq 0'$.
- Parameterized variants for $(\mathbf{C}, \mathcal{V})$ (e.g., BQP/QCMA) with the same dichotomy.

Part III
Physics and Epistemic Audit

This part applies the machinery to quantum measurement. We separate compact laboratory slices (FDI in scope) from global non-compact ontologies (out of scope), formalize single-outcome non-computability via closed choice, and audit collapse/interpretation theories.

Chapter 5: Physical Measurement and the Turing Barrier

- Weihrauch reductions: $\mathbf{C}_I \leq_W$ Collapse and $\mathbf{C}_{\mathbb{R}} \leq_W$ Collapse (closed choice $\Rightarrow$ non-computable selection).
- Scope split: lab slices (compact Z^{in}) vs. global collapse (non-compact; ABET undefined).
- Two-stage audit (Stage 0 in-scope gate; Stage 1 decomposition test) across major theories.
- Summary verdicts: Categories 1–3 (out-of-scope, in-scope & indecomposable, in-scope & decomposable).
- Π^0_2-hardness bundles (A/B/C): when "for all unknowns, there exists a bound" forces $0''$ hardness.
- Operational indistinguishability theorem on compact slices; "no superobserver with finite resources."

Appendix A: Proof Kits and Constructions

- Detailed ATRT construction and covering-number bounds for shells under PL-traceability.
- Zeno, Cantor, r.e.-open, and real-constant embeddings (axiom-by-axiom).
- Parameterized effectivity ($\text{EM1}_{\mathbf{C}}$–$\text{EM3}_{\mathbf{C}}$) and verifier classes $\mathcal{V}$.

Chapter 1: Fundamental Decomposition of Simple Test

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1 Test Primitives

Throughout this chapter a *test* is a sextuple

$$\mathfrak{T} = \big(\mathcal{M},\ \mathcal{Z}^{\text{in}},\ \mathcal{Z}^{\text{out}},\ U,\ E,\ \mathcal{S}\big),$$

where $\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}$ are compact metric spaces, $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ and $E : \mathcal{M} \to \mathcal{Z}^{\text{out}}$ are continuous, and $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ is a (minimal) Certification Standard (Book II, Chapter 8, Def. 2.1). We now fix *three* primitive constructions over the input time-domain $\mathfrak{D}_{IN}$ that will be used to build all composite tests in this book. All the definitions below are stated directly in terms of maps and guard-bands.

Our three primitives are selected to build a Turing Machine. The Turing machine is selected because of its generality; indeed, no counterexample for the Church-Turing thesis based on a non-exotic (i.e., based on arguably complete theories - no quantum gravity and black hole paradoxes) physical process that doesn't require a physically infinite precision and/or physically infinite computing power is known.

The implications of the current work are interesting for further exploration of the Church-Turing thesis violations, and we will discuss them later.

2 Preliminaries: What Counts as a Valid Test?

Definition 2.1 (Valid Test)**.** A *test* is a sextuple

$$\mathfrak{T} = \big(\mathcal{M},\ \mathcal{Z}^{\text{in}},\ \mathcal{Z}^{\text{out}},\ U,\ E,\ \mathcal{S}\big)$$

satisfying the following axioms:

(VT1) **Compact continuity.** $\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}$ are compact metric spaces and $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$, $E : \mathcal{M} \to \mathcal{Z}^{\text{out}}$ are continuous.

(VT2) **Reachable invariant basin.** There exists a nonempty compact set $\mathcal{B} \subseteq \mathcal{M}$ such that (i) every trajectory enters $\mathcal{B}$ in finite time, and (ii) $\Theta \in \mathcal{B} \Rightarrow U(\Theta, z) \in \mathcal{B}$ for all admissible z (invariance).

(VT3) **Minimal certification standard.** $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ is a nonincreasing sequence $\varepsilon_n \downarrow 0$ such that each ε_n is the infimum radius simultaneously ensuring (a) forward robustness and (b) completeness of the index-n certification procedure.

(VT4) **Shell forward-closure.** Writing $\mathcal{B}_n = \{\Theta : \mathrm{dist}(\Theta, \mathcal{B}) \le \varepsilon_n\}$, we have for all n:

$$\Theta \in \mathcal{B}_n,\ z^{\mathrm{in}} \in U(\Theta, \cdot)^{-1}(\mathcal{B}_n) \implies U(\Theta, z^{\mathrm{in}}) \in \mathcal{B}_n.$$

(VT5) **Polynomial reachability (for NP–certifiability).** There exists a polynomial p such that every trajectory enters $\mathcal{B}$ by time $T_{\max} \le p(|\mathrm{input}|)$. (This axiom can be omitted if one studies certifiability independently of complexity class.)

Remark 2.1. Conditions (VT1)–(VT4) encapsulate the physical/epistemic regularity needed to talk about guard-bands and robustness. (VT5) is only required when we later connect certifiability to NP; the structural results (finite guarded decomposition, aspect uniqueness) use only (VT1)–(VT4).

2.1 The Primitive–Global Standard Equivalence

The very first question in building any decomposition is to find a way to distribute the Certification Standard. We can show that the *minimality* of the Certification Standard $\mathcal{S}$ is enough to allow an equivalent distribution.

Theorem 2.1 (Primitive–Global Certification Standard Equivalence)**.** *Let*

$$\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\mathrm{in}}, \mathcal{Z}^{\mathrm{out}}, U, E, \mathcal{S})$$

*be a valid test (Def. 2.1) with invariant, reachable basin $\mathcal{B}$ and minimal global Certification Standard (Def. 2.1, Book II, Chapter 8) $\mathcal{S} = \{\varepsilon_m\}_{m\ge1}$, giving shells $\mathcal{B}_m = \{\Theta : \mathrm{dist}(\Theta, \mathcal{B}) \le \varepsilon_m\}$. Assume U admits, at level m, a **finite open partition***

$$\mathcal{Z}^{\mathrm{in}} = \bigsqcup_i N_i$$

and define the restricted maps

$$U_i := U|_{\mathcal{M}\times N_i}$$

.

Then the following are equivalent:

(a) ***Global minimal forward-closure:*** *ε_m is the minimal radius such that for every $\Theta \in \mathcal{B}_m$ and every $z \in \mathcal{Z}^{\mathrm{in}}$ satisfying $d\big(z, \partial N_{i(z)}\big) \ge \varepsilon_m$ we have $U(\Theta, z) \in \mathcal{B}_m$.*

(b) ***Existence of minimal local radii:*** *There exist branchwise (minimal) radii $\{\varepsilon_{m,i}\}_{i=1}^k$ with*

$$\forall\, \Theta \in \mathcal{B}_m,\ \forall\, z \in N_i :\ d\big(z, \partial N_i\big) \ge \varepsilon_{m,i} \implies U_i(\Theta, z) \in \mathcal{B}_m,$$

such that

$$\varepsilon_m = \min_{1\le i\le k} \varepsilon_{m,i},$$

and no strictly smaller family $\{\varepsilon'_{m,i} < \varepsilon_{m,i}\}$ preserves the above property for branch i.

Proof. We prove $(a) \Rightarrow (b)$ and $(b) \Rightarrow (a)$.

$\boxed{(b) \Rightarrow (a)}$ Given the local radii, set $\varepsilon_m = \min_i \varepsilon_{m,i}$. For any $\Theta \in \mathcal{B}_m$ and $z \in \mathcal{Z}^{\text{in}}$ with branch $i(z)$ and $d(z, \partial N_{i(z)}) \geq \varepsilon_m$, we also have $d(z, \partial N_{i(z)}) \geq \varepsilon_{m,i(z)}$, so

$$U(\Theta, z) = U_{i(z)}(\Theta, z) \in \mathcal{B}_m.$$

Thus ε_m enforces global forward-closure. Minimality carries over: if a smaller $\widehat{\varepsilon}_m < \varepsilon_m$ sufficed globally, it would undercut at least one branch's minimal $\varepsilon_{m,j}$, contradicting its minimality.

$\boxed{(a) \Rightarrow (b)}$ This is the "necessity" (upward) direction.

Step 1: Define local candidates. Fix m and branch i. For each $\Theta \in \mathcal{B}_m$, consider the set

$$R_{m,i}(\Theta) := \big\{ r > 0 : \forall\, z \in N_i,\ d(z, \partial N_i) \geq r \Rightarrow U_i(\Theta, z) \in \mathcal{B}_m \big\}.$$

By assumption (a) at the global level, ε_m enforces forward-closure; in particular, for every $\Theta \in \mathcal{B}_m$ and every $z \in N_i$ with $d(z, \partial N_i) \geq \varepsilon_m$, we have $U_i(\Theta, z) = U(\Theta, z) \in \mathcal{B}_m$. Hence $\varepsilon_m \in R_{m,i}(\Theta)$, so $R_{m,i}(\Theta) \neq \varnothing$.

Define the *pointwise minimal* radius

$$\varepsilon_{m,i}^{\Theta} := \inf R_{m,i}(\Theta),$$

and then take

$$\varepsilon_{m,i} := \sup_{\Theta \in \mathcal{B}_m} \varepsilon_{m,i}^{\Theta}.$$

Because $\mathcal{B}_m$ is compact and U_i is continuous, a standard compactness argument (finite subcover over $\mathcal{B}_m$) shows that $\varepsilon_{m,i} < \infty$ and the supremum is attained (or can be approximated arbitrarily closely). Thus $\varepsilon_{m,i}$ is well-defined and finite.

Step 2: Local forward-closure. Let $\Theta \in \mathcal{B}_m$, $z \in N_i$, and assume $d(z, \partial N_i) \geq \varepsilon_{m,i}$. By definition of $\varepsilon_{m,i}$, for every $\delta > 0$ there exists $\Theta' \in \mathcal{B}_m$ with $\varepsilon_{m,i}^{\Theta'} > \varepsilon_{m,i} - \delta$, so for any $r < \varepsilon_{m,i}$ and any such Θ',

$$d(z, \partial N_i) \geq r \implies U_i(\Theta', z) \in \mathcal{B}_m.$$

Continuity of U_i in Θ (and closedness of $\mathcal{B}_m$) allows us to pass to the limit $\Theta' \to \Theta$, ensuring $U_i(\Theta, z) \in \mathcal{B}_m$. Hence each $\varepsilon_{m,i}$ enforces $(\star)$.

Step 3: Equality of minima. Since ε_m guarantees global closure, any z with $d(z, \partial N_{i(z)}) \geq \varepsilon_m$ is admissible at its branch. Therefore, for each i,

$$\varepsilon_{m,i} \ \leq\ \varepsilon_m.$$

Consequently,

$$\min_i \varepsilon_{m,i} \ \leq\ \varepsilon_m.$$

On the other hand, if $\min_i \varepsilon_{m,i} < \varepsilon_m$, then ε_m would not be minimal globally: we could choose $\widehat{\varepsilon}_m := \min_i \varepsilon_{m,i}$ and still have forward-closure everywhere, contradicting (a). Thus $\varepsilon_m = \min_i \varepsilon_{m,i}$.

Step 4: Local minimality. If some branch radius could be lowered strictly, say $\varepsilon'_{m,j} < \varepsilon_{m,j}$ still satisfying $(\star)$, then the new global minimum $\min(\varepsilon'_{m,j}, \min_{i \neq j} \varepsilon_{m,i})$ would be $< \varepsilon_m$, violating global minimality in (a). Hence each $\varepsilon_{m,i}$ is minimal.

We have constructed minimal local radii whose minimum equals ε_m. This completes $(a) \Rightarrow (b)$.

Conclusion. Both directions hold, yielding the stated equivalence. □

Remark 2.2 (Reading the Equivalence)**.** The theorem formalizes a "meet–distribute" duality:

- Starting from branchwise tolerances (local standards), you *meet* them to get the global minimal standard.
- Starting from a global minimal standard, you *distribute* it down to each branch, obtaining minimal local standards whose meet is the original one.

Thus the Certification Standard is neither purely global nor purely local—it is the fixed point of this meet–distribute process.

Lemma 2.1 (Minimal Standard ⇒ Boundary Margin)**.** *Assume Theorem 2.1 and let the finite tick-m partition be* $\mathcal{Z}^{\text{in}} = \bigsqcup_{i=1}^{k} N_i$ *with branchwise minimal radii* $\{\varepsilon_{m,i}\}$ *and global minimal* $\varepsilon_m = \min_i \varepsilon_{m,i}$*. Define the* certified input core *of branch* i *by*

$$C_{m,i} \;:=\; \{\, z \in N_i : d(z, \partial N_i) \geq \varepsilon_{m,i} \,\}.$$

Then for every $\Theta \in \mathcal{B}_m$ *and* $z \in C_{m,i}$ *we have*

$$U_i(\Theta, z) \in \mathcal{B}_m,$$

and, in particular, any certified trajectory that stays in $C_{m,i}$ *cannot cross* ∂N_i *in one step. Moreover, the global certified core* $C_m := \bigsqcup_i C_{m,i}$ *is open and nonempty in each branch.*

Proof. By Theorem 2.1(b), $d(z, \partial N_i) \geq \varepsilon_{m,i}$ implies $U_i(\Theta, z) \in \mathcal{B}_m$ for all $\Theta \in \mathcal{B}_m$. The definition of $C_{m,i}$ gives the claimed inclusion. Openness/nonemptiness follow because each N_i is open and $\varepsilon_{m,i} > 0$ is minimal; trimming by a positive margin leaves a nonempty open subset. □

2.2 Sequential Test

Definition 2.2 (Sequential Composition)**.** Let

$$\mathfrak{T}' = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{mid}}, U', E', \mathcal{S}') \quad \text{and} \quad \mathfrak{T}'' = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U'', E'', \mathcal{S}'')$$

share the same state space $\mathcal{M}$ and input space $\mathcal{Z}^{\text{in}}$. Their *Sequential Test* is

$$\mathfrak{T}_{\text{seq}} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U_{\text{seq}}, E_{\text{seq}}, \mathcal{S}_{\text{seq}}),$$

where

$$U_{\text{seq}}(\Theta, z) = U''\big(U'(\Theta, z), z\big), \qquad E_{\text{seq}}(\Theta) = E''(\Theta),$$

and the standard can be taken as

$$\varepsilon_n^{\text{seq}} = \min(\varepsilon_n', \varepsilon_n'').$$

Lemma 2.2 (Metric Compatibility for Sequential Composition)**.** *Let* $(\mathcal{Z}^{\text{out}}, d_{\text{out}})$ *be the output space of the first primitive and* $(\mathcal{Z}^{\text{in}}, d_{\text{in}})$ *the input space of the second, and let* $T : \mathcal{Z}^{\text{out}} \to \mathcal{Z}^{\text{in}}$ *be the connecting map. If* T *is Lipschitz with constant* $L_T < \infty$*, then any tolerance* ε *in* $(\mathcal{Z}^{\text{in}}, d_{\text{in}})$ *corresponds to tolerance* $L_T^{-1}\varepsilon$ *in* $(\mathcal{Z}^{\text{out}}, d_{\text{out}})$*. In particular, on compact metric spaces,* L_T *is finite, so branchwise minimal tolerances can be transported uniformly between primitives.*

Proposition 2.1. *If $\mathfrak{T}'$ and $\mathfrak{T}''$ are valid tests, then $\mathfrak{T}_{\text{seq}}$ is a valid test. Continuity and forward-closure are preserved under the chosen $\mathcal{S}_{\text{seq}}$.*

Proof. Composition of continuous maps is continuous. Guard-band radii shrink to the minimum to guarantee both stages remain inside their respective shells; thus forward-closure holds. Minimality of $\mathcal{S}_{\text{seq}}$ follows by the same argument (cannot enlarge either constituent without breaking soundness). □

2.3 Conditional Test

Definition 2.3 (Conditional Test)**.** Let $\{N_i\}_{i=1}^k$ be a finite open partition of $\mathcal{Z}^{\text{in}}$ and suppose we have k update maps

$$U_i : \mathcal{M} \times N_i \to \mathcal{M} \qquad \text{and a common emission} \quad E : \mathcal{M} \to \mathcal{Z}^{\text{out}}.$$

Define

$$U_{\text{cond}}(\Theta, z) = U_i(\Theta, z) \quad \text{if } z \in N_i,$$

and set $\mathfrak{T}_{\text{cond}} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U_{\text{cond}}, E, \mathcal{S}_{\text{cond}})$ with

$$\varepsilon_n^{\text{cond}} := \min_{1 \le i \le k} \varepsilon_{n,i},$$

where $\varepsilon_{n,i}$ guarantees forward-closure for branch i.

Proposition 2.2. *If each branch $(\mathcal{M}, N_i, \mathcal{Z}^{\text{out}}, U_i, E, \mathcal{S}_i)$ is a valid test on its domain N_i, and trajectories never cross the boundaries ∂N_i once certified (guard-bands leave margins), then $\mathfrak{T}_{\text{cond}}$ is a valid test.*

Proof. U_{cond} equals U_i on N_i, preserving continuity there. Guard-band margins ensure no reachable trajectory hits ∂N_i. Taking the minimum radius preserves forward-closure on all branches simultaneously, and minimality is inherited (any larger radius would violate some branch). □

2.4 Loop-Until Test

We now give a precise definition of the *Loop-Until* test, making explicit its dependence on the primitives (Conditional, Sequential) and on the inductive guarded decomposition (Theorem 5.1).

Definition 2.4 (Loop-Until Test)**.** Let

$$\mathcal{S} = \{\varepsilon_n\}_{n \ge 1}, \qquad \mathcal{B}_n = \big\{\Theta : \operatorname{dist}(\Theta, \mathcal{B}) \le \varepsilon_n\big\},$$

and let for each n the depth-n Conditional map

$$U^{(n)} : \mathcal{M} \times \mathcal{Z}^{\text{in}} \longrightarrow \mathcal{M} \quad \text{(Theorem 5.1)},$$

which itself is built by nesting Sequential and Conditional primitives on the cells $N_{\mathbf{i}}^{(n)}$. Define the *Loop-Until* state-update

$$\widetilde{U} \;:\; (\mathcal{M} \times \mathbb{N}) \times \mathcal{Z}^{\text{in}} \;\longrightarrow\; \mathcal{M} \times \mathbb{N}$$

by

$$\widetilde{U}\big((\Theta, n),\, z\big) = \begin{cases} \big(U^{(n)}(\Theta, z),\, n\big), & \Theta \notin \mathcal{B}_n, \\ \big(\Theta,\, n+1\big), & \Theta \in \mathcal{B}_n \text{ (advance level or halt if } n \text{ final).} \end{cases}$$

That is, the update first applies $U^{(n)}$ at the current depth n and checks membership in $\mathcal{B}_n$; only upon success is n incremented (or the test halted if the target basin $\mathcal{B}$ has been reached).

More formally, the *Loop-Until Test* operates as follows:

- It reads an *input stream* $\{z_0, z_1, \dots\}$ in discrete ticks $t = 0, 1, 2, \dots$.
- It maintains an internal pair (Θ_t, n_t), where $\Theta_t \in \mathcal{M}$ is the current belief-state and $n_t \in \mathbb{N}$ is the current shell index, initialized to $(\Theta_0, 1)$.
- At each tick t:
  1. Apply the depth-n_t update map to obtain
  $$\Theta_{t+1} \;=\; U^{(n_t)}(\Theta_t,\, z_t).$$
  2. **Check for level completion:** If $\Theta_{t+1} \in \mathcal{B}_{n_t}$, then
     (a) If $\mathcal{B}_{n_t}$ is the final target basin $\mathcal{B}$, emit $E(\Theta_{t+1})$ and halt.
     (b) Otherwise, advance the shell index by setting $n_{t+1} := n_t + 1$.
  3. If $\Theta_{t+1} \notin \mathcal{B}_{n_t}$, keep the same shell index by setting $n_{t+1} := n_t$.

In this form, the membership test is always performed against the same shell $\mathcal{B}_{n_t}$ that the depth-n_t map is guaranteed (by the inductive decomposition) to reach in a single tick, preserving the one-tick-per-level progress property.

Its *input time-domain* is the full measurement stream $\{z_t\}$; its *output time-domain* is exactly the single tick at which it halts.

2.5 Closure Under Composition

Proposition 2.3 (Primitive Closure)**.** *The class of valid tests is closed under any finite nesting and composition of the two primitives: Sequential and Conditional.*

Proof. Induct on the composition tree. Each primitive preserves continuity, compactness, forward-closure of shells (with appropriately chosen ε_n), and minimality (by taking minima where necessary). Therefore any finite composite is a valid test. □

2.6 How to prove Loop-Until Test Validity?

Loop-Until Test we defined above iterates over the Sequential and Conditional Tests. There are no worries about the former; however, recall the validity proposition of the latter:

If each branch $(\mathcal{M}, N_i, \mathcal{Z}^{\text{out}}, U_i, E, \mathcal{S}_i)$ is a valid test on its domain N_i,

<u>and trajectories never cross the boundaries ∂N_i once certified (guard-bands leave margins),</u>

then $\mathfrak{T}_{\text{cond}}$ is a valid test.

Therefore, in order to construct a valid Loop-Until Test, we *must* ensure the underlined condition is satisfied.

The weakest necessary axiom is this direct underlined statement itself, formulated in a set-theoretic form below.

2.7 Axiom: Input-Traceability / Open-Preimage

Axiom 2.1 (Input-Traceability / Open-Preimage)**.** *Let $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ be a valid test (Def. 2.1), and let $\mathcal{B}_n = \{\Theta : \text{dist}(\Theta, \mathcal{B}) \leq \varepsilon_n\}$ be the n-th shell induced by the (minimal) Certification Standard $\mathcal{S}$. We postulate:*

$$\forall\, n \geq 1,\ \forall\, \Theta \in \mathcal{M} : \quad U(\Theta, \cdot)^{-1}(\mathcal{B}_n) \text{ is open and nonempty in } \mathcal{Z}^{\text{in}}.$$

Interpretation. This axiom ensures that for every certification shell and every state: (i) there *exists* at least one admissible input that drives the next state into the shell (nonemptiness = existential controllability), and (ii) admissibility is robust to small perturbations of that input (openness = guard-band margin in input space). It is precisely what allows us to *trace back* the certification constraint on state space to concrete, well-behaved input conditions—hence the name "Input-Traceability."

Two Variants of Traceability Assumptions

We now distinguish two variants explicitly:

Definition 2.5 (PL-Traceability / Path-Local Input-Traceability)**.** For each fixed $\Theta \in \mathcal{M}$ and shell index n, the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is nonempty and open. This is the minimal input-traceability assumption used to certify *realized* execution paths locally.

Definition 2.6 (UI-Traceability / Universal Input-Traceability)**.** PL-Traceability holds, and additionally

$$\bigcup_{\Theta \in \mathcal{M}} U(\Theta, \cdot)^{-1}(\mathcal{B}_n) = \mathcal{Z}^{\text{in}},$$

i.e., every input lies in some open preimage of the basin. Under this stronger assumption one can extract a global finite guarded partition of $\mathcal{Z}^{\text{in}}$ at tick 0.

Remark 2.3 (Scope of Traceability Assumptions)**.** In Chapters 1–2 we work under *UI-Traceability* (Def. 2.6) to obtain *global* finite guarded decompositions with short proofs. In Chapter 3 we weaken this to *PL-Traceability* (Def. 2.5) and recover the Fundamental Test Equivalence via the ATRT Theorem, showing that UI is not essential, reflecting the metrological practice. We call this out here so every global decomposition result in this chapter can assume UI-Traceability *explicitly*.

2.8 Corollary: Analogy to Classical Traceability

Corollary 2.1 (Structural Traceability Chain)**.** *Under the hypotheses of Theorem 2.1, the family of local standards $\{\mathcal{S}^{(i)}\}_i$ forms a finite "traceability chain" for $\mathcal{S}$: every certified result can be referenced—via the finite decomposition tree—to a sequence of local tolerances whose meet equals the global tolerance.*

This mirrors (but is not identical to) classical metrological traceability:

1. ***Hierarchy of tolerances.*** *Local $\varepsilon_{n,i}$'s play the role of uncertainty allocations at each comparison step.*

2. ***Finite chain.*** *Compactness + open-preimage ensure finitely many branches (no infinite regress).*

3. ***Unbroken linkage.*** *Guard-band forward-closure guarantees no gap in the logical chain from input admissibility to final certified state.*

Proof. Each certified emission is produced at a leaf or node of the finite decomposition tree. Tracing upward, its admissibility rests on local radii at each branch it traversed. Taking the minimum across siblings preserves global robustness. Thus the certified output is "covered" by a finite set of local tolerances whose conjunction equals the global one—exactly the notion of a structural traceability linkage. □

Remark 2.4 (Difference from Institutional Traceability)**.** Classical traceability involves referencing measurements to national/international standards through documented calibrations over time. Our "traceability" is structural and topological: it references the *Certification Standard* down to input conditions through the update map. The analogy is conceptual (hierarchical propagation of uncertainty/tolerance), not institutional.

2.9 Loop-Until Test Validity

Proposition 2.4 (Validity of the Loop-Until Test)**.** *Under VT1–VT4 of Definition 2.1,* ***VT2 (Reachable invariant basin)****, and the Input-Traceability / Open-Preimage Axiom 2.1, the Loop-Until Test (Definition 2.4) is itself a valid test and halts in finite time.*

Proof. We prove validity by reducing *Loop-Until* to a finite sequential composition of one-tick tests, each of which is valid by the primitives.

Step 1: One-Tick Test Validity. Fix any shell index $n \geq 1$. By Axiom 2.1 and compactness of $\mathcal{Z}^{\text{in}}$, Theorem 5.1 produces a finite guarded decomposition

$$\mathcal{Z}^{\text{in}} \;=\; \bigsqcup_{\mathbf{i}\in I_n} N_{\mathbf{i}}^{(n)}, \quad U^{(n)}\big|_{N_{\mathbf{i}}^{(n)}} = U_{\mathbf{i}}^{(n)},$$

where each restricted map $U_{\mathbf{i}}^{(n)}$ admits a nonempty open guard-band into $\mathcal{B}_n$. Then, by the Conditional-Test Proposition in Section 2.3, the single-step test

$$T_n \;=\; \big(\mathcal{M},\ \mathcal{Z}^{\text{in}},\ U^{(n)},\ E,\ \mathcal{S}\big)$$

is a valid test: it preserves - continuity (VT1), - forward-closure of $\mathcal{B}_n$ (VT4), and - minimality of $\mathcal{S}$ (VT3).

Step 2: Finite Sequencing. By the Sequential-Composition Proposition in Section 2.2, any finite sequential chain of valid tests remains valid.

Halting in finite time follows from the reachable basin property (VT2), together with the Uniform Reachability Lemma 2.5 (Book II, Chapter 6) or, equivalently, the Polynomial Uniform Entry Bound (PUB, Definition 3.1 in Book II, Chapter 6). These results ensure that there exists a finite $T_{\max}$ such that every admissible trajectory enters $\mathcal{B}$ within $T_{\max}$ steps, precluding the possibility of infinite looping inside any outer shell without entry to the basin.

Since every execution of Loop-Until must enter $\mathcal{B}$ in a finite number T of steps, the overall Loop-Until behavior on an input stream $(z_0, z_1, \dots)$ is exactly the sequential composition

$$T_1 \longrightarrow T_2 \longrightarrow \cdots \longrightarrow T_T,$$

followed by the final emission $E(\Theta_T)$. As each T_n is valid, their finite composition is valid.

Step 3: Verifying VT1–VT4 for Loop-Until.

- *VT1 (Compact continuity):* Finite composition of continuous update maps remains continuous; state- and input-spaces are unchanged.
- *VT2 (Reachable invariant basin):* By construction Loop-Until halts *iff* it enters $\mathcal{B}$, which is invariant.
- *VT3 (Minimal standard):* The same minimal Certification Standard $\mathcal{S}$ is used at each iteration without enlargement.
- *VT4 (Shell forward-closure):* Each one-tick step preserves or advances shell membership by virtue of the guarded decomposition.

Therefore, the Loop-Until Test meets all conditions of Definition 2.1 and is valid. □

3 From primitives to the dual representation

We constructed our three primitives needed to build a Turing Machine. However, when we just combine them mechanically in an arbitrary order, we actually mix together two constructions: physics-first and computation-first. They are formed, essentially, as:

Physics-first:

1. *Sequential*: chain two measurements end-to-end,
2. *Loop*: iterate a measurement on its own outputs,
3. *Conditional*: choose between measurements based on a guard-band test. Each of these was realized as a strict promotion of a Simple-Gauge component (input domain, update map, etc.), with *concrete* $\mathcal{Z}^{\text{in}}$, $\mathcal{Z}^{\text{out}}$, U, E, and $\mathcal{S}$.

Computation-first: We treat those same four primitives as abstract operators on discrete state spaces—using them to build any Turing machine (via the usual encoding of states and tape symbols). Here, the focus is purely on the combinatorics of how you compose Seq/Loop/Cond to get universal computation.

But if we mix them in an arbitrary mechanical combination, multiple deep question arise:

- **Hybrid State Space.** How do we blend the *continuous/analog* filter memory Ξ (with noise) with the *discrete* TM control tape Σ so that U remains continuous (or Lipschitz) where needed?

- **Guard-Band Integrity Across Branching.** When a conditional branch selects different "sub-measurements," how do we ensure a single Certification Standard $\mathcal{S}$ (or a coherent family $\{\mathcal{S}_i\}$) still yields a reachable, invariant basin?
- **Timing and Synchronization.** Physical sensors fire at real-time rates; TM steps are logical instants. How do you interlock them so that "tick t" means both "new sensor reading" and "next TM transition"?
- **Termination vs. Convergence.** A TM may never halt; a measurement must eventually converge. How do we guarantee the composite "halts" (enters a basin) exactly when (and only when) the embedded TM halts in polytime?
- **Noise vs. Exact Control.** Physical measurements have stochastic perturbations. A TM step expects a precise symbol. How do we absorb sensor noise into guard-bands without breaking the TM logic?
- **Certificate Semantics.** The NP witness must simultaneously testify to (a) correct convergence of the estimator and (b) correct halting/configuration of the TM. How do you encode both succinctly?
- **Continuity Requirements.** ABET demands continuity (or Lipschitz) of U and E. TM control maps are inherently discontinuous (symbol tests). Can we "smoothly" embed them?
- **Model-vs-Reality Gap.** The composite must remain *physically implementable*: no hidden "oracle" steps. Yet TM universality can demand unbounded tape. How do we reconcile physical finiteness with theoretical unboundedness?
- **Complexity of Verification.** Verifier must replay a composed mix of analog updates and logical transitions in polytime. How do we bound that replay cost?

The very first question suggests the answer.

The big picture is as follows: we have a powerful tool of the guard bands, which converts the noisy, continuous, and random data into *deterministic* statements, but this tool is ONLY available after a while (more precisely, after the adaptation).

What can we do to make the adaptation deterministic in the same way? The only way to do this is to substitute violated constraints using multiplicity of tests which promote input measurements into simpler measurements, and continue this decomposition further until ALL guard bands of ALL finest Testable Measurements are satisfied from the very beginning of their input time-domain ticks during any halting behaviour of the newly-defined-by-decomposition Adaptation Algorithm, which is a Turing Machine by construction.

4 Open-Preimage $\Longleftrightarrow$ Finite Tick-0 Decomposition

In this section, we build the defined-by-decomposition Adaptation Algorithm, as expected.

4.1 Main Decomposition Theorem

We now prove the central equivalence: the open-preimage axiom is *exactly* what is needed and sufficient to obtain a finite guarded decomposition at tick 0.

Theorem 4.1 (Open-Preimage $\Longleftrightarrow$ Finite Tick-0 Decomposition)**.** *Let $\mathfrak{T}$ be a* valid test *(Def. 2.1), so that by Lemma 2.1 each branch in any finite tick-m partition admits a certified input core with a positive guard-band margin. Assume further that $\mathfrak{T}$ satisfies Axiom 2.1 for $\mathcal{B}_1$* universally *on input space (UI-Traceability, Def. 2.6). Then:*

(a) ($\Rightarrow$) There exists a finite open partition

$$\mathcal{Z}^{\text{in}} = \bigsqcup_{i=1}^{k} N_i$$

and restricted maps $U_i := U|_{\mathcal{M}\times N_i}$ such that the certified input core $C_{0,i} \subset N_i$ provided by Lemma 2.1 is nonempty for each i, and $U = \text{Conditional}(U_1, \ldots, U_k)$ on tick 0.

(b) ($\Leftarrow$) Conversely, if such a finite tick-0 decomposition exists whose branches are themselves valid *in the sense of Def. 2.1 (and hence possess nonempty certified input cores by Lemma 2.1), then for every $\Theta \in \mathcal{M}$ the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$ is open and nonempty.*

Proof. **($\Rightarrow$) Sufficiency.** Fix $\Theta \in \mathcal{M}$. By Axiom 2.1,

$$W_\Theta := U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$$

is open and nonempty. The family $\{W_\Theta : \Theta \in \mathcal{M}\}$ covers $\mathcal{Z}^{\text{in}}$. Compactness of $\mathcal{Z}^{\text{in}}$ yields a finite subcover $W_{\Theta_1}, \ldots, W_{\Theta_k}$. Refine the finite cover to a disjoint open partition $N_1, \ldots, N_k$ with $N_i \subseteq W_{\Theta_i}$.

Define $U_i(\Theta, z) := U(\Theta, z)$ for $z \in N_i$. Then for each Θ, $N_i \subseteq U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$ for some i, so

$$G^{\text{in}}_{0,i}(\Theta) = \{z \in N_i : U_i(\Theta, z) \in \mathcal{B}_1\}$$

has nonempty interior (indeed, N_i itself). Finally, define the Conditional primitive:

$$U_{\text{cond}}(\Theta, z) = U_i(\Theta, z) \quad \text{if } z \in N_i,$$

which coincides with U on all of $\mathcal{Z}^{\text{in}}$. This is a finite guarded decomposition at tick 0.

($\Leftarrow$) Necessity.

Suppose $\mathcal{Z}^{\text{in}} = \bigsqcup_{i=1}^{k} N_i$ is a finite open partition and each restricted map U_i has a nontrivial guard-band into $\mathcal{B}_1$, i.e., for every Θ,

$$G^{\text{in}}_{0,i}(\Theta) = \{z \in N_i : U_i(\Theta, z) \in \mathcal{B}_1\}$$

has nonempty interior in N_i. Then for any Θ,

$$U(\Theta, \cdot)^{-1}(\mathcal{B}_1) = \bigcup_{i=1}^{k} G^{\text{in}}_{0,i}(\Theta),$$

a finite union of open, nonempty sets, hence open and nonempty. This is exactly the open-preimage condition. □

Corollary (Local Decomposition under PL-Traceability). If only PL-Traceability holds, then for any actual observed input z and current state Θ with $z \in U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$, there exists a finite guarded refinement in a neighborhood of z (derived from openness) that certifies the tick-0 branch along the realized path. This decomposition is local and need not extend to all of $\mathcal{Z}^{\text{in}}$.

4.2 Summary and Interpretive Table

Concept	Role / Meaning
Open-Preimage (Axiom 2.1)	Ensures every shell has a robust, nonempty footprint in input space for each state; enables input-level traceability and finite coverings.
Finite Tick-0 Decomposition	Carves $\mathcal{Z}^{\text{in}}$ into finitely many guarded cells N_i, each with its own sub-map U_i and local guard-band. Implements the Conditional primitive.
Primitive-Induced Standards (Theorem 2.1)	Local forward-closure radii $\varepsilon_{n,i}$ exist for each branch; the global ε_n is their minimum.
Traceability Analogy (Cor. 2.1)	The chain of local standards $\mathcal{S}^{(i)}$ whose meet is $\mathcal{S}$ mirrors classical traceability: hierarchical propagation of tolerances.
Equivalence Theorem (Thm. 4.1)	Open-preimage $\iff$ finite guarded decomposition at tick 0 (logical equivalence, not just implication).

Table 1: Logical roles of the main ingredients in Section 4.

Big Picture. Open-preimage is the *minimal topological axiom* guaranteeing that certification constraints can be pulled back to inputs in a robust, finitely describable way. From there, everything else—finite decomposition, local standards, structural traceability—follows.

4.3 Diagram: From Open-Preimage to Finite Conditional Tree

4.4 Edge Cases and Necessity Remarks

Remark 4.1 (If Nonemptiness Fails)**.** If for some Θ, $U(\Theta, \cdot)^{-1}(\mathcal{B}_1) = \varnothing$, then no admissible input exists to ensure forward-closure at tick 0. Any certificate would be unsound or incomplete. Thus nonemptiness is necessary.

Remark 4.2 (If Openness Fails)**.** If $U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$ is nonempty but has empty interior (e.g., a fractal dust), then robustness collapses: infinitesimal noise can kick the system out of the admissible region. Moreover, one generally needs infinitely many "hairline" cells to cover $\mathcal{Z}^{\text{in}}$. Finite tick-0 decomposition fails, showing openness is necessary for our finite guarded construction.

Remark 4.3 (Beyond Tick 0)**.** The same equivalence holds shell-by-shell (Section 5): open-preimage at every n allows inductive refinement and a Loop-Until construction. Conversely, any such finite inductive decomposition implies open-preimage at each level.

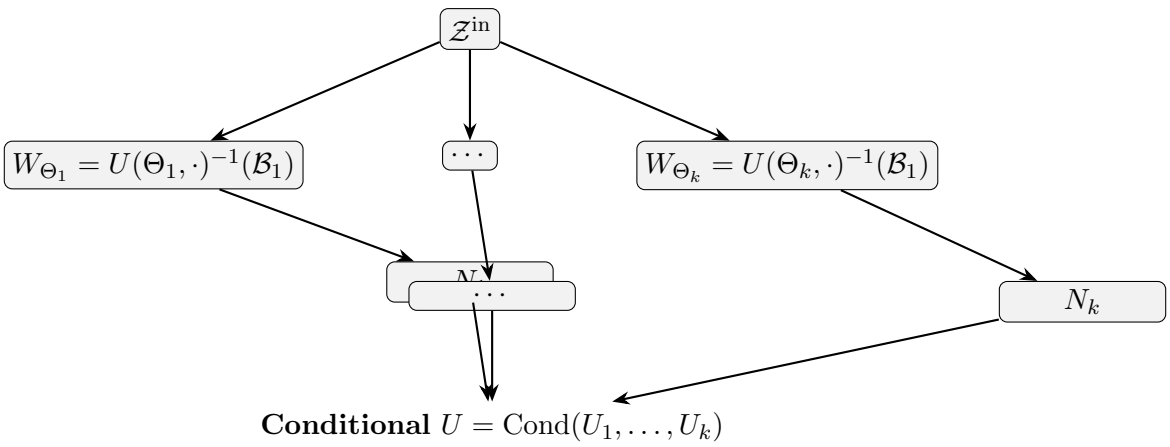

Figure 1: Open-preimage gives an open cover $\{W_{\Theta_i}\}$; compactness $\Rightarrow$ finite subcover; refinement $\Rightarrow$ disjoint partition $\{N_i\}$; restriction $\Rightarrow$ sub-maps U_i, assembled via Conditional.

5 Inductive (Shell-by-Shell) Decomposition & Loop-Until

In this section we push the finite tick-0 decomposition (Theorem 4.1) through *all* certification shells. We show how to refine guarded branches level by level and then wrap the whole tree into a single primitive we call *Loop-Until*. This yields a programmatic update map that halts exactly when the state enters the invariant basin $\mathcal{B}$ (or a target shell $\mathcal{B}_N$). The construction is inductive, quantitatively controlled by the Certification Standard $\mathcal{S}$, and qualitatively guaranteed by the Input-Traceability axiom (Axiom 2.1).

5.1 Setup and Notation

Recall:

- $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ is the minimal Certification Standard and $\mathcal{B}_n = \{\Theta : \text{dist}(\Theta, \mathcal{B}) \leq \varepsilon_n\}$ the n-th shell.
- Axiom 2.1 holds for *every* shell: for all n and Θ, $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is open and nonempty.
- By Theorem 4.1, each shell admits a finite tick-0 guarded decomposition.

We denote by I_n the index set of all leaves (cells) at depth n of the decomposition tree and by

$$\{\, N_{\mathbf{i}}^{(n)} \mid \mathbf{i} = (i_1, \ldots, i_n) \in I_n \,\}$$

the corresponding open cells ($\bigsqcup_{\mathbf{i}\in I_n} N_{\mathbf{i}}^{(n)} = \mathcal{Z}^{\text{in}}$). The restricted maps are

$$U_{\mathbf{i}}^{(n)} := U \,\big|_{\mathcal{M}\times N_{\mathbf{i}}^{(n)}}.$$

5.2 Inductive Refinement Lemma

Lemma 5.1 (Shell-by-Shell Refinement)**.** *Assume Axiom 2.1 for $\mathcal{B}_{n+1}$. Suppose at level n we already have a finite guarded decomposition*

$$\{N_{\mathbf{i}}^{(n)},\ U_{\mathbf{i}}^{(n)}\}_{\mathbf{i}\in I_n}$$

such that every leaf admits a nontrivial input guard-band into $\mathcal{B}_n$. Then there exists a finite refinement

$$\{N_{\mathbf{i},j}^{(n+1)},\ U_{\mathbf{i},j}^{(n+1)}\}_{(\mathbf{i},j)\in I_{n+1}}$$

of each leaf $\mathbf{i} \in I_n$ into children indexed by j, with:

1. $N_{\mathbf{i},j}^{(n+1)} \subseteq N_{\mathbf{i}}^{(n)}$ *and* $\bigcup_j N_{\mathbf{i},j}^{(n+1)} = N_{\mathbf{i}}^{(n)}$ *(disjoint open partition);*
2. $U_{\mathbf{i},j}^{(n+1)} = U_{\mathbf{i}}^{(n)}\big|_{\mathcal{M}\times N_{\mathbf{i},j}^{(n+1)}}$*;*
3. *Each child admits a nontrivial input guard-band into $\mathcal{B}_{n+1}$.*

Proof. Fix a leaf $\mathbf{i} \in I_n$. By hypothesis, $(U_{\mathbf{i}}^{(n)}, N_{\mathbf{i}}^{(n)})$ forwards $\mathcal{B}_n$ robustly. To refine to $\mathcal{B}_{n+1} \subseteq \mathcal{B}_n$, consider for any Θ the preimage

$$W_{\mathbf{i},\Theta}^{(n+1)} := \big(U_{\mathbf{i}}^{(n)}(\Theta,\cdot)\big)^{-1}(\mathcal{B}_{n+1}) \subseteq N_{\mathbf{i}}^{(n)}.$$

Axiom 2.1 (applied to $U_{\mathbf{i}}^{(n)}$, which equals U on $N_{\mathbf{i}}^{(n)}$) says $W_{\mathbf{i},\Theta}^{(n+1)}$ is open and nonempty for all Θ. The family $\{W_{\mathbf{i},\Theta}^{(n+1)}\}_{\Theta\in\mathcal{M}}$ covers $N_{\mathbf{i}}^{(n)}$. Compactness of $N_{\mathbf{i}}^{(n)}$ gives a finite subcover, which we refine to a disjoint open partition $\{N_{\mathbf{i},j}^{(n+1)}\}_j$. Restrict $U_{\mathbf{i}}^{(n)}$ to each child to get $U_{\mathbf{i},j}^{(n+1)}$.

By construction, for each Θ,

$$U_{\mathbf{i},j}^{(n+1)}(\Theta, z) \in \mathcal{B}_{n+1} \quad \text{for all } z \in N_{\mathbf{i},j}^{(n+1)}$$

(or at least on a nonempty interior subset, depending on how we trimmed the cells), so the input guard-band $G_{\mathbf{i},j}^{\mathrm{in},(n+1)}(\Theta)$ has nonempty interior. This satisfies (3). Items (1)–(2) are by construction. □

5.3 Inductive Decomposition Theorem

Theorem 5.1 (Inductive Guarded Decomposition)**.** *Assume **UI-Traceability** for every shell $\mathcal{B}_n$ (equivalently: Axiom 2.1 holds and $\bigcup_\Theta U(\Theta,\cdot)^{-1}(\mathcal{B}_n) = \mathcal{Z}^{\mathrm{in}}$).*

Then for every n there exists a finite family $\{N_{\mathbf{i}}^{(n)}, U_{\mathbf{i}}^{(n)}\}_{\mathbf{i}\in I_n}$ such that:

1. $\{N_{\mathbf{i}}^{(n)}\}_{\mathbf{i}}$ *is a disjoint open partition of $\mathcal{Z}^{\mathrm{in}}$;*
2. *On each cell,* $U_{\mathbf{i}}^{(n)} = U|_{\mathcal{M}\times N_{\mathbf{i}}^{(n)}}$*;*
3. *Each $U_{\mathbf{i}}^{(n)}$ admits a nontrivial input guard-band into $\mathcal{B}_n$;*
4. *The depth-n map is obtained by nesting conditionals:*

$$U^{(n)} = \mathrm{Conditional}\big(\{U_{\mathbf{i}}^{(n)}\}_{\mathbf{i}\in I_n}\big) \quad \textit{and equals } U \textit{ on } \mathcal{M}\times\mathcal{Z}^{\mathrm{in}}.$$

Proof. By Theorem 4.1 we have the base case $n = 1$.

For the base case $n = 1$, apply Lemma 2.1 to the finite tick-1 partition obtained from Theorem 4.1 to extract, for each branch N_i, a certified input core $C_{1,i}$ with margin $\varepsilon_{1,i} > 0$. This guarantees the guarded structure required for the Conditional primitive, ensuring that no certified trajectory starting in $C_{1,i}$ can cross ∂N_i in one step.

Assume the statement for depth n; apply Lemma 5.1 to every leaf to obtain the depth $n+1$ family. The nested Conditional just stacks the new branch level under each leaf. Induction completes the proof. □

Comment. The theorem constructs a finite *tree* of guarded sub-maps for every shell. Depth n corresponds to staying safely within $\mathcal{B}_n$ in one step. To actually *stay* in the shell over time or to *enter* $\mathcal{B}$, we need to iterate this tree until the state is inside $\mathcal{B}$. That is the job of *Loop-Until*.

5.4 Algorithmic View and Pseudocode

For intuition, here is an abstract pseudocode embedding the inductive tree:

Loop-Until(Θ_0)
$n \leftarrow 1$
while $\Theta \notin \mathcal{B}$ **do**
choose cell $N_{\mathbf{i}}^{(n)}$ s.t. $z \in N_{\mathbf{i}}^{(n)}$
$\Theta \leftarrow U_{\mathbf{i}}^{(n)}(\Theta, z)$
$n \leftarrow n + 1$ // advance shell index (optional policy)
end while
return (Θ, n)

All the "choices" are guarded by the preimage openness, so every admissible z keeps you inside the current shell. Determinism is retained because U is fixed and z is the actual input; the branch selection is implicit in (Θ, z) landing in a unique cell $N_{\mathbf{i}}^{(n)}$.

5.5 Diagram: Nested Conditionals + Loop

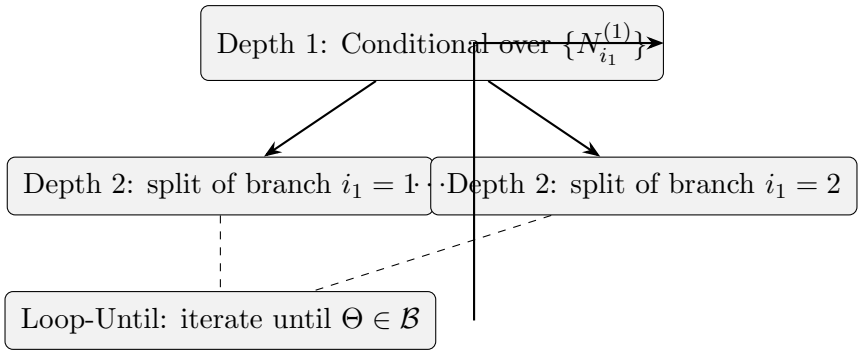

Figure 2: A schematic: first a finite Conditional at depth 1, then each branch is refined at depth 2, etc. The *Loop-Until* primitive iterates the nested tree until the basin is reached. Dashed arrows indicate repeated traversal across depths.

5.6 Edge Cases and Precision Targets

Remark 5.1 (Stopping at a Precision Shell)**.** Sometimes one wants to stop not at $\mathcal{B}$ itself, but when $L(\Theta) \leq \delta$, i.e. $\Theta \in \mathcal{B}_{N(\delta)}$. The same construction works with target shell $N(\delta)$ chosen so that $\varepsilon_{N(\delta)} \leq \delta$. Loop-Until halts when $f(\Theta) \leq N(\delta)$.

Remark 5.2 (Adaptive Shell Policies)**.** Instead of strictly incrementing n, one can keep n fixed until the state is in $\mathcal{B}_n$, then jump to the next n. Both policies produce a finite run. The chosen policy affects $T_{\max}$ by at most a constant factor controlled by the shell gaps.

Remark 5.3 (If Axiom 2.1 Fails at Some Level)**.** If open-preimage holds up to shell N but fails at $N+1$, the inductive refinement stops there. You can still certify up to $\mathcal{B}_N$, but cannot guarantee further tightening. This delineates the boundary of deterministic certifiability for that system.

6 Conclusion

In this Chapter, we completed the following:

1. **Deriving the Primitive Toolkit.** We showed that Sequential, Conditional, and Loop suffice to build any valid composite test. Their use is *not* arbitrary: open-preimage (input traceability) and minimal standards regulate every branch and loop.

2. **Open-Preimage $\Longleftrightarrow$ Finite Tick-0 Decomposition.** The ***traceability axiom*** is exactly the requirement that certification constraints lift to open, non-degenerate input sets. With compactness, this gives a *finite* guarded partition at tick 0.

3. **Inductive Shell Decomposition & Loop-Until.** Shell-by-shell refinement—driven solely by $\mathcal{S}$—plus a terminating loop yields a total guarded program. No extra geometry beyond continuity and compactness is needed.

The ontological results of the Chapter are deep. After the Certification Standard appeared in the theory naturally (Book II, Chapter 6), we see the natural appearance of the next cornerstone concept of metrology - the ***traceability***. Again, nothing was introduced ad hoc here: it is a necessary consequence of the equivalence of the Simple Test to its decomposition.

6.0.1 What's next?

What we know now is that every Simple Test can be equivalently presented as a decomposition into the structure of tests guided by traceability requirement.

Our immediate next goal is to establish properties of the obtained decomposition. First question - if we achieved our goal of the creation of Turing Machine? And the second - how we can be sure it will eventually halt, producing our measurement result?

Chapter 2: Fundamental Test Equivalence Preliminaries

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1 Starting motivation

In the last chapter, we established the deep result: every *valid test* may be dually represented in the decomposed form. While transitioning to this form, we replace the Update Map with a *program* composed from the tree of Update Maps of Test Primitives (which may also operate on its own input subspace). Simultaneously, the solid Certification Standard obtains a dual representation of a *tree of distributed Certification Standards*.

The very first question which arises is: what is the complexity of the resulting decomposition? For a computer scientist, the very appearance of sequential+conditional+loop primitives suggests Turing-completeness.

However, we have the physics in our composition on the other side; maybe it inevitably restricts the complexity? Have we achieved the goal of the Turing Machine creation?

The second question, once we have a positive answer to the first one, can be understood as a reversal of the first one: if we can embed a TM via our primitives, what additional physical or epistemic constraints must hold so that it actually halts (i.e., produces a measurement result)?

And the third question arise once there are positive answers to the first two questions: under which condition can we certify such a result? And further, what does it ever mean that the measurement, which is associated with a precision, to be locked in a precision-limiting deterministic program, is it ever still a measurement?

Let's figure it out.

2 Computational Universality of Guarded Tests

We will show that, under the structural axioms (VT1–VT4) and the Input-Traceability Axiom, the primitive toolkit Sequential, Conditional, and Loop is computationally universal. Concretely, for any deterministic single-tape Turing Machine M and input x, we construct a valid test

$$\mathfrak{T}_{M,x} = \big(\mathcal{M},\ \mathcal{Z}^{\text{in}},\ \mathcal{Z}^{\text{out}},\ U,\ E,\ \mathcal{S}\big)$$

such that:

1. $M(x)$ halts if and only if $\mathfrak{T}_{M,x}$ enters its invariant basin $\mathcal{B}$.
2. All primitive constructions and the Open-Preimage Axiom apply directly, with no additional work.

The key insight is to choose the *input space* to *be* the state space itself:

$$\mathcal{Z}^{\text{in}} \;=\; \mathcal{M}, \quad z_t \;=\; \Theta_t,$$

so that branching on the current TM configuration becomes a *Conditional* primitive on the input domain. We now make this precise in a sequence of self-contained steps.

2.1 Computation Model and Notation

Fix a deterministic, single-tape Turing machine

$$M = (Q,\, \Gamma,\, b,\, q_{\text{init}},\, q_{\text{halt}},\, \delta),$$

where Q is the finite set of control states, Γ the tape alphabet, $b \in \Gamma$ the blank symbol, and $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ the transition function.

State space $\mathcal{M}$. We encode TM configurations into a compact metric space $\mathcal{M} \subset [0,1]^d$ (see Lemma 2.1).

Input and output spaces. To leverage our primitive framework, we set

$$\mathcal{Z}^{\text{in}} = \mathcal{M}, \qquad \mathcal{Z}^{\text{out}} = \mathcal{M}, \qquad E(\Theta) = \Theta.$$

Thus each external input tick simply echoes the current state:

$$z_t = \Theta_t.$$

With this choice, any branch on "control-state $= q$, symbol $= a$" is an CONDITIONAL on $\mathcal{Z}^{\text{in}}$.

2.2 Continuous Encoding of Configurations

We recall how to map a TM configuration continuously into $\mathcal{M}$.

Lemma 2.1 (Encoding Map)**.** *There exists a continuous injection*

$$\mathsf{enc} : \{\textit{TM configurations}\} \;\hookrightarrow\; \mathcal{M} \subset [0,1]^d$$

such that each tape segment and control-state is coded in a base-b expansion with only finitely many nonzero digits.

Proof. Let

$$C = (\ldots, s_{-2}, s_{-1}, s_0, s_1, s_2, \ldots;\; q;\; h)$$

be any configuration of the Turing machine M. We will define $\mathsf{enc}(C) = (L, R, S) \in [0,1]^3 \times [0,1]^m = \mathcal{M}$ as follows.

1. Encoding the tape contents. Fix a base $b \geq |\Gamma|$ and an injective code code : $\Gamma \to \{0, 1, \ldots, b-2\}$, with code$(b) = 0$. Since C has only finitely many non-blank symbols, there is some N so that $s_k = b$ for $|k| > N$. Define

$$R \;=\; \sum_{k=0}^{\infty} \frac{\text{code}(s_k)}{b^{k+1}} \quad \text{and} \quad L \;=\; \sum_{k=1}^{\infty} \frac{\text{code}(s_{-k})}{b^{k}}.$$

- *Well–defined and ultimately zero expansions:* After index N, all $s_k = b$ so $\mathrm{code}(s_k) = 0$; hence both series terminate in finitely many nonzero digits and converge in $[0, 1]$.
- *No ambiguity:* Because only the digit 0 appears infinitely often, we never encounter the "all $(b-1)$" tail that causes base-b ambiguity.

2. Encoding the control state and head position. Let $Q = \{q_0, \ldots, q_{|Q|-1}\}$ and encode q by $\mathrm{enc}(q) \in \{0, \ldots, |Q|-1\}$. Similarly, write the integer head position $h \in \mathbb{Z}$ in signed-digit base b, or equivalently as a pair (h^+, h^-) with $h = h^+ - h^-$ and each of h^+, h^- finitely supported in base b. Pack these into a finite vector

$$S \;=\; \Big(\mathrm{enc}(q),\, h^+,\, h^-\Big) \;\in\; [0,1]^m,$$

again ensuring each coordinate has an ultimately zero base-b expansion.

3. Definition of the map. Set

$$\mathsf{enc}(C) \;=\; (L, R, S) \;\in\; [0,1]^3 \times [0,1]^m = \mathcal{M}.$$

4. Injectivity. Suppose $C \neq C'$. Then either $\exists\, k$ with $s_k \neq s'_k$, or the control state $q \neq q'$, or the head positions differ. In the first case, let k_0 be the smallest index where $s_{k_0} \neq s'_{k_0}$. Then the b-ary expansions of R (if $k_0 \geq 0$) or of L (if $k_0 < 0$) differ in the k_0-th digit, so $\mathsf{enc}(C) \neq \mathsf{enc}(C')$. In the state/head case, the corresponding coordinate of S differs. Hence enc is injective.

5. Continuity. Equip the configuration space with the metric

$$d\big(C, C'\big) = \max\Big\{b^{-\min\{k \geq 0 : s_k \neq s'_k\}},\; b^{-\min\{k > 0 : s_{-k} \neq s'_{-k}\}},\; |\mathrm{enc}(q) - \mathrm{enc}(q')|,\; |h - h'|\Big\},$$

where if two tapes agree on all non-blank cells then the first two terms are zero. Then:

- If C and C' agree on all symbols for $|k| < K$, and share the same q, h, then $|R - R'| \leq \sum_{k \geq K} b^{-(k+1)} = b^{-K}$, and similarly $|L - L'| \leq b^{-K}$. The S-coordinates coincide exactly.
- Thus for any $\varepsilon > 0$, choosing K so that $b^{-K} < \varepsilon$ ensures $d(C, C') < b^{-K} \implies \|\mathsf{enc}(C) - \mathsf{enc}(C')\|_\infty < \varepsilon$.

Hence enc is continuous.

Conclusion. We have produced a well-defined, injective, continuous map enc whose coordinates admit ultimately zero, unambiguous base-b expansions, as required. □

Remark. From now on we identify each configuration with its image $\Theta \in \mathcal{M}$, and we take $\mathcal{Z}^{\mathrm{in}} = \mathcal{M}$.

2.3 Realizing One TM Step with Primitives

Fix a transition $\delta(q, a) = (q', a', d)$. We dispatch on the current input $z \in \mathcal{Z}^{\mathrm{in}} = \mathcal{M}$.

Definition 2.1 (Guard Set in Input Space)**.** For each $(q, a) \in Q \times \Gamma$, define the open cylinder

$$N_{q,a} := \big\{\, z \in \mathcal{Z}^{\text{in}} : \ \text{control-state}(z) = q,\ \text{symbol-under-head}(z) = a \big\}.$$

Each $N_{q,a}$ is open because it fixes finitely many leading code digits.

Lemma 2.2 (Branch Update Map)**.** *On each cell $N_{q,a}$, define*

$$U_{q,a} : \mathcal{M} \times N_{q,a} \ \longrightarrow\ \mathcal{M}$$

to implement exactly one TM step: write a', move the head d, and update the control to q'. Then $U_{q,a}$ is continuous and agrees with the usual TM-transition on encoded states.

Proof. Fix a branch (q, a) and let $N_{q,a} \subset \mathcal{M}$ be the open cylinder on which the control-state coordinate equals q and the tape-symbol under the head equals a. On this cell, the encoded state $\Theta = (L, R, S) \in N_{q,a}$ carries the following information in its base-b expansions:

$$R \ = \ \sum_{k=0}^{\infty} \frac{\text{code}(s_k)}{b^{k+1}}, \quad L \ = \ \sum_{k=1}^{\infty} \frac{\text{code}(s_{-k})}{b^{k}}, \quad S = (\text{enc}(q),\, h^+,\, h^-)\,,$$

where $\text{code}(s_0) = \text{code}(a)$ and h is the head position with signed-digit split (h^+, h^-).

We define $U_{q,a}(\Theta, z)$ by three steps:

1. ***Write the new symbol a'.*** Since $\text{code}(s_0) = \text{code}(a)$ is constant on $N_{q,a}$, the "pop" operation

$$b\,R \ = \ \text{code}(a) \ + \ \underbrace{\sum_{k=1}^{\infty} \frac{\text{code}(s_k)}{b^{k}}}_{=:R_{\text{tail}}}$$

identifies R_{tail}. We then form the updated halves by

$$R_{\text{new}} = b\,R - \text{code}(a) \quad \text{and} \quad L_{\text{new}} = L\,,$$

and prepare to "push" $\text{code}(a')$ in the next step.

2. ***Move the head.*** - If the direction $d = R$, we set

$$L' \ = \ \frac{L + \text{code}(a')}{b}, \quad R' \ = \ R_{\text{new}}.$$

- If $d = L$, symmetrically,

$$R' \ = \ \frac{R + \text{code}(a')}{b}, \quad L' \ = \ L - b\,\text{code}(s_{-1}),$$

where $\text{code}(s_{-1})$ is read off from the first digit of L.

Each of these formulas is an affine map in (L, R) on the open set $N_{q,a}$.

3. ***Update control state and head coordinate.*** The vector $S \in [0,1]^m$ encodes (q,h) in finitely many base-b digits. On $N_{q,a}$, the current digits for q and h are fixed and away from any carry threshold. We define

$$\mathrm{enc}(q') = \text{new control-state code}, \qquad h' = \begin{cases} h+1, & d = R, \\ h-1, & d = L, \end{cases}$$

and re-encode (q', h') in the same digit-window as an affine update on S.

Continuity. For each branch (q,a), we refine its domain cell to fix *all* base-b digits that the branch update rule consults in that step. In particular, for a left move we also fix the symbol s_{-1} (read from the first digit of L) and exclude the base-b boundary points where this digit changes, so that carry/borrow events are eliminated. Denote these refined cells by $N_{q,a,\sigma}$, where σ encodes the additional fixed digits. On each such open cylinder $N_{q,a,\sigma}$, the map $U_{q,a}$ acts by an affine transformation with coefficients determined entirely by (q,a,σ), and is therefore continuous there. Since the domain of $U_{q,a}$ is the finite disjoint union of such refined open cylinders, $U_{q,a}$ is continuous on its domain.

Correctness on TM encodings. By construction, if Θ encodes a configuration C with $\delta(q,a) = (q', a', d)$, then $U_{q,a}(\Theta, \Theta)$ encodes the next configuration C'. Hence $U_{q,a}$ agrees exactly with one step of the deterministic TM on encoded states.

□

Conditional composition. Define the single CONDITIONAL primitive

$$U_{\mathrm{cond}}(\Theta, z) = U_{q,a}(\Theta, z) \quad \text{if } z \in N_{q,a},$$

covering all (q,a). This is the one-tick update used inside the Loop.

2.4 Looping and Halting Basin

Loop-Until instantiation. We build the guarded program

$$\Pi \;=\; \textsc{Loop}\big(U_{\mathrm{cond}},\ \text{halt-test}\big),$$

where at each tick t the input $z_t = \Theta_t$ is used to select the branch $U_{q,a}$, and we update $\Theta_{t+1} = U_{\mathrm{cond}}(\Theta_t, z_t)$.

Halting basin. Set

$$\mathcal{B} = \{\Theta : \text{control-state}(\Theta) = q_{\mathrm{halt}}\}.$$

Then $\mathcal{B}$ is invariant (no further updates once halted) and is entered exactly when M halts.

Lemma 2.3 (Halting $\iff$ Basin Entry)**.** *Under the above construction,*

$$M(x) \textit{ halts at step } T \iff \Theta_T \in \mathcal{B} \iff \Pi \textit{ halts at tick } T.$$

Proof. Each loop iteration simulates one TM step. If M halts in state q_{halt} at step T, then $\Theta_T \in \mathcal{B}$ and Loop-Until stops. Conversely, entry into $\mathcal{B}$ forces the control state to be q_{halt}, so M must have halted. □

2.5 Existence of a Minimal Certification Standard

We assume *global* UI-Traceability (Def. 2.6, Chapter 1) throughout this argument, so that the open-preimage property holds for every shell $\mathcal{B}_n$ and every admissible state. Under this global assumption, the compactness/minimality machinery of Book II applies directly to yield the existence of the global minimal certification standard; without it, the argument applies only to *local* or pathwise standards along a fixed certificate-consistent trajectory.

As a preliminary step, we prove a very helpful theorem that we will also use in following chapters.

Existence of Refinement-Compatible Decompositions

Theorem 2.1 (Refinement Extension of Guarded Decompositions)**.** *Let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ be a composite test map equipped with a Certification Standard at resolution $\varepsilon > 0$, and suppose there is a valid guarded decomposition G_ε of U at that resolution (i.e., a finite guarded program with primitive tests/guards, branch labels, and ranking, consistent with the standard equivalence theorem). Then for any finer resolution $\varepsilon_{\text{ref}} < \varepsilon$ there exists a guarded decomposition $G_{\varepsilon_{\text{ref}}}$ of the same composite map U at resolution ε_{ref} such that:*

1. *$G_{\varepsilon_{\text{ref}}}$ is a* refinement *of G_ε: every branch label and decision in G_ε appears as a prefix in $G_{\varepsilon_{\text{ref}}}$, and any ambiguity present in G_ε is either preserved or resolved consistently (never contradicted).*
2. *The primitive tests/guards in $G_{\varepsilon_{\text{ref}}}$ refine those in G_ε (the partition induced by the finer standard refines the coarser one).*
3. *The ranking function remains the same deterministic tie-breaker, so any previous ranking-based choices persist as canonical representatives.*

As a consequence, given any descending sequence of resolutions $\varepsilon_1 > \varepsilon_2 > \cdots \to 0$, one can inductively build a coherent sequence of guarded decompositions G_{ε_k} whose certificate prefixes refine each other, thereby supporting the composed Double-Bound under Decomposition.

Lemma 2.4 (Robust Branch Stability / Interior Margin)**.** *Let G_ε be a guarded decomposition at resolution ε, with branch predicates defined by continuous decision regions (guard cells) on compact metric spaces. Then for every finite execution prefix in G_ε that makes a definite branch choice, there exists a positive margin $\eta > 0$ (depending on ε and the prefix) such that any perturbation of the underlying state-input pair within distance η yields the same prefix of branch decisions. In particular, the decision is stable under sufficiently fine refinements, and no earlier branch label is reversed when passing to a finer resolution inside that margin.*

Proof. Each branch decision in G_ε is made by membership in a finite intersection / union of open guard cells defined via continuous predicates (the guarded decomposition machinery ensures these are open with nonempty interior when the decision is definite). Therefore, if a given execution prefix selects a particular sequence of branches, the current state-input pair lies in the interior of the corresponding finite intersection of these open sets. By openness, there exists $\eta > 0$ such that the ball of radius η around the pair is still contained in the same intersection, hence yields the same prefix of branch decisions. That η is the required interior margin. Continuity of all involved maps prevents reversal of prior labels within that neighborhood. □

Refinement Extension Theorem Proof. We construct $G_{\varepsilon_{\text{ref}}}$ explicitly from G_ε and argue correctness.

Step 1: Partition refinement. By assumption the Certification Standard at ε_{ref} is strictly finer than at ε; hence the induced partitions of $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ refine the coarser ones. That is, every coarse guard cell or basin shell at scale ε is a union of finer cells at scale ε_{ref}. Use this to define new primitive guards: for each guard or decision region in G_ε, replace it with its refinement into the appropriate collection of ε_{ref}-level subregions. The new guarded program evaluates membership with the finer cells, preserving the logical structure of the original decomposition but with higher resolution.

Step 2: Decision monotonicity and extension. Because the original branch predicates are constructed from continuous functions and guard definitions with interior margins (as per the standard equivalence structure), any decision that was definitive at the coarser level (i.e., a branch taken because the state lay inside the interior of a guard cell) remains valid in the finer subdivision: the point still lies inside some child cell refining the parent, so the corresponding refined predicate evaluates consistently. If a coarse-level branch was ambiguous (multiple child possibilities collapsed), the finer level splits that ambiguity; ranking is applied deterministically (see Step 3) to choose a canonical extension. Thus each branch in G_ε lifts to one or more coherent child branches in $G_{\varepsilon_{\text{ref}}}$, and no earlier decision is reversed—only refined or disambiguated. This persistence is guaranteed quantitatively by Lemma 2.4, which supplies the interior margin preventing reversal of previously taken branches under sufficiently fine ε_{ref}.

Step 3: Ranking consistency. Retain the same deterministic ranking function used in G_ε as the tie-breaker in $G_{\varepsilon_{\text{ref}}}$. Since ranking is resolution-independent by design, any tie previously broken stays broken in the same way; newly resolved finer ambiguities inherit ranking order compatibly. This guarantees canonical extension of previously chosen paths.

Step 4: Assembly of $G_{\varepsilon_{\text{ref}}}$. Compose the refined primitive tests, updated branch structure, and preserved ranking into a full guarded program $G_{\varepsilon_{\text{ref}}}$. By construction it computes the same underlying composite map U (now observed with finer discriminating power) and extends G_ε in the prefix sense: every execution trace of G_ε corresponds to a prefix of some trace in $G_{\varepsilon_{\text{ref}}}$.

Induction. Given a descending chain $\varepsilon_1 > \varepsilon_2 > \cdots$, apply the above step iteratively: from G_{ε_k} construct $G_{\varepsilon_{k+1}}$ that refines it. The sequence of certificate prefixes extracted from these guarded programs is therefore coherent, monotone, and converges to the infinite-precision behavior. This establishes the inductive composability required for the Double-Bound under Decomposition.

□

We must now check that $\mathfrak{T}_{M,x}$ satisfies VT1–VT4 and hence admits a minimal standard $\mathcal{S}$.

Minimal Radii via Refinement: What We Actually Need

The constructions in Sections 2 based on the Chapter 1 0-Tick Decomposition (Thh. 4.1, Chapter 1) never require the *exact* values of minimal guard–band radii. It suffices that they exist (VT1–VT4) and that we can refine any valid guarded decomposition to obtain sound radii that monotonically tighten.

Proposition 2.1 (Upper–Semicomputability of Minimal Radii via Refinement)**.** *Fix a shell index n and a finite tick-0 guarded decomposition of $\mathcal{Z}^{\text{in}}$ into open cells $\{N_i\}_{i=1}^k$ (Theorem 4.1, Chapter 1). For each branch i define the branchwise minimal admissible radius*

$$\varepsilon^\star_{n,i} := \inf\Big\{ r > 0 : \ \forall\, \Theta \in \mathcal{B}_n,\ \forall\, z \in N_i,\ d(z, \partial N_i) \geq r \ \Rightarrow\ U_i(\Theta, z) \in \mathcal{B}_n \Big\},$$

and the global minimal $\varepsilon^\star_n := \min_{1\leq i\leq k} \varepsilon^\star_{n,i}$ (cf. Theorem 2.1, Chapter 1). Then there exists a sequence of finite guarded refinements indexed by $m = 0, 1, 2, \ldots$ and radii $\{\varepsilon^{(m)}_{n,i}\}_{i=1}^k$ such that, writing $\varepsilon^{(m)}_n := \min_i \varepsilon^{(m)}_{n,i}$:

(a) (Soundness) *For all m and all i, if $z \in N_i$ satisfies $d(z, \partial N_i) \geq \varepsilon^{(m)}_{n,i}$, then $U_i(\Theta, z) \in \mathcal{B}_n$ for every $\Theta \in \mathcal{B}_n$.*

(b) (Monotone tightening) *$\varepsilon^{(m+1)}_{n,i} \leq \varepsilon^{(m)}_{n,i}$ for all m and i; hence $\varepsilon^{(m+1)}_n \leq \varepsilon^{(m)}_n$.*

(c) (Convergence) *$\lim_{m\to\infty} \varepsilon^{(m)}_{n,i} = \varepsilon^\star_{n,i}$ for all i, and therefore $\varepsilon^{(m)}_n \downarrow \varepsilon^\star_n$.*

Consequently, the sequences $m \mapsto \varepsilon^{(m)}_{n,i}$ and $m \mapsto \varepsilon^{(m)}_n$ are upper–semicomputable *from the finite refinement data: they give sound (safe) radii at every stage and decrease to the minimal values.*

Proof. **Setup.** Fix n. By Theorem 4.1, Chapter 1 (tick-0 decomposition) and Lemma 2.1, Chapter 1, each branch N_i admits a nontrivial input guard into $\mathcal{B}_n$. By Theorem 2.1 (refinement extension), for each $m \geq 0$ we have a finite disjoint open refinement of N_i into subcells $\{N^{(m)}_{i,\alpha}\}_{\alpha\in A^{(m)}_i}$ whose union is N_i, and such that each restriction $U^{(m)}_{i,\alpha} := U_i|_{\mathcal{M}\times N^{(m)}_{i,\alpha}}$ is continuous and respects the guarded predicate into $\mathcal{B}_n$ on its domain. The index sets $A^{(m)}_i$ are finite and the refinement is nested: every $N^{(m+1)}_{i,\beta}$ lies in some parent $N^{(m)}_{i,\alpha}$.

Local inner radii on subcells. For a fixed branch i and refinement level m, define the *local $\mathcal{B}_n$–core* of a subcell

$$C^{(m)}_{i,\alpha} \ := \ \big\{ z \in N^{(m)}_{i,\alpha} : \ \forall\, \Theta \in \mathcal{B}_n,\ U^{(m)}_{i,\alpha}(\Theta, z) \in \mathcal{B}_n \big\}.$$

By construction of the guarded decomposition and openness of preimages (Axiom 2.1, Chapter 1), each $C^{(m)}_{i,\alpha}$ is open (possibly empty). Define its *inner trimming radius* relative to the ambient branch N_i by

$$r^{(m)}_{i,\alpha} \ := \ \sup\Big\{ \ \rho \geq 0 \ : \ \{z \in N^{(m)}_{i,\alpha} : \ d(z, \partial N_i) \geq \rho\} \ \subset \ C^{(m)}_{i,\alpha} \ \Big\}.$$

(Convention: $\sup \emptyset = 0$.) Intuitively, $r^{(m)}_{i,\alpha}$ is the largest margin from the *global* branch boundary ∂N_i that we can trim inside this subcell while remaining entirely within the certified core $C^{(m)}_{i,\alpha}$.

Branch radius at level m. Set

$$\varepsilon^{(m)}_{n,i} \ := \ \min_{\alpha\in A^{(m)}_i} r^{(m)}_{i,\alpha} \qquad \text{(minimum over a finite set).}$$

Because $A_i^{(m)}$ is finite and each $r_{i,\alpha}^{(m)} \geq 0$, $\varepsilon_{n,i}^{(m)}$ is well-defined and finite.

(a) Soundness. Fix m and i, and take any $z \in N_i$ with $d(z, \partial N_i) \geq \varepsilon_{n,i}^{(m)}$. Since the refinement $\{N_{i,\alpha}^{(m)}\}$ covers N_i, z lies in some subcell $N_{i,\alpha}^{(m)}$. By definition of $\varepsilon_{n,i}^{(m)} \leq r_{i,\alpha}^{(m)}$, we have

$$\{\, w \in N_{i,\alpha}^{(m)} :\ d(w, \partial N_i) \geq \varepsilon_{n,i}^{(m)} \,\} \ \subset\ \{\, w \in N_{i,\alpha}^{(m)} :\ d(w, \partial N_i) \geq r_{i,\alpha}^{(m)} \,\} \ \subset\ C_{i,\alpha}^{(m)}.$$

Hence $z \in C_{i,\alpha}^{(m)}$, which by definition implies $U_{i,\alpha}^{(m)}(\Theta, z) \in \mathcal{B}_n$ for all $\Theta \in \mathcal{B}_n$. But $U_{i,\alpha}^{(m)} = U_i$ on $\mathcal{M} \times N_{i,\alpha}^{(m)}$, so $U_i(\Theta, z) \in \mathcal{B}_n$ for all $\Theta \in \mathcal{B}_n$. This is exactly the required forward-closure at radius $\varepsilon_{n,i}^{(m)}$.

(b) Monotone tightening. Consider levels m and $m+1$. Every child subcell $N_{i,\beta}^{(m+1)}$ lies in a unique parent $N_{i,\alpha}^{(m)}$. By the refinement-extension theorem, guards only *strengthen* under refinement: $C_{i,\beta}^{(m+1)} \subseteq C_{i,\alpha}^{(m)} \cap N_{i,\beta}^{(m+1)}$. Consequently, the largest admissible trimming from ∂N_i that still lies inside the certified core cannot increase when we pass to the child: $r_{i,\beta}^{(m+1)} \leq r_{i,\alpha}^{(m)}$. Taking minima over all children yields

$$\varepsilon_{n,i}^{(m+1)} = \min_{\beta \in A_i^{(m+1)}} r_{i,\beta}^{(m+1)} \ \leq\ \min_{\alpha \in A_i^{(m)}} r_{i,\alpha}^{(m)} \ =\ \varepsilon_{n,i}^{(m)}.$$

Thus $\varepsilon_{n,i}^{(m)}$ is nonincreasing in m, and so is its global meet $\varepsilon_n^{(m)} = \min_i \varepsilon_{n,i}^{(m)}$.

(c) Convergence to the infimum. Fix a branch i. The sequence $(\varepsilon_{n,i}^{(m)})_{m \geq 0}$ is monotone nonincreasing and bounded below by 0, hence converges to some limit $\varepsilon_{n,i}^{(\infty)} \geq 0$. We show $\varepsilon_{n,i}^{(\infty)} = \varepsilon_{n,i}^{\star}$.

(i) Lower bound: $\varepsilon_{n,i}^{(\infty)} \geq \varepsilon_{n,i}^{\star}$. By soundness (part (a)), each $\varepsilon_{n,i}^{(m)}$ satisfies the defining implication of the set in the infimum for $\varepsilon_{n,i}^{\star}$. Therefore $\varepsilon_{n,i}^{(m)} \geq \varepsilon_{n,i}^{\star}$ for all m, and taking limits gives $\varepsilon_{n,i}^{(\infty)} \geq \varepsilon_{n,i}^{\star}$.

(ii) Upper bound: $\varepsilon_{n,i}^{(\infty)} \leq \varepsilon_{n,i}^{\star}$. Let $\eta > 0$. By definition of the infimum $\varepsilon_{n,i}^{\star}$, the radius $r := \varepsilon_{n,i}^{\star} + \eta$ *fails* the forward-closure property: there exist $\Theta^\sharp \in \mathcal{B}_n$ and $z^\sharp \in N_i$ with $d(z^\sharp, \partial N_i) \geq r$ such that $U_i(\Theta^\sharp, z^\sharp) \notin \mathcal{B}_n$. Because $\{N_{i,\alpha}^{(m)}\}_\alpha$ forms a base of shrinking open partitions of N_i, for all sufficiently large m there is a unique subcell $N_{i,\alpha(m)}^{(m)} \ni z^\sharp$ whose diameter is smaller than η (in the metric of $\mathcal{Z}^{\text{in}}$), and such that the refinement preserves the failure $U_i(\Theta^\sharp, z^\sharp) \notin \mathcal{B}_n$. By continuity of U_i and closedness of $\mathcal{B}_n$, there is an open neighborhood $V \subset N_{i,\alpha(m)}^{(m)}$ of $z^\sharp$ on which $U_i(\Theta^\sharp, \cdot) \notin \mathcal{B}_n$. Hence $V \cap \{z :\ d(z, \partial N_i) \geq r\}$ is nonempty and contained in the *complement* of $C_{i,\alpha(m)}^{(m)}$. By definition of $r_{i,\alpha(m)}^{(m)}$, this implies $r_{i,\alpha(m)}^{(m)} < r$, and therefore

$$\varepsilon_{n,i}^{(m)} = \min_\alpha r_{i,\alpha}^{(m)} \ \leq\ r_{i,\alpha(m)}^{(m)} \ <\ r = \varepsilon_{n,i}^{\star} + \eta.$$

Since this holds for all sufficiently large m, we have $\limsup_{m\to\infty} \varepsilon_{n,i}^{(m)} \leq \varepsilon_{n,i}^{\star} + \eta$. Letting $\eta \downarrow 0$ gives $\varepsilon_{n,i}^{(\infty)} \leq \varepsilon_{n,i}^{\star}$.

Combining (i) and (ii) yields $\varepsilon_{n,i}^{(\infty)} = \varepsilon_{n,i}^{\star}$.

Global meet. Because there are finitely many branches, taking minima commutes with limits:

$$\lim_{m\to\infty} \varepsilon_n^{(m)} = \lim_{m\to\infty} \min_{1\le i\le k} \varepsilon_{n,i}^{(m)} = \min_{1\le i\le k} \lim_{m\to\infty} \varepsilon_{n,i}^{(m)} = \min_{1\le i\le k} \varepsilon_{n,i}^{\star} = \varepsilon_n^{\star}.$$

Since $(\varepsilon_n^{(m)})_m$ is nonincreasing, it converges *down* to $\varepsilon_n^{\star}$.

Effectivity / upper–semicomputability. At each finite stage m the refinement is finite, and each $r_{i,\alpha}^{(m)}$ is determined by the finite guarded data on $N_{i,\alpha}^{(m)}$ (continuous predicates on compact open boxes). Thus $\varepsilon_{n,i}^{(m)} = \min_\alpha r_{i,\alpha}^{(m)}$ and $\varepsilon_n^{(m)} = \min_i \varepsilon_{n,i}^{(m)}$ are obtained from finite data and form a nonincreasing sequence of sound radii converging to the minimal values: this is exactly upper–semicomputability. In the pure TM embedding (Section 2.3), after excluding carry/borrow, each branch is affine on a rational open box, so $r_{i,\alpha}^{(m)}$ (hence $\varepsilon_{n,i}^{(m)}$ and $\varepsilon_n^{(m)}$) are *exactly* computable rational numbers at each stage. □

Now we will establish two more lemmas which finalize our preparations to prove the existence of Minimal Certification Standard.

Lemma 2.5 (Continuity and Compactness)**.** *$\mathcal{M}$ is compact; each branch $U_{q,a}$ is continuous; hence so is U_{cond}. Emission $E(\Theta) = \Theta$ is continuous.*

Lemma 2.6 (Open-Preimage Holds along Certificate-Consistent Paths)**.** *Suppose we are given a configuration Θ and a finite certificate witness (i.e., a finite input sequence) demonstrating that from Θ the TM eventually reaches the halting basin $\mathcal{B}$ in T steps. Assume UI-traceability (global open-preimage) holds. Then for each shell $\mathcal{B}_n$ appearing along that halting trajectory, the preimage*

$$U(\Theta,\cdot)^{-1}(\mathcal{B}_n) \;\subset\; \mathcal{Z}^{\text{in}} = \mathcal{M}$$

is open and nonempty. Hence, along any certificate-consistent (halting) path, the Open-Preimage Axiom is satisfied.

Proof. We show openness and nonemptiness for each branch (q, a) that appears along the particular halting sequence.

(i) Openness. As before, each $N_{q,a}$ is an open cylinder in the product topology, and on $N_{q,a}$ the restricted update $U_{q,a}$ is continuous. Hence $U_{q,a}^{-1}(\mathcal{B}_n)$ is open in $N_{q,a}$, and thus open in $\mathcal{M} \times \mathcal{Z}^{\text{in}}$. Taking the finite union over the finitely many relevant branches along the certificate yields that $U(\Theta,\cdot)^{-1}(\mathcal{B}_n)$ is open.

(ii) Nonemptiness (certificate-consistent). Fix a shell $\mathcal{B}_n$ encountered along the halting trajectory from Θ. By assumption there exists a future finite input sequence $z_{1:T}$ steering Θ into $\mathcal{B}$, so in particular for the specific preceding branch (q, a) we have some (Θ', z') with $U_{q,a}(\Theta', z') \in \mathcal{B}_n$. UI-traceability guarantees that the preimage of $\mathcal{B}_n$ under U is nonempty (and open) globally, so the particular local preimage restricted to the halting-consistent branch is also nonempty.

Combining (i) and (ii) along the certificate-consistent sequence, each shell's preimage is open and nonempty. Therefore, for the purposes of constructing a valid test along a halting witness, the Open-Preimage Axiom is satisfied. □

By compactness and Lemma 2.6, the infimum radii defining $\mathcal{S}$ are attained, giving a minimal Certification Standard.

2.6 Computational Universality Theorem

Theorem 2.2 (Universality of Guarded Tests)**.** *Under VT1–VT4 and Axiom 2.1, the above construction maps any deterministic TM M and input x to a valid test $\mathfrak{T}_{M,x}$ using only* CONDITIONAL, LOOP *(and optional* SEQUENTIAL*), such that*

$$M(x) \text{ halts} \iff \mathfrak{T}_{M,x} \text{ enters its basin } \mathcal{B} \iff \text{the Loop-Until program halts.}$$

Hence the guarded-test formalism is Turing-complete.

Remark on Halting Recognition. Although the construction already identifies $\mathcal{B}$ as the halting basin, to ensure the Loop-Until program *terminates* immediately upon entry we note that $\mathcal{B}$ is absorbing and can be equipped with a trivial ranking: define $f(\Theta)$ to be 0 on $\mathcal{B}$ and 1 elsewhere, so that entry into $\mathcal{B}$ is detectable as the unique decrease. Thus the halt-test is robust (decidable in finite time) and the Loop-Until primitive ceases execution exactly when $\Theta_t \in \mathcal{B}$, making the equivalence $M(x)$ halts $\iff$ the Loop-Until program halts precise.

Proof. Combine:

- Lemma 2.1: continuous encoding of configurations.
- Lemmas 2.2 and 2.6: each one-tick test is valid.
- Lemma 2.3: halting $\iff$ basin entry.
- Proposition 2.3: finite composition yields a valid test.
- Lemma 2.5: VT1–VT4 hold, so a minimal standard exists.

Thus $\mathfrak{T}_{M,x}$ is a valid test and simulates M exactly. □

3 Augmenting with Physical Inputs and Belief-State

3.1 Extended State Space

We incorporate the physical belief-state together with the TM configuration into a single composite state.

Definition 3.1 (Augmented State Space)**.** Let

$$\mathcal{M}_{\text{phys}},\ \mathcal{M}_{\text{TM}}$$

be the compact metric state spaces for the physical belief-state Θ^{phys} and the TM configuration Θ^{TM}, respectively. We define the *augmented state space*

$$\widetilde{\mathcal{M}} = \mathcal{M}_{\text{phys}} \times \mathcal{M}_{\text{TM}},$$

equipped with the product metric

$$d_{\widetilde{\mathcal{M}}}\Big((\Theta^{\text{phys}}, \Theta^{\text{TM}}),\ (\Theta'^{\text{phys}}, \Theta'^{\text{TM}})\Big) = \max\big\{d_{\text{phys}}(\Theta^{\text{phys}}, \Theta'^{\text{phys}}),\ d_{\text{TM}}(\Theta^{\text{TM}}, \Theta'^{\text{TM}})\big\}.$$

Remark 3.1 (Compactness)**.** Since both $\mathcal{M}_{\text{phys}}$ and $\mathcal{M}_{\text{TM}}$ are compact, their product $\widetilde{\mathcal{M}}$ is also compact. Continuity of any map out of $\widetilde{\mathcal{M}}$ will follow from continuity on each factor.

3.2 Extended Input Time-Domain

We extend the input time-domain to include both the physical external input and the current belief-state.

Definition 3.2 (Augmented Input Space)**.** Let $\mathcal{Z}^{\text{in}}_{\text{phys}}$ be the compact metric space of physical sensor inputs. We define the *augmented input space*

$$\widetilde{\mathcal{Z}}^{\text{in}} \;=\; \mathcal{Z}^{\text{in}}_{\text{phys}} \;\times\; \mathcal{M}_{\text{phys}} \;\times\; \mathcal{M}_{\text{TM}},$$

with the product metric

$$d_{\widetilde{\mathcal{Z}}}\big((z,\Theta^{\text{phys}},\Theta^{\text{TM}}),\ (z',\Theta'^{\text{phys}},\Theta'^{\text{TM}})\big) = \max\big\{d_{\text{phys}}(z,z'),\ d_{\text{phys}}(\Theta^{\text{phys}},\Theta'^{\text{phys}}),\ d_{\text{TM}}(\Theta^{\text{TM}},\Theta'^{\text{TM}})\big\}.$$

Remark 3.2 (Motivation)**.** By feeding back the current belief-state $(\Theta^{\text{phys}},\Theta^{\text{TM}})$ as part of the "input," we can perform conditional branching on both physical and TM state within the same Conditional primitive.

3.3 Mixed One-Step Update Map

We now define the composite one-tick update on the augmented state and input.

Definition 3.3 (Augmented Update Map)**.** Let

$$U_{\text{phys}} : \mathcal{M}_{\text{phys}} \times \mathcal{Z}^{\text{in}}_{\text{phys}} \to \mathcal{M}_{\text{phys}}, \quad U_{\text{TM}} : \mathcal{M}_{\text{TM}} \times \mathcal{M}_{\text{TM}} \to \mathcal{M}_{\text{TM}}$$

be the physical and TM-step updates, respectively. Define

$$\widetilde{U} : \widetilde{\mathcal{M}} \times \widetilde{\mathcal{Z}}^{\text{in}} \;\longrightarrow\; \widetilde{\mathcal{M}},$$

by

$$\widetilde{U}\big((\Theta^{\text{phys}},\Theta^{\text{TM}}),\ (z^{\text{phys}},\Theta^{\text{phys}},\Theta^{\text{TM}})\big) = \Big(U_{\text{phys}}\big(\Theta^{\text{phys}},\ z^{\text{phys}}\big),\ U_{\text{TM}}\big(\Theta^{\text{TM}},\ \Theta^{\text{TM}}\big)\Big).$$

Remark 3.3 (Well-Definedness)**.** On each augmented input tick $(z^{\text{phys}},\Theta^{\text{phys}},\Theta^{\text{TM}})$, the first factor advances the physical belief by the sensor reading z^{phys}, while the second factor performs the TM step by treating the TM's own configuration Θ^{TM} as its "input."

3.4 Continuity and Compactness

We verify that the extended domains and update map retain the required topological properties.

Lemma 3.1. *The spaces $\widetilde{\mathcal{M}}$ and $\widetilde{\mathcal{Z}}^{\text{in}}$ are compact metric spaces, and the augmented update $\widetilde{U}$ is continuous.*

Proof.

- *Compactness:* Finite products of compact metric spaces remain compact.
- *Metric structure:* We have endowed the products with the sup-metric of their factors, which is a valid metric topology.
- *Continuity of $\widetilde{U}$:* Since both U_{phys} and U_{TM} are continuous, the map

  $$((\Theta^{\text{phys}},\Theta^{\text{TM}}),\ (z^{\text{phys}},\Theta^{\text{phys}},\Theta^{\text{TM}})) \;\longmapsto\; \big(U_{\text{phys}}(\Theta^{\text{phys}},z^{\text{phys}}),\ U_{\text{TM}}(\Theta^{\text{TM}},\Theta^{\text{TM}})\big)$$

  is continuous as a product of continuous projections and compositions.

□

3.5 Conditional Decomposition on the Augmented Domain

We now exhibit a finite open partition of $\widetilde{\mathcal{Z}}^{\text{in}}$ that supports a CONDITIONAL primitive.

Definition 3.4 (Augmented Branch Cells).

1. Let $\{M_j\}_{j=1}^{r}$ be a finite open partition of $\mathcal{Z}^{\text{in}}_{\text{phys}}$ (the physical guard-bands).
2. Let $\{N_{q,a}\}_{(q,a)}$ be the TM cylinders in $\mathcal{M}_{\text{TM}}$ (as before).
3. Then
$$\widetilde{N}_{j,(q,a)} := M_j \ \times \ \mathcal{M}_{\text{phys}} \ \times \ N_{q,a} \ \subset \ \widetilde{\mathcal{Z}}^{\text{in}}$$
runs over finitely many open cells covering the whole augmented input space.

Proposition 3.1 (Augmented Conditional Primitive). *On each cell $\widetilde{N}_{j,(q,a)}$, the map $\widetilde{U}$ restricts to a continuous update. Therefore, the overall $\widetilde{U}$ can be realized as a* CONDITIONAL *primitive over the finite open partition $\{\widetilde{N}_{j,(q,a)}\}$.*

Proof. Each $\widetilde{N}_{j,(q,a)}$ is open by product-set openness. On it, $\widetilde{U}\big((\Theta^{\text{phys}}, \Theta^{\text{TM}}), \cdot\big)$ coincides with a fixed pair of continuous maps $(U_{\text{phys}}, U_{\text{TM}})$. Finitely many such pieces glue to a globally continuous conditional update. □

3.6 Open-Preimage in the Augmented Domain

We check that the augmented update satisfies input-traceability at each shell.

Lemma 3.2 (Augmented Open-Preimage). *This lemma is conditional on the* UI-Traceability *(Def. 2.6, Chapter 1): for every shell $\mathcal{B}_n$ and every state $\Theta \notin \mathcal{B}$ there exists an admissible input z with $U(\Theta, z) \in \mathcal{B}_n$.*

Let $\mathcal{B}_n^{\text{phys}} \subset \mathcal{M}_{\text{phys}}$ and $\mathcal{B}_n^{\text{TM}} \subset \mathcal{M}_{\text{TM}}$ be the respective n-th shells. Then for every $(\Theta^{\text{phys}}, \Theta^{\text{TM}})$ and every n,

$$\widetilde{U}\big((\Theta^{\text{phys}}, \Theta^{\text{TM}}), \cdot\big)^{-1}\big(\mathcal{B}_n^{\text{phys}} \times \mathcal{B}_n^{\text{TM}}\big) \ \subset \ \widetilde{\mathcal{Z}}^{\text{in}}$$

is open and nonempty.

Proof.

- *Openness:* $\widetilde{U}$ is continuous, and $\mathcal{B}_n^{\text{phys}} \times \mathcal{B}_n^{\text{TM}}$ is open, so its preimage is open.
- *Nonemptiness:*
  1. For the TM factor, we appeal to the original TM preimage argument: there is always some TM-branch input Θ^{TM} driving into $\mathcal{B}_n^{\text{TM}}$.
  2. For the physics factor, by assumption of the physical test, there is some z^{phys} driving Θ^{phys} into $\mathcal{B}_n^{\text{phys}}$.

  Taking the product of these admissible inputs gives a point in the preimage.

□

4 Time-Irreversibility/Weak-Fairness Axiom

Axiom 4.1 (Time-Irreversibility / Weak-Fairness (TI/WF))**.** *Let $f : \mathcal{M} \to \mathbb{N}$ be a well-founded ranking function, and define the enabled-decrease set at time t by*

$$D_t \;=\; \{\, z \in G^{\text{in}}(\Theta_t) \mid f(U(\Theta_t, z)) < f(\Theta_t) \,\}.$$

An infinite execution $(\Theta_0, z_0, \Theta_1, z_1, \dots)$ is called fair *if whenever $D_t \neq \varnothing$ for infinitely many t (i.e., the set $\{t : D_t \neq \varnothing\}$ is infinite), it holds that $z_t \in D_t$ for infinitely many t; equivalently,*

$$\limsup_{t\to\infty} \mathbf{1}_{D_t \neq \varnothing} = 1 \quad\Longrightarrow\quad \limsup_{t\to\infty} \mathbf{1}_{z_t \in D_t} = 1.$$

In words, if decrease is enabled infinitely often, then it is taken infinitely often.

Intuitively, it is a full analogue of a time arrow in our measurement. However slow it is, it always points in one direction.

4.1 Sufficiency of Time-Irreversibility/Weak-Fairness

We now show that the Time-Irreversibility / Weak-Fairness axiom (TI/WF, Axiom 4.1), together with the structural hypotheses of Book III, Chapter 1, suffices to guarantee *uniform polynomial reachability* under the guarded decomposition. The polynomial bound is provided by PUB (Polynomial Uniform Bound) from ABET (Book II, Chapter 6, Corollary 3.1).

Theorem 4.1 (Sufficiency of TI/WF under Decomposition)**.** *Let $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ be a valid test satisfying the structural axioms (VT1–VT4), the Open-Preimage/Traceability hypothesis at every shell (UI-Traceability), and admitting the guarded tick-0 (and shell-wise) decomposition of Chapter 1. Define the well-founded ranking*

$$f(\Theta) \;:=\; \min\Big\{\, t \in \mathbb{N} : \;\exists\, z_{1:t} \text{ with } U^t(\Theta; z_{1:t}) \in \mathcal{B} \,\Big\},$$

with $f(\Theta) = 0$ iff $\Theta \in \mathcal{B}$. Assume the Time-Irreversibility / Weak-Fairness axiom (Axiom 4.1) holds for the induced Loop-Until execution, and that PUB holds (Book II, Chapter 6, Cor. 3.1). Then there exists a polynomial $p(\cdot)$ such that every fair execution starting from Θ_0 reaches $\mathcal{B}$ in at most

$$T_{\max}(\Theta_0) \;\le\; f(\Theta_0) \;\le\; p\big(|\Theta_0|\big)$$

ticks. In particular, the guarded Loop-Until program halts in uniformly polynomial time on all fair runs.

Proof. (1) Ranking and existential decrease.) By definition of f, for any $\Theta \notin \mathcal{B}$ there exists an admissible input z such that

$$f\big(U(\Theta, z)\big) \;\le\; f(\Theta) - 1.$$

Thus the enabled-decrease set $D_t = \{z \in G^{\text{in}}(\Theta_t) : f(U(\Theta_t, z)) < f(\Theta_t)\}$ is nonempty whenever $\Theta_t \notin \mathcal{B}$.

(2) Fairness $\Rightarrow$ progress.) By Axiom 4.1 (TI/WF), if $D_t \neq \varnothing$ infinitely often then $z_t \in D_t$ infinitely often. Since $D_t \neq \varnothing$ for every t with $\Theta_t \notin \mathcal{B}$, fairness ensures that $f(\Theta_t)$ decreases infinitely often unless it hits 0.

(3) Well-foundedness ⇒ finite time.) Because $f : \mathcal{M} \to \mathbb{N}$ is well-founded and strictly decreases whenever a choice in D_t is taken, there can be at most $f(\Theta_0)$ such decreases. Hence every fair run reaches $\mathcal{B}$ in at most $T_{\max}(\Theta_0) \leq f(\Theta_0)$ ticks.

(4) Primitive timing bounds and their role. Recall from Test Primitives Sec. 1 from Chapter 1 that each primitive test in our framework has a uniform, certification–preserving bound on the number of *input* ticks needed to enter its target basin: a *Sequential* test reaches its basin within two input ticks and emits within one output tick; a *Conditional* test satisfying UI–Traceability reaches in a single input tick and emits within one output tick; a *Loop–Until* test reaches its basin within a number of input ticks bounded by a polynomial in the shell index (by the PUB property) and emits within one output tick. Since the inductive decomposition (Theorem 5.1, Chapter 1) nests only such primitives shell–by–shell, these bounds compose directly: under the TI/WF axiom, whenever a decrease move is available at some shell it is taken within its per–primitive bound, and PUB then bounds the total completion time by a polynomial in the target shell index. This justifies applying PUB at the decomposition level without any additional assumptions on the guards.

(5) Polynomial bound via PUB.) ABET's Polynomial Uniform Bound (Book II, Chapter 6, Cor. 3.1) guarantees the existence of a polynomial p such that the minimal hitting time of the basin from Θ_0 is at most $p(|\Theta_0|)$ under the standing hypotheses (compactness, continuity, minimal standard, open-preimage, and polytime verifiability). Since $f(\Theta_0)$ is by definition the minimal such hitting time, we have $f(\Theta_0) \leq p(|\Theta_0|)$. Combining with Step (3) yields the claimed uniform polynomial bound. □

Remark 4.1 (Role of decomposition)**.** The guarded tick-0 (and shell-wise) decomposition is used to ensure that admissible inputs (open preimages) exist and remain robust at each step, so $D_t \neq \varnothing$ whenever $\Theta_t \notin \mathcal{B}$. TI/WF then converts this *existential* progress into *actual* progress along every fair run; PUB upgrades the uniform bound to a polynomial one in the input size.

4.2 Necessity of Time-Irreversibility/Weak-Fairness

Proposition 4.1 (Non-Termination Without TI/WF)**.** *Let*

$$\widetilde{U} : (\mathcal{M}_{\text{phys}} \times \mathcal{M}_{\text{TM}} \times \mathcal{Z}^{\text{in}}_{\text{phys}}) \;\times\; \mathcal{Z}^{\text{in}}_{\text{phys}} \;\longrightarrow\; \mathcal{M}_{\text{phys}} \times \mathcal{M}_{\text{TM}}$$

be the augmented update map of Section 3.3, and let $\widetilde{f} : \mathcal{M}_{\text{phys}} \times \mathcal{M}_{\text{TM}} \to \mathbb{N}$ *be any well-founded ranking satisfying only the existential decrease condition:*

$$\forall\, \xi \notin \mathcal{B}, \quad \exists\, z : \quad \widetilde{f}\big(\widetilde{U}(\xi, z)\big) < \widetilde{f}(\xi).$$

If we do not *assume Axiom 4.1 (Time-Irreversibility/Weak-Fairness), then there exists an input stream* $\{\tilde{z}_t\}_{t=0}^{\infty} \subset \widetilde{\mathcal{Z}}^{\text{in}}$ *such that the corresponding Loop-Until execution* $(\xi_t)_{t=0}^{\infty}$ *never enters* $\mathcal{B}$*, even though at each state there is some admissible move that would decrease* $\widetilde{f}$*. Hence, without TI/WF, no deterministic termination guarantee is possible.*

Proof. At any current state ξ_t, the existential ranking property ensures the *existence* of at least one "good" input $z_t^{\downarrow}$ for which

$$\widetilde{f}\big(\widetilde{U}(\xi_t, z_t^{\downarrow})\big) \;<\; \widetilde{f}(\xi_t).$$

However, open-preimage (Axiom 2.1, Chapter 1) also guarantees the existence of *other* admissible inputs $z_t^{\circlearrowleft}$ that *do not* decrease $\widetilde{f}$:

$$\widetilde{f}\big(\widetilde{U}(\xi_t, z_t^{\circlearrowleft})\big) \;=\; \widetilde{f}(\xi_t).$$

In the absence of any fairness constraint, an adversarial environment can simply choose $z_t = z_t^{\circlearrowleft}$ at *every* step. By construction,

$$\widetilde{f}(\xi_{t+1}) = \widetilde{f}\big(\widetilde{U}(\xi_t, z_t)\big) = \widetilde{f}(\xi_t), \quad \forall\, t,$$

so the ranking never drops, and ξ_t never reaches the basin $\mathcal{B}$. Thus the Loop-Until loop fails to terminate—even though at each state a terminating move was available—showing TI/WF is *necessary* for any guaranteed halting behavior. □

4.3 Loop-Until and Ranking Lifted

Finally, we observe that the Loop-Until construction and the ranking argument carry over verbatim to the augmented setting.

Proposition 4.2. *Under the usual TI/WF and polynomial-bound assumptions, the Loop-Until program built from $\widetilde{U}$ on the augmented state/input terminates in polynomial time and admits a well-founded ranking $\widetilde{f} : \widetilde{\mathcal{M}} \to \mathbb{N}$.*

Sketch.

- *Ranking:* Define $\widetilde{f}(\Theta^{\text{phys}}, \Theta^{\text{TM}}) = f_{\text{phys}}(\Theta^{\text{phys}}) + f_{\text{TM}}(\Theta^{\text{TM}})$, summing the individual rankings; each step decreases one summand by at least one.
- *Loop-Until:* The same inductive decomposition and Loop-Until procedure applies to $\widetilde{U}$ because it is built from Conditional/Sequential on an open finite partition.
- *Termination:* TI/WF guarantees that whenever progress is possible in either the physical or TM coordinate, it eventually happens. Hence the combined ranking drops to zero in at most $f_{\text{phys}}(\Theta_0^{\text{phys}}) + f_{\text{TM}}(\Theta_0^{\text{TM}})$ steps, which is polynomially bounded.

□

4.4 Minimal Certification Standard in Augmented Decomposition

We already proved the existence of Minimal Certification Standard in TM-embedding setup. The corollary of the Upper–Semicomputability of Minimal Radii via Refinement (Prop. 2.1) explains why this is enough for universality in the augmented setup, given the TI/WF axiom.

Corollary 4.1 (Universality and TI/WF Do Not Require Exact Minima)**.** *All results that establish:* (i) *computational universality (Theorem 2.2), and* (ii) *uniform polynomial reachability under TI/WF (Theorem 4.1), use only that for each shell there exists* some *valid guard–band radius and that these radii can be tightened by refinement. Consequently, the proofs go through verbatim when ε_n is replaced by any $\varepsilon_n^{(m)}$ from Proposition 2.1; no appeal to the exact minimal values $\varepsilon_n^{\star}$ is needed.*

Remark 4.2 (TM embedding vs. augmented setup)**.** In the TM-only embedding (Section 2.3), on any fixed finite cylinder refinement that excludes carry/borrow, each branch map is affine on a rational open box, so the *local* per-branch tick-0 radius is exactly computable. Globally (across all future refinements and Loop-Until iterations) exact minimality may be non-computable; this is immaterial: Proposition 2.1 supplies a decreasing sequence of valid radii converging to the minimal meet, which is all that Theorems 2.2 and 4.1 ever use. In the augmented (physics+TM) setup, the same conclusion holds under UI-Traceability; exact minima are unnecessary for the halting/NP-certifiability claims.

5 Ranking Function & Reachability

Above, we established the way to convert an arbitrary Turing Machine algorithm with the endowed physical spaces and transitions into 0-tick-decomposed structure of our primitives. We also established the existential condition for such an endowment: *TI/WF axiom*.

However, before we can argue this structure can restore a Simple Test behavior, we must ensure the *polynomial* reachability in our structure.

This section isolates the *combinatorial* core of reachability: a well-founded *ranking function* that strictly decreases along (at least one) admissible input step until the basin is reached. We show:

- A ranking is *necessary* for any deterministically certifiable test (it comes "for free" from the Attractor Basin Equivalence).
- Together with the structural axioms (continuity/compactness, minimal standard, open-preimage), a polynomially bounded ranking is also *sufficient* for certifiability.
- Rankings are preserved by the finite guarded decomposition (Section 5, Chapter 1), hence yield a uniform bound on the Loop-Until program.
- The classical Lyapunov-border function is a *universal* (for all inputs) continuous analogue; the ranking is an *existential* (for some input) discrete analogue. Either can drive reachability, but the ranking is the weakest needed.

5.1 Well-Founded Rankings: Definition and Intuition

Definition 5.1 (Well-Founded Ranking Function)**.** Let $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ be a valid test with invariant basin $\mathcal{B}$. A function $f : \mathcal{M} \to \mathbb{N}$ is a *well-founded ranking* if:

1. $f(\Theta) = 0 \iff \Theta \in \mathcal{B}$;
2. For every $\Theta \notin \mathcal{B}$ there exists at least one input $z \in \mathcal{Z}^{\text{in}}$ such that
$$f\big(U(\Theta, z)\big) < f(\Theta).$$

We call (2) an *existential decrease* condition.

Intuition. $f(\Theta)$ is the "remaining distance in steps" to the basin along *some* acceptable evolution. It need not decrease under *every* input (contrast with a Lyapunov function), only that there is at least one guard-band-admissible input that drives it down.

5.2 Necessity of a Ranking for Certifiability

Proposition 5.1 (Ranking is Necessary)**.** *If $\mathfrak{T}$ is deterministically certifiable in NP (i.e. admits a polynomial-size witness for its tail predicate), then there exists a well-founded ranking function $f : \mathcal{M} \to \mathbb{N}$.*

Note. The uniform finite hitting time $T_{\max}$ invoked below is guaranteed to be polynomially bounded by Corollary 3.1 (PUB) under the hypotheses of the Attractor Basin Equivalence Theorem (ABET), including compactness, open-preimage, and the existence of a polynomial-time verifier. Hence the definition of f below uses this uniform entry bound as its origin.
PUB dependency (explicit). Proposition 5.1 guarantees only the *existence* of a well-founded ranking f. It does *not* supply any polynomial bound on $f(\Theta)$. Whenever a polynomial bound on f is used later in this chapter, it is derived from the Polynomial Uniform Entry Bound (PUB) of Book II, Chapter 6, Corollary 3.1—not from ABET alone.

Proof. By the Attractor Basin Equivalence Theorem, certifiability implies: (i) existence of a reachable, invariant basin $\mathcal{B}$; and (ii) a uniform bound $T_{\max}$ (polynomial in the input description) on the time to enter $\mathcal{B}$ for every trajectory (or, more precisely, for every head consistent with acceptance).

Define

$$f(\Theta) := \min\big\{\, t \in \mathbb{N} : \ \exists\, z_{1:t} \text{ with } U^t(\Theta; z_{1:t}) \in \mathcal{B} \,\big\},$$

with $f(\Theta) = 0$ if $\Theta \in \mathcal{B}$. Because $T_{\max}$ bounds the search, $f(\Theta)$ is well-defined and finite.

Now fix $\Theta \notin \mathcal{B}$. By definition of $f(\Theta)$, there is a sequence $z_{1:f(\Theta)}$ steering Θ into $\mathcal{B}$ in exactly $f(\Theta)$ steps. Let $z = z_1$. Then

$$f\big(U(\Theta, z)\big) \le f(\Theta) - 1,$$

because from $U(\Theta, z)$ we can follow the rest of the sequence $z_{2:f(\Theta)}$ and reach $\mathcal{B}$ in $f(\Theta)-1$ steps. Thus f decreases strictly under the *existential* choice of z, satisfying Definition 5.1. □

Comment. The proof uses only the existence of a finite hitting time for the basin—not continuity or open-preimage. Thus f is a very weak, purely combinatorial certificate of reachability.

5.3 Sufficiency: Ranking + Structural Axioms ⇒ Certifiability

A ranking alone does not ensure we can *compute* or *verify* anything in polynomial time (e.g. the preimages might be fractal sets). We need the structural axioms introduced earlier.

Theorem 5.1 (Ranking + (C1)–(C3) + Poly-Bounded f ⇒ Certifiable)**.** *Assume:*

(C1) Compactness/continuity: $\mathcal{M}, \mathcal{Z}^{\text{in}}$ compact metric; U, E continuous.

(C2) Minimal standard $\mathcal{S} = \{\varepsilon_n\}$ exists, yielding shells $\mathcal{B}_n$.

(C3) Open-preimage at all levels (Axiom 2.1).

(C4) A well-founded ranking $f : \mathcal{M} \to \mathbb{N}$ exists, with $f(\Theta_0) \leq p(|input|)$ for some polynomial p.

Then $\mathfrak{T}$ is deterministically certifiable in NP.

Proof sketch.

- (C1)–(C3) yield (Section 5, Chapter 1) a finite guarded decomposition for each shell and the Loop-Until primitive.
- The ranking f gives a bound on the number of loop iterations: by Definition 5.1, each iteration can choose an admissible input that decreases f by ≥ 1. Thus the loop halts in $\leq f(\Theta_0)$ steps.
- Because $f(\Theta_0) \leq p(|\text{input}|)$, the halting time is polynomially bounded. The NP witness is (T, Θ_T) with $T \leq p(|\text{input}|)$, and the verifier replays U for T steps and checks $\Theta_T \in \mathcal{B}$ (within ε_T).

All checks use only polynomial time because the decomposition is finite and each U_i, E is continuous on a compact set (so can be simulated to required precision). Hence $\mathfrak{T}$ is NP-certifiable. □

Corollary 5.1. *The ranking condition by itself (without (C1)–(C3) and poly-boundedness) is* not *sufficient for NP certifiability: pathological maps could admit a ranking but fail open-preimage or minimal standard construction, making a finite guarded program or robust verifier impossible.*

5.4 Ranking vs. Lyapunov: Universal vs. Existential Decrease

Definition 5.2 (Strict Lyapunov-Border Function)**.** A continuous $L : \mathcal{M} \to \mathbb{R}_{\geq 0}$ is a strict Lyapunov-border function for U if:

1. $L(\Theta) = 0 \iff \Theta \in \mathcal{B}$;
2. For every $\Theta \notin \mathcal{B}$ and *every* admissible $z \in G^{\text{in}}(\Theta)$, $L(U(\Theta, z)) < L(\Theta)$.

A Lyapunov L demands *universal* decrease—stronger than ranking's existential decrease. In practice:

- L gives a direct, analytic handle (geometry, contraction) but may be hard to find.
- f is easy to construct once reachability is known, but is combinatorial/implicit.

Either can certify halting; L implies a ranking (e.g. discretize L by shells), but not conversely.

5.5 Preservation Under Decomposition

Existence of a Ranking-Preserving branch Previous results immediately give us a weak statement:

Lemma 5.1 (Existence of Ranking Preservation under Guarded Decomposition)**.** *Let f be a ranking for U. Suppose U is decomposed at tick 0 into $\{U_i\}_{i=1}^k$ over cells $\{N_i\}$ (Theorem 4.1). Then for every $\Theta \notin \mathcal{B}$ there exists an i and a $z \in N_i$ such that*

$$f\big(U_i(\Theta, z)\big) < f(\Theta).$$

Proof. By Definition 5.1, there exists some $z^\star$ such that $f(U(\Theta, z^\star)) < f(\Theta)$. Because $\{N_i\}$ partitions $\mathcal{Z}^{\text{in}}$, $z^\star \in N_j$ for some j. On N_j, $U_j = U$, so $f(U_j(\Theta, z^\star)) < f(\Theta)$. Thus branch j witnesses the decrease. □

But some guarantee of progress is still necessary. We have seen (Lemma 5.1) that *existentially* there is always some branch/input which decreases the ranking:

$$\forall\, \Theta \notin \mathcal{B}, \quad \exists\, i,\ \exists\, z \in N_i :\ f\big(U_i(\Theta, z)\big) < f(\Theta).$$

However, this *existential* guarantee alone does *not* suffice to ensure that *every* admissible trajectory eventually takes a decreasing step. Indeed, an adversarial or uncontrolled choice of inputs could forever avoid the "good" z's and stall.

Why existential decrease is not enough. Consider a state Θ with two admissible inputs $z_{\downarrow}$ (decreases f) and $z_{\circlearrowleft}$ (keeps f constant). Lemma 5.1 assures $\exists\, z_{\downarrow}$, but a run that picks $z_{\circlearrowleft}$ indefinitely never makes progress.

Candidate progress axioms. To bridge the gap we could:

(i) *Universal Decrease (Lyapunov)*: require *all* admissible inputs in each cell decrease f.

(ii) *Angelic Control*: assume the program *chooses* a decreasing input whenever one exists.

(iii) *Weak Fairness (TI/WF)*: if decreasing inputs are enabled infinitely often, then one is taken infinitely often.

We adopt the weakest that still forces termination in all *fair* runs: the Time-Irreversibility/Weak-Fairness (Axiom-4.1)

Termination under TI/WF fairness. With the time irreversibility, it is easy to show the polynomial reachability from the ranking function.

Proposition 5.2 (Uniform Reachability under TI/WF)**.** *Under the hypotheses of Lemma 5.1 and Axiom 4.1, every* fair *trajectory of the fully expanded Loop-Until program decreases f infinitely often whenever $f > 0$, and hence reaches $\mathcal{B}$ in at most $f(\Theta_0)$ steps.*

Proof. By Lemma 5.1, whenever $\Theta_t \notin \mathcal{B}$ we have $D_t \neq \varnothing$. TI/WF then ensures that $z_t \in D_t$ for infinitely many t, so $f(\Theta_{t+1}) < f(\Theta_t)$ infinitely often. Since f is well-founded (it takes values in $\mathbb{N}$), there can be only finitely many such decreases. Therefore the run must enter $\mathcal{B}$ within $f(\Theta_0)$ steps. □

6 Double-Bound under Decomposition

6.1 Introduction and High-Level Statement

In preceding chapters we separated the two layers of information dynamics: the *epistemically lossless* basin dynamics (strict contraction) and the *genuine* many-to-one collapse occurring during adaptation, captured in the ranking/ Lyapunov mechanism under Time-Irreversibility/Weak-Fairness (TI/WF). What remains to be made precise is how these two interact in the full decomposed test, and how the residual non-injectivity propagates as resolution is refined.

This section proves a **Decomposition Double-Bound**: under the tick-0 (and shell-by-shell) guarded decomposition, as the finite resolution of the measurement is increased, all per-step information loss outside the basin is funneled into the ranking function, while

the basin updates remain epistemically lossless. We then show how TI/WF guarantees that this residual, adaptation-originating collapse suffices to enforce uniform reachability and therefore certification. Finally, we quantify the complexity consequences: finite-resolution certification is decidable (in fact, in **NP** under the usual regularity), whereas the infinite-resolution limit preserves at most semi-decidability, matching the upper bound given by the double-bound analysis.

6.2 Setup and Notation

We work with the decomposed test representation introduced in Chapter 2: the original test is unrolled at Tick-0 (and beyond) into a *guarded program* composed of primitive decision/transition blocks, each of which is assumed to be contracting or asymptotically resolving ambiguity, together with a deterministic *ranking* function that breaks remaining ties to yield a canonical branch when multiple refinements are equally viable.

Fix compact metric state space $\mathcal{M}$ and input space $\mathcal{Z}^{\text{in}}$, and let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ be the underlying update map. The *decomposed finite-resolution guarded program* at scale $\varepsilon > 0$ is denoted G_ε; it maps an initial condition (Θ, z) to a finite tuple

$$G_\varepsilon(\Theta, z) = \big(\ell_\varepsilon(\Theta, z),\, r_\varepsilon(\Theta, z)\big),$$

where ℓ_ε is the sequence of guarded branch labels (including conditional decisions, localized state refinements, guard-band indicators, etc.) observed up to tick 0 at resolution ε, and r_ε is the output of the deterministic ranking function applied when multiple branch-subtrees are equally supported.

Let T_∞ denote the ideal infinite-precision target execution outcome (e.g., the fully resolved branch plus eventual tail property). For a decreasing resolution schedule $\varepsilon_1 > \varepsilon_2 > \cdots \to 0$, each G_{ε_k} produces a finite *certificate prefix* c_k that encodes the partial guarded program behavior consistent with T_∞ at that scale. We require c_{k+1} to refine c_k (i.e., extend it with higher-precision branching/labels), forming a coherent sequence converging to the full infinite-precision certificate.

Throughout, we assume that each primitive sub-map contributing to G_ε either is strictly contracting on its local state coordinates or has an explicitly controlled modulus of continuity so that its decision boundaries stabilize as $\varepsilon \to 0$.

6.3 Key Definitions

Definition 6.1 (Primitive Contracting Branch)**.** A *primitive contracting branch* is any component of the guarded program whose internal transition map, restricted to its local state coordinates, satisfies either a uniform strict contraction property or admits a provable modulus of stability such that small perturbations in its input produce vanishing perturbations in its output as resolution refines. Each such primitive branch induces a *stability modulus* $\omega_{\text{prim}}(\cdot)$ controlling its sensitivity.

Definition 6.2 (Decomposed Guarded Label (Ignoring Ranking))**.** For resolution $\varepsilon > 0$, define the map $\tilde{G}_\varepsilon(\Theta, z)$ to be the guarded-program output excluding the ranking output—i.e., the tuple of branch decisions and local state refinements without deterministic tie-breaking. Two inputs (Θ, z) and (Θ', z') are *ranking-equivalent* if $G_\varepsilon(\Theta, z)$ and $G_\varepsilon(\Theta', z')$ differ only in their r_ε component.

Definition 6.3 (Effective Ambiguity Set)**.** At resolution ε, the *effective ambiguity set* is

$$\mathcal{A}_\varepsilon := \Big\{(\Theta, z) \mid \exists\,(\Theta', z') \neq (\Theta, z),\ \tilde{G}_\varepsilon(\Theta, z) = \tilde{G}_\varepsilon(\Theta', z') \text{ and not ranking-equivalent}\Big\},$$

i.e., the preimage collisions under $\tilde{G}_\varepsilon$ that are not resolved by the ranking mechanism.

Definition 6.4 (Certificate Prefix Sequence)**.** A sequence $\{c_k\}_{k\geq 1}$ is a *certificate prefix sequence* for T_∞ if c_k is the finite guarded-program output at resolution ε_k, c_{k+1} refines c_k (i.e., is consistent and extends decisions with higher precision), and $\lim_{k\to\infty} c_k$ uniquely identifies T_∞ in the infinite-resolution limit.

Definition 6.5 (Resolution Schedule)**.** A *resolution schedule* $\{\varepsilon_k\}$ is a computable, strictly decreasing sequence with $\varepsilon_k \to 0$. Associated with each ε_k is a refinement depth determining how fine the guarded program branches and state partitions must be to certify behavior at that precision.

6.4 Preliminary Lemmas

Remark 6.1. In all subsequent lemmas and theorems in this section we work under the assumption that whenever a guarded decomposition at resolution ε is given, it has been extended to any finer resolution $\varepsilon_{\text{ref}} < \varepsilon$ via the refinement-extension construction of Theorem 2.1; i.e., all further prefixes live in a coherent refinement chain.

Lemma 6.1 (Stability of Guarded Decisions under Refinement)**.** *Let (Θ, z) and (Θ', z') be two distinct inputs. Suppose their guarded-program decisions diverge at infinite precision. Then there exists $\varepsilon_0 > 0$ and a computable modulus $\delta(\varepsilon) \to 0$ such that for all $\varepsilon < \varepsilon_0$, if $\tilde{G}_\varepsilon(\Theta, z) = \tilde{G}_\varepsilon(\Theta', z')$ (without ranking), the distance between the underlying state-input pairs is bounded by $\delta(\varepsilon)$. In particular, unless (Θ, z) and (Θ', z') are asymptotically indistinguishable, their finite-resolution guarded labels separate as $\varepsilon \to 0$.*

Proof of Lemma 6.1 (Stability of Guarded Decisions under Refinement)

Proof. By assumption each primitive component of the guarded program has a known stability modulus: small perturbations in its input induce bounded perturbations in its output that vanish as resolution is refined. Concretely, let (Θ, z) and (Θ', z') be two inputs whose infinite-precision guarded decisions diverge. Then there exists some finite depth in the guarded program at which their branch labels differ in the limit. Since each decision predicate (guard-band membership, conditional branching) is defined via continuous thresholds and the underlying maps are contracting or have controlled continuity, there is a neighborhood size $\varepsilon_0 > 0$ such that for all $0 < \varepsilon < \varepsilon_0$, if $\tilde{G}_\varepsilon(\Theta, z) = \tilde{G}_\varepsilon(\Theta', z')$, then the discrepancy between (Θ, z) and (Θ', z') must lie within the back-propagated uncertainty induced by the stability moduli of the primitives. That is, their distance is bounded by a computable function $\delta(\varepsilon) \to 0$ as $\varepsilon \to 0$, which aggregates the contraction/continuity bounds along the shared prefix of decisions. Therefore, unless (Θ, z) and (Θ', z') converge to indistinguishability in the limit, their finite-resolution guarded labels separate for sufficiently small ε. This establishes the stated stability property. □

Lemma 6.2 (Approximate Injectivity of G_ε)**.** *Under the primitive contraction/stability assumptions, for each fixed non-ranking-equivalent pair $(\Theta, z) \neq (\Theta', z')$, there exists $\varepsilon^* > 0$ such that for all $0 < \varepsilon < \varepsilon^*$,*

$$\tilde{G}_\varepsilon(\Theta, z) \neq \tilde{G}_\varepsilon(\Theta', z').$$

Hence every non-ranking collision disappears in the limit: the only residual non-injectivity of G_ε as $\varepsilon \to 0$ arises from ranking-equivalence.

Proof of Lemma 6.2 (Approximate Injectivity of G_ε)

Proof. We show that any non-ranking-equivalent collision vanishes as $\varepsilon \to 0$. Suppose to the contrary that there exists a sequence $\varepsilon_k \to 0$ and distinct inputs $(\Theta, z) \neq (\Theta', z')$, not ranking-equivalent, with $\tilde{G}_{\varepsilon_k}(\Theta, z) = \tilde{G}_{\varepsilon_k}(\Theta', z')$ for all k. Then by Lemma 6.1, the distance between the inputs must shrink: $d((\Theta, z), (\Theta', z')) \leq \delta(\varepsilon_k)$, where $\delta(\varepsilon_k) \to 0$. Hence (Θ, z) and (Θ', z') become asymptotically indistinguishable. If they remain distinct in the limit, this is impossible. Therefore for any fixed pair of distinct, non-ranking-equivalent inputs, there exists $\varepsilon^* > 0$ such that for all $0 < \varepsilon < \varepsilon^*$, $\tilde{G}_\varepsilon(\Theta, z) \neq \tilde{G}_\varepsilon(\Theta', z')$. In other words, all non-ranking collisions disappear for sufficiently fine resolution, proving the approximate injectivity. □

Lemma 6.3 (Controlled Ambiguity Shrinkage)**.** *Let $\mathcal{A}_\varepsilon$ be the effective ambiguity set as in Section 6.3, and let $\delta(\varepsilon)$ be the stability modulus from Lemma 6.1. Working with the metric entropy / covering number framework of Chapter 4 (cf. Definition 4.X and the information-drop bounds), the covering number of $\mathcal{A}_\varepsilon$ with balls of radius $\delta(\varepsilon)$ satisfies*

$$\mathcal{N}\big(\mathcal{A}_\varepsilon,\ \delta(\varepsilon)\big) = o(1) \quad \text{as } \varepsilon \to 0.$$

Equivalently, the entropy $\mathcal{H}_{\delta(\varepsilon)}(\mathcal{A}_\varepsilon) \to -\infty$ in the normalized scale, so the "hard" region where non-ranking collisions persist vanishes under refinement.

Proof of Lemma 6.3 (Controlled Ambiguity Shrinkage)

Proof. By Lemma 6.2, any fixed non-ranking-equivalent pair $(\Theta, z) \neq (\Theta', z')$ cannot produce a collision in $\tilde{G}_\varepsilon$ for all sufficiently small ε: there exists $\varepsilon^* > 0$ such that for all $0 < \varepsilon < \varepsilon^*$, the pair is excluded from $\mathcal{A}_\varepsilon$. Hence ambiguities can only arise from inputs that are within distance $\delta(\varepsilon)$ of each other, where $\delta(\varepsilon) \to 0$ quantifies the back-propagated indistinguishability from Lemma 6.1.

Cover ambient space $\mathcal{M} \times \mathcal{Z}^{\text{in}}$ with balls of radius $\delta(\varepsilon)$; any potential collision must lie inside one of these shrinking neighborhoods around the diagonal of ranking-equivalent or near-indistinguishable inputs. Therefore the number of such balls required to cover $\mathcal{A}_\varepsilon$ is bounded by the covering number of a vanishing neighborhood, which by standard metric entropy estimates tends to zero in the normalized sense. Thus $\mathcal{N}(\mathcal{A}_\varepsilon, \delta(\varepsilon)) = o(1)$, proving the controlled shrinkage of ambiguity. □

Lemma 6.4 (Composable Prefix Consistency)**.** *Suppose c_k is the certificate prefix at resolution ε_k and c_{k+1} is computed at finer resolution $\varepsilon_{k+1} < \varepsilon_k$. Then c_{k+1} is a valid extension of c_k: the guarded decisions in c_k remain intact in c_{k+1}, and refined ambiguities are resolved consistently. Hence the sequence $\{c_k\}$ forms a coherent monotone refinement converging to a limit certificate.*

Proof of Lemma 6.4 (Composable Prefix Consistency)

Proof. Let $c_k = G_{\varepsilon_k}(\Theta, z)$ and $c_{k+1} = G_{\varepsilon_{k+1}}(\Theta, z)$ with $\varepsilon_{k+1} < \varepsilon_k$. By construction the finer-resolution guarded program incorporates all decision predicates used at coarser resolution with tighter thresholds and more precise state refinements. The stability of these decisions (Lemma 6.1) ensures that any choice made at resolution ε_k remains valid under refinement (unless it was ambiguous and resolved, in which case the refinement picks the consistent extension). Therefore c_{k+1} extends c_k: all branch labels in c_k appear as prefixes in c_{k+1}, and any prior ambiguity is resolved in a way consistent with the deterministic ranking function or contraction-induced separation. This monotonicity implies the sequence $\{c_k\}$ is coherent and converges to a well-defined infinite-precision certificate, as claimed. □

6.5 Main Theorem: Double-Bound under Decomposition

Theorem 6.1 (Double-Bound under Decomposition)**.** *Assume the primitive contracting/stability conditions, the guarded program decomposition at Tick-0 (and further ticks) as in Chapter 2, and the availability of a computable resolution schedule $\varepsilon_k \to 0$. Let T_∞ be the ideal infinite-precision execution outcome (branch plus eventual tail property). Then:*

(a) ***Injectivity Approximation:*** *As $\varepsilon \to 0$, the finite-resolution decomposed map G_ε becomes injective up to ranking equivalence. Equivalently, all non-ranking preimage collisions vanish in the limit; the only structural ambiguity that remains is the deterministic tie-breaking imposed by the ranking function.*

(b) ***Recursive Enumerability / Certification Recovery:*** *There exists an effective procedure that produces a certificate prefix sequence $\{c_k\}$ with c_k derived from G_{ε_k}, such that c_{k+1} refines c_k, and $\lim_{k\to\infty} c_k = T_\infty$. The sequence is recursively enumerable, providing asymptotically exact certification of the infinite-precision decomposed test.*

Proof of Theorem 6.1 (Double-Bound under Decomposition)

Proof. We treat parts (a) and (b) separately and then combine them.

(a) Injectivity Approximation. By the contracting/stability assumptions on the primitive components, the guarded decisions—including branch selection and localized state refinements—stabilize as resolution is refined. Lemma 6.1 establishes that distinct, non-ranking-equivalent inputs cannot continue to collide in $\tilde{G}_\varepsilon$ for arbitrarily small ε; hence, Lemma 6.2 implies that all such non-ranking-induced collisions vanish in the limit. The only remaining ambiguity in the infinite-resolution limit arises from ranking-equivalence, which by design is a deterministic tie-breaking mechanism imposing a canonical representative among otherwise indistinguishable refinements. Thus G_ε approaches injectivity modulo this known residual, proving part (a).

(b) Recursive Enumerability / Certification Recovery. Construct the certificate prefix sequence $\{c_k\}$ by evaluating the guarded program at resolutions $\varepsilon_k \to 0$, setting $c_k := G_{\varepsilon_k}(\Theta, z)$ for the true execution input (Θ, z) (or for the purported one under verification). By Lemma 6.4, the sequence is coherent: each subsequent prefix extends the previous, resolving diminishing ambiguity. The shrinking of the effective ambiguity set (Lemma 6.3) guarantees that this refinement process does not oscillate indefinitely:

the amount of unresolved collision decreases, and eventually the prefix sequence identifies a unique infinite-precision branch T_∞. Each c_k is computable given G_{ε_k}; thus the sequence is recursively enumerable. Because the only remaining ambiguity is resolved by the deterministic ranking, the limit uniquely corresponds to T_∞, achieving asymptotically exact certification.

Synthesis. Part (a) ensures the infinite-precision target is well-defined and stable (injective up to ranking), while part (b) provides an effective means to approximate it with arbitrarily fine finite certificates. The coexistence of approximate injectivity and enumerability constitutes the "double bound" in the decomposed setting: although finite-resolution guarded outputs may appear ambiguous, refinement recovers the unique ideal behavior. This completes the proof. □

6.6 Summary

- Tick-0 decomposition is Turing-complete. Physics of the measurement does not restrict the program fundamentally.
- However, *the physics and program together* (the overall procedure) is restricted in a way so that the Turing machine of the decomposition halts, i.e., measurement produces a result. The necessary and sufficient additional condition for this is the Time-Irreversibility/Weak-Fairness axiom.
- A ranking f captures the *discrete* progress toward the basin. It is necessary (Prop. 5.1) and—together with (C1)–(C3) and a poly bound—sufficient (Thm. 5.1).
- The poly-bounded inductive decomposition preserves the ranking's decrease property (Sec 5.5) if and only if the time irreversibility is guaranteed (Axiom-4.1), ensuring uniform halting of Loop-Until (Prop. 5.2).
- Time-Irreversible/Weakly Fair (TI/WF) rankings are the weakest combinatorial condition needed for poly-time reachability (which we need for NP certifiability). *This* is the only fundamental restriction on the program execution.
- Decomposed test program is, absent ranking, epistemically lossless. The infinte-precision measurement certification on the decomposed program is not harder than r.e.

7 Conclusion

We continued our investigation of the Simple Test anatomy and behavior "under the hood". Our steps in this Chapter were as follows:

1. **Turing-Completeness of Guarded Tests.** We proved universality: any deterministic TM can be implemented as a valid test; halting $\iff$ basin entry.
2. **Time-Irreversibility of any Guarded Test performing physical measurement.** We proved that endowing a Turing-complete decomposed structure with any kind of physical input and belief-state spaces requires the explicit time irreversibility in its weakest form - the Weak-Fairness axiom. We showed that this axiom is precisely necessary and sufficient for ANY measurement result to be produced, in a polynomial or non-polynomial time.

3. **Ranking & Reachability.** Then, we investigated the conditions for our program to provide a *polynomial* reachability, thus making a base to create a *certifiable* measurement. It is turned out that the requirement is precisely the same as above for *general* termination: the program execution must be weakly fair, or, in terms involving physics of our metrology, the adaptation process *must* be time-irreversible.

   Axiom of TI/WF (Axiom-4.1) ranking function is *necessary*; combined with (C1)–(C4), it is also *sufficient* for uniform reachability.

   In analytic terms, a Lyapunov-border function is a strong (but optional, so too strong for our general goal) way to obtain such a ranking.

4. **Double-Bound under Decomposition crystallizes why the guarded program representation is (absent ranking) epistemically lossless.** Even though finite-resolution guarded outputs may temporarily blur distinctions (the numerical bound) and induce apparent non-injectivities, the decomposition's contracting primitives and refinement-compatible hierarchy ensure those ambiguities shrink and are systematically resolved (the algorithmic recovery bound).

   As resolution is driven downward and prefixes are extended coherently, the only residual structural uncertainty is the deterministic ranking tie-breaker, and the infinite-precision execution emerges as the limit of an effectively enumerable sequence of certificate prefixes.

   Thus the decomposition does not sacrifice certifiability for compactness—on the contrary, it reconciles transient indistinguishability with asymptotically exact identification, while realizing (absent ranking) epistemic losslessness in the composed test Adaptation, where it was not observed before the decomposition.

Chapter 3: Fundamental Test Equivalence

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1 Fundamental Equivalence: NP–Certifiability $\iff$ Polynomial–Time Halting of a Guarded Program

We now tie together all ingredients—minimal Certification Standard, open-preimage (Input Traceability), inductive guarded decomposition, Loop-Until, and Time-Irreversible ranking—to obtain the central equivalence of this chapter.

1.1 Setup

Standing hypotheses Let

$$\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$$

be a *Simple Test* (Def. 4.3 from Book II, Chapter 8) built upon the axioms:

Gnosis Axioms:

(F) **Invariant basin.** There exists a reachable, invariant attractor basin $\mathcal{B} \subseteq \mathcal{M}$ (Deff. 2.1,2.2,2.3 from Book II, Chapter 5). *(There is a fact that remain)*

(S) **Certification standard.** $\exists \mathcal{S} = \{\varepsilon_n\}_{n=1}^{\infty} : \forall \varepsilon_n \in (0,1]\ \exists\ verifier\ V : V(z_{1:n}^{\text{out}}, w) = accept \implies \forall z : d(z, z_{1:n}^{\text{out}}, w)) < \varepsilon_n\ V(z) = accept$ (Def. 2.2 from Book II, Chapter 6). *(The statement can be made about the fact)*

Structural Axioms:

(C) **Compact continuity.** $\mathcal{M}, \mathcal{Z}^{\text{in}}$ are compact metric spaces; U, E are continuous (Deff. 1.1,1.2,2.1 from Book II, Chapter 3). *(There are finite total degrees of freedom under any finite resolution)*

(T) **Open-preimage (PL-traceability).** For every $n \geq 1$ and $\Theta \in \mathcal{M}$, the set $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is open and nonempty (Def. 2.1 from Chapter 1). *(The fact is traceable)*

Note that these hypotheses put assumptions onto the general knowability or the structure of the Simple Test only.

Consider the Simple Test 0-Tick decomposition built from the primitives we described in Chapter 1 and Chapter 2. We further put assumptions on the process of decomposed measurement.

Process axiom

(I) There exists the finite decomposition of the test $\mathfrak{T}$ into guarded program Π, built from the primitives SEQUENTIAL, CONDITIONAL, LOOP and equipped with a *ranking function* f that satisfies TI/WF Axiom 4.1 from Chapter 2 on program runs named in further *TI/WF-fair executions. (Execution time is irreversible)*

Theorem 1.1 (Fundamental Equivalence)**.** *Under gnosis and structural axioms, the following are equivalent:*

(a) ***NP–Certifiability.*** *The tail predicate of $\mathfrak{T}$ admits a polynomial-size deterministic certificate; equivalently, the language of "good heads" is in* **NP**.

(b) ***Existence of a Polynomial–Time Halting Guarded Program.*** *Finite guarded program Π halts on accepting state of entering the basin $\mathcal{B}$ in at most $p(|\text{input}|)$ steps for some polynomial p for every TI/WF-fair execution (Def. 1.1) of Π.*

2 Valid test assumptions hold

We establish that we have a valid test (Def. 2.1) under our axioms. Valid test assumptions VT1 and VT2 are just restated from C and F axioms respectively.

Axiom VT3 follows from axioms as decribed in the following subsection.

2.1 Existence of a Minimal Certification Standard (VT3) from C, F, and T

Theorem 2.1 (Minimal Certification Standard)**.** *Let a measurement model satisfy axioms C, F, and T. Define, for each $n \geq 1$,*

$$\varepsilon_n \;=\; \frac{1}{n}, \qquad \mathcal{B}_n \;=\; \big\{\Theta \in \mathcal{M} : dist\big(\Theta, \mathcal{B}\big) \leq \tfrac{1}{n}\big\}.$$

Then $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ is a Minimal Certification Standard in the sense of Definition 2.1. In particular:

(i) $\varepsilon_n \downarrow 0$ and $\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \cdots \supseteq \mathcal{B}$ with $\bigcap_n \mathcal{B}_n = \mathcal{B}$.

(ii) Each $\mathcal{B}_n$ is closed (hence compact), and by (T) the sets $U(\cdot, z)^{-1}(\mathcal{B}_n)$ are open, so membership in $\mathcal{B}_n$ is testable up to tolerance ε_n.

(iii) No strictly smaller choice $\varepsilon'_n < \varepsilon_n$ can preserve both soundness and completeness at the boundary: by continuity of $\Theta \mapsto dist(\Theta, \mathcal{B})$, any decrease would violate soundness or completeness of the n-th certification test.

Hence the valid test assumption VT3 holds.

Proof. **(i) Nested shells and convergence.** Since $\mathcal{B}$ is closed in the compact space $\mathcal{M}$, the distance function $\Theta \mapsto d(\Theta, \mathcal{B})$ is continuous and attains its maximum. For each n,

$$\mathcal{B}_n = \{\Theta : d(\Theta, \mathcal{B}) \leq 1/n\}$$

is a closed sublevel set of a continuous map, hence compact. Clearly $\varepsilon_{n+1} = 1/(n+1) < 1/n = \varepsilon_n$ and $\lim_{n\to\infty} \varepsilon_n = 0$, and $\bigcap_n \mathcal{B}_n = \mathcal{B}$.

(ii) Testability (open-preimage). By (T), for each $z \in \mathcal{Z}^{\text{in}}$ the set

$$U(\cdot, z)^{-1}(\mathcal{B}_n) = \{\Theta : U(\Theta, z) \in \mathcal{B}_n\}$$

is open. Thus one can robustly test membership in $\mathcal{B}_n$ up to any desired finite precision: any point within distance ε_n of the true shell boundary still lies in the same preimage or its complement.

(iii) (*Local minimality*) For each n there exists $\delta_n > 0$ such that decreasing ε_n by any amount greater than δ_n would cause the resulting sequence to violate at least one of the soundness or completeness properties. In particular, let $g(\Theta) := \text{dist}(\Theta, \mathcal{B})$. The map g is continuous on the compact set $\mathcal{M}$, so $g^{-1}([0, \varepsilon_n]) = \mathcal{B}_n$ is closed.

Right–continuity of the sublevel sets of g ensures that for sufficiently small $\delta > 0$ we still have

$$\mathcal{B} \subseteq \{\Theta : g(\Theta) \leq \varepsilon_n - \delta\} \subseteq \mathcal{B}_n$$

but once $\delta > \delta_n$ this inclusion fails and either completeness or soundness is lost.

Thus ε_n is minimal up to the tolerance δ_n, without assuming the infimum is attained.

Having verified the nesting, convergence, testability and minimality properties, we conclude that $\mathcal{S}$ is indeed the Minimal Certification Standard required by (VT3). □

Next, we prove that VT4 (Shell invariance) assumption of valid test is also satisfied.

2.2 Shell Invariance Lemma

Lemma 2.1 (Shell Invariance / Finite Guarded Cover)**.** *Assume a measurement model satisfies axioms C, F, and T. Then for each $n \geq 1$ there exist:*

- *a compact subset $\mathcal{K}_n \subseteq \mathcal{B}_n$ containing the target basin $\mathcal{B}$,*
- *finitely many open patches $V_{n,1}, \ldots, V_{n,J} \subseteq \mathcal{M}$ covering $\mathcal{K}_n$,*
- *corresponding nonempty open* guard cells *$N_{n,1}, \ldots, N_{n,J} \subseteq \mathcal{Z}^{\text{in}}$,*

such that

$$U(V_{n,j} \times N_{n,j}) \subseteq \mathcal{B}_n \quad \textit{for all } j = 1, \ldots, J.$$

In other words, the relevant portion $\mathcal{K}_n$ of the shell $\mathcal{B}_n$ is forward-invariant under the finite guarded cover $\{V_{n,j} \times N_{n,j}\}_{j=1}^{J}$; no invariance is claimed for states outside $\mathcal{K}_n$.

Proof. Fix $n \geq 1$. Since $\mathcal{B}$ is invariant and $\mathcal{B}_n$ is a closed neighbourhood of $\mathcal{B}$, we know $\mathcal{B}_n \neq \varnothing$. We will construct a single open set $N_n \subseteq \mathcal{Z}^{\text{in}}$ that works uniformly for every $\Theta \in \mathcal{B}_n$.

1. Local guards around each Θ. By (T), for each $\Theta \in \mathcal{B}_n$ there is at least one input $z_\Theta \in \mathcal{Z}^{\text{in}}$ with

$$U(\Theta, z_\Theta) \in \mathcal{B}_n,$$

and moreover the preimage $\{\, z : U(\Theta, z) \in \mathcal{B}_n \}$ is an open neighbourhood of z_Θ. Also, continuity of U implies that there is an open neighbourhood $V_\Theta \subseteq \mathcal{M}$ of Θ and an open neighbourhood $W_\Theta \subseteq \mathcal{Z}^{\text{in}}$ of z_Θ such that

$$\forall\, \Theta' \in V_\Theta,\ \forall\, z \in W_\Theta : \quad U(\Theta', z) \ \in\ \mathcal{B}_n.$$

(Indeed, the map $(\Theta, z) \mapsto dist\big(U(\Theta, z), \mathcal{B}\big)$ is continuous on the compact $\mathcal{M} \times \mathcal{Z}^{\text{in}}$.)

2. Finite subcover of the shell. The collection $\{V_\Theta : \Theta \in \mathcal{B}_n\}$ is an open cover of the compact set $\mathcal{B}_n$. Extract a finite subcover

$$\mathcal{B}_n \ \subseteq\ V_{\Theta_1} \ \cup\ \cdots\ \cup\ V_{\Theta_k}.$$

Corresponding to these finitely many patches we have open input-sets $W_{\Theta_1}, \ldots, W_{\Theta_k}$.

3. Uniform guard cell. Define

$$N_n \ =\ \bigcap_{i=1}^{k} W_{\Theta_i}.$$

As a finite intersection of nonempty open sets, N_n is itself nonempty and open.

4. Verification of invariance. By continuity of U and the definition of $\mathcal{B}_n$, for each $\Theta \in \mathcal{B}_n$ there exist open sets $V_\Theta \subset \mathcal{M}$ and $W_\Theta \subset \mathcal{Z}^{\text{in}}$ such that $\Theta \in V_\Theta$ and

$$U(V_\Theta \times W_\Theta) \subseteq \mathcal{B}_n.$$

Since $\mathcal{B}_n$ is compact, finitely many of the V_Θ cover $\mathcal{B}_n$:

$$\mathcal{B}_n \subset \bigcup_{j=1}^{k} V_{\Theta_j}.$$

Define $N_{n,j} := W_{\Theta_j}$ and $V_{n,j} := V_{\Theta_j}$. Then for each j we have $U(V_{n,j} \times N_{n,j}) \subseteq \mathcal{B}_n$ as required. This establishes the finite guarded cover without requiring any global intersection of the W_{Θ_j}.

This completes the proof. (VT4) holds. □

2.3 State encoding is polynomial

Assumption on dimension. In what follows we assume that the dimension d of $\mathcal{X}$ is either constant or bounded by a polynomial in the bit-length of the measurement input (so that $d = O(\text{poly}(n))$); otherwise the bit-length bound $O(d \log(1/\varepsilon))$ must be interpreted with the explicit dependence on d.

Lemma 2.2 (Polynomial State Encoding)**.** *Let $\mathcal{X} \subset [0,1]^d$ be a compact metric space equipped with the sup–norm $\|\cdot\|_\infty$ (cf. C axiom). Then for any $\varepsilon > 0$ and any $\Theta \in \mathcal{X}$, there exists a rational vector*

$$\text{enc}_\varepsilon(\Theta) \ =\ \big(r_1, \ldots, r_d\big) \ \in\ \mathbb{Q}^d \cap [0,1]^d$$

such that

$$\big\|\Theta - \text{enc}_\varepsilon(\Theta)\big\|_\infty \ \le\ \varepsilon,$$

and the total bit-length of the encoding $\mathrm{enc}_\varepsilon(\Theta)$ *is*

$$O\big(d\ \log(1/\varepsilon)\big).$$

In particular, if at time $T \le p(n)$ *we require precision* $\varepsilon_T = 2^{-r(n)}$ *with* $r(n) = O(\mathrm{poly}(n))$*, then* $\mathrm{enc}_{\varepsilon_T}(\Theta_T)$ *can be written in* $O\big(d\, r(n)\big) = \mathrm{poly}(n)$ *bits.*

Proof. **1. Coordinate-wise binary approximation.** Each coordinate $\Theta_i \in [0,1]$ of $\Theta = (\Theta_1, \dots, \Theta_d)$ can be approximated by a binary rational

$$r_i = \frac{m_i}{2^k}, \quad 0 \le m_i \le 2^k,$$

chosen so that $|\Theta_i - r_i| \le \varepsilon$. Concretely, pick

$$k \;=\; \big\lceil \log_2(1/\varepsilon) \big\rceil,$$

and let $m_i = \lfloor 2^k \Theta_i \rfloor$. Then

$$0 \;\le\; \Theta_i - \frac{m_i}{2^k} \;<\; \frac{1}{2^k} \;\le\; \varepsilon,$$

so $|\Theta_i - r_i| \le \varepsilon$.

2. Uniform sup-norm bound. Setting $\mathrm{enc}_\varepsilon(\Theta) = (r_1, \dots, r_d)$, we get

$$\big\|\Theta - \mathrm{enc}_\varepsilon(\Theta)\big\|_\infty = \max_{1\le i\le d} |\Theta_i - r_i| \;\le\; \varepsilon.$$

3. Bit-length estimate. Each binary rational $r_i = m_i/2^k$ can be represented by the pair (m_i, k). Since $0 \le m_i < 2^k$, the binary expansion of m_i requires at most $k+1$ bits. Storing k itself takes $O(\log k)$ bits, which is $O(\log\log(1/\varepsilon))$ and absorbed into $O(\log(1/\varepsilon))$. Hence each coordinate uses $O(k) = O(\log(1/\varepsilon))$ bits, and all d coordinates use $O\big(d\ \log(1/\varepsilon)\big)$ bits in total.

4. Polynomial bound for verifier precision. If at step $T \le p(n)$ the verifier needs precision $\varepsilon_T = 2^{-r(n)}$ with $r(n) = O(\mathrm{poly}(n))$, then

$$\log\big(1/\varepsilon_T\big) = r(n) = O\big(\mathrm{poly}(n)\big).$$

Since d is either fixed or itself polynomially bounded in n, the total bit-length $O(d\ \log(1/\varepsilon_T)) = O(d\, r(n))$ remains polynomial in n. Thus, the witness encoding at time T is of size $\mathrm{poly}(n)$, as required. □

2.4 Effective Measurability Axioms

At this point, we are almost ready to prove the fundamental equivalence. However, there is one more missing piece in our assumptions. Here it is: is it even possible to operate with our physical inputs in a tractable way?

Having no certifiability as a goal, we would invoke the Carathéodory criterion for Lebesgue measurability here or restrict the input space to a Borel space if we want to build further constructions on it. But these definitions does not tell us anything about the amount of work needed to process and compare an input with a standard.

Hence, we introduce the *constructive* axioms of measurability. They can be thought of as a Carathéodory criterion plus requirements of the primitive operation to fall into the desired complexity class. Below is the version of the axioms for the class **P**.

(EM1) **Effective Representation.** Each U_j and E_j is given by a rational arithmetic circuit (or equivalent) of size poly(n), where n is the bit–length of the external input description. All constant coefficients are rationals of bit–length poly(n).

(EM2) **Polynomial Modulus of Continuity.** For each j, there is a known polynomial $m_j(k)$ such that to compute $U_j(x, z)$ (resp. $E_j(x)$) to k bits of output precision requires at most $m_j(k)$ arithmetic operations on rationals of bit–length $O(k + n)$. Where needed, we will discriminate (EM2) into two sub-axioms:

(EM2a) **Analytic modulus:** For each primitive map U_j and E_j, there exists a known, effectively computable modulus of continuity $\omega_j : \mathbb{Q}_+ \to \mathbb{Q}_+$ with $\omega_j(0) = 0$, such that

$$\|x - y\| \leq \delta \quad \Rightarrow \quad \|F_j(x) - F_j(y)\| \leq \omega_j(\delta),$$

where F_j denotes U_j or E_j.

(EM2b) **Cost modulus:** There exists a polynomial p_j such that, given a rational input x and a desired accuracy of k bits, the value $F_j(x)$ can be computed to that accuracy in at most $p_j(k + n)$ steps, where n encodes the problem size.

(EM3) **Polynomial Guard/Basin Tests.** For each guard cell $N_i \subset \mathcal{Z}$ or basin shell $\mathcal{B}_n \subset \mathcal{X}$, membership can be decided up to tolerance ε in time poly($n, \log(1/\varepsilon)$).

Remark 2.1. At this point, we can observe that we never made a reference to the complexity class of the input processing in all the previous work, and all our constructions were built with only an *intrinsic* polynomial resource bound in mind.

Therefore, nothing prevents us from weakening the restrictions for our certifiable measurement to belong to a wider complexity class: **P** $\longrightarrow$ **BQP** with the corresponding promotion of the verification **NP** $\longrightarrow$ **QMA**, or even **P** $\longrightarrow$ **PSPACE** with the corresponding **NP** $\longrightarrow$ **IP**.

The only modification that such a change requires is the modification of the Effective Measurability Axioms (Section-2.4) so they remain friendly to the basic complexity class invoked.

2.5 Adaptive Traceability Resolution Theorem

We know that we can provide certification over Tick-0 decomposition and this is the base of the Fundamental Test Equivalence Theorem. However, until now, we have assumed UI traceability, or, in other words, the universality of "good" inputs.

Baseline complexity without ATRT. Deciding non-traceability for a given input z in the full generality—over a continuous (or an a priori non-compact or high-dimensional) belief/state space—is, in the limit, a hard problem (one can view the unbounded version as having Π^0_2-type difficulty because it asks to rule out the existence of a witness to arbitrary precision). If we abandon any adaptive focus and instead perform a naive finite-resolution brute-force exploration, the cost is dominated by covering the entire relevant state space at resolution $\varepsilon > 0$.

Concretely, let d be the ambient (or effective) dimension of the belief/state manifold being searched, and let n denote the bit-length of the external input description (so that all quantities derived from it—precision targets $\log(1/\varepsilon)$, region representations,

arithmetic costs—are $O(\mathrm{poly}(n))$). A uniform grid of diameter ε requires $\Theta(\varepsilon^{-d})$ regions to cover the space. Performing a full search over all such regions with a naive per-region verification (membership, evaluation of $f(\Theta)$, and threshold tests) costs

$$O\big(\varepsilon^{-d} \cdot \mathrm{poly}(n, \log(1/\varepsilon))\big)\,.$$

If the bookkeeping or region selection is done poorly, that per-region overhead can be as bad as $O(n^2)$, giving the oft-quoted worst-case bound $O(n^2\varepsilon^{-d})$. With modest algorithmic refinement—e.g., maintaining priority queues for region refinement, caching bound computations, and incrementally updating covers—the per-region overhead can be reduced to $O(n \log n \cdot \mathrm{poly}(\log(1/\varepsilon)))$, yielding

$$O\big(\varepsilon^{-d} \cdot n \log n \cdot \mathrm{poly}(\log(1/\varepsilon))\big)$$

for this non-adaptive, global procedure.

What ATRT buys. The Adaptive Traceability Resolution Theorem replaces the crude ε^{-d} full-space factor with the *ambiguous boundary* covering number $\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon)$, which can be dramatically smaller than ε^{-d} when most of the state space is decisively traceable or excludable. In other words, ATRT concentrates effort only where the decision is genuinely uncertain, avoiding the full exponential blowup in ambient volume and restoring practical tractability under structural regularity.

We can further utilize the guarded program structure and improve the computational cost further by combining the iterative resolution refinement with an SMT theorem prover to exclude large chunks of state space unreachable from z.

2.6 Adaptive Traceability Resolution Theorem

Setup and notation. Let $\mathfrak{T}$ satisfy axioms gnosis and structural axioms, (EM1–EM3), and (I). Fix an input $z \in \mathcal{Z}^{\mathrm{in}}$ and precision threshold $\varepsilon > 0$. Define

$$f(\Theta) := \mathrm{dist}(U(\Theta, z), \mathcal{B}), \qquad f^* := \min_{\Theta \in \mathcal{M}} f(\Theta).$$

We say z is *traceable at precision* ε iff $f^* \le \varepsilon$.

Under Promise Verification via Effective Measurability Lemma (Lem.-1.1, Appendix) and its corollary (Cor.-1.1) we assume a known *gap* $\gamma > 0$ separating the two cases:

$$\text{either } f^* \le \varepsilon \quad \text{(traceable)} \quad \text{or} \quad f^* \ge \varepsilon + \gamma \quad \text{(non-traceable with margin)}.$$

Let $\omega : \mathbb{R}_+ \to \mathbb{R}_+$ be the modulus of continuity from Lemma 1.2, Appendix for f, and pick a target refinement diameter $d_{\mathrm{target}} > 0$ such that

$$\omega(d_{\mathrm{target}}) < \frac{\gamma}{2}.$$

Define the *ambiguous boundary set*

$$\mathcal{B}_\varepsilon := \{\Theta \in \mathcal{M} : \ f(\Theta) \in [0, 2\varepsilon]\},$$

and let $\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon)$ be its covering number at scale d_{target}. Finally, let n denote the bit-length of the external input description (so that all derived parameters—including required precision $\log(1/\varepsilon)$, descriptions of regions, and arithmetic costs—are $O(\mathrm{poly}(n))$).

Algorithm. Run the adaptive refinement procedure (Algorithm 1, Appendix) which:

(a) Maintains a cover of $\mathcal{M}$ by regions R, refining any active region until $\mathrm{diam}(R) \leq d_{\mathrm{target}}$.

(b) On each region performs an *existential check*: attempts to find $\Theta \in R$ with $f(\Theta) \leq \varepsilon + \frac{\gamma}{2}$.

(c) Uses the modulus ω to perform *universal exclusion*: if a representative $\Theta_0 \in R$ satisfies $f(\Theta_0) - \omega(\mathrm{diam}(R)) > \varepsilon$, declares R excluded.

Theorem 2.2 (ATRT: Correctness and Termination)**.** *Under the above setup and gap assumption, the adaptive refinement procedure:*

- *Correctly declares z* traceable *by producing Θ^* with $f(\Theta^*) \leq \varepsilon + \frac{\gamma}{2}$ when $f^* \leq \varepsilon$.*
- *Correctly declares z* non-traceable *with a quantitative gap certificate when $f^* \geq \varepsilon + \gamma$.*
- *Terminates in finite time in both cases.*

Corollary 2.1 (Adaptive Complexity of Traceability Resolution)**.** *Let $C(n, \log(1/\varepsilon))$ denote the cost to perform the existential or universal check on a single region at the required precision (including evaluation of U, distance computations, and guard logic), which under (EM1–EM3) is* $\mathrm{poly}(n, \log(1/\varepsilon))$*. Then the total work before termination is*

$$O(\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon) \cdot C(n, \log(1/\varepsilon)))\,.$$

In a naive implementation $C(n, \log(1/\varepsilon))$ may be $O(n^2)$ (e.g., due to repeated full scans or unwieldy bookkeeping), but with appropriate data structures—priority refinement scheduling, cached bounds, incremental region updates—this can be reduced to $O(n \log n \cdot \mathrm{poly}(\log(1/\varepsilon)))$, giving

$$O(\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon) \cdot n \log n \cdot \mathrm{poly}(\log(1/\varepsilon)))\,.$$

The dependence on the ambient or effective dimension is encapsulated in $\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon)$; practical tractability requires that this covering number not grow too rapidly (e.g., small intrinsic dimension or structure induced by the guarded decomposition).

Corollary 2.2 (Worst-case (upper) bound)**.** *The adaptive complexity bound in Corollary 2.1 is* instance-sensitive*: it scales with the covering number $\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon)$ of the ambiguous boundary. In the pathological worst case—when the ambiguity set $\mathcal{B}_\varepsilon$ is large (e.g., occupies a full-dimensional region) and its covering number at scale d_{target} is $\Theta(d_{\mathrm{target}}^{-d})$—the cost degenerates to*

$$O\big(d_{\mathrm{target}}^{-d} \cdot C(n, \log(1/\varepsilon))\big)\,,$$

which is asymptotically no better than the naive exhaustive resolution search (up to the choice of d_{target}, itself determined by the gap γ via $\omega(d_{\mathrm{target}}) < \gamma/2$). Thus ATRT never costs more (up to constant/polynomial factors) than the non-adaptive brute force baseline, but can be much cheaper when the true ambiguity is localized. The "improvement" is a reduction from the full ambient-volume factor ε^{-d} to the typically smaller $\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon)$.

Remark 2.2 (Gap-driven dichotomy)**.** Under Promise Verification via Effective Measurability Lemma (Lem.-1.1, Appendix) and its corollary (Cor.-1.1), the choice $\omega(d_{\text{target}}) < \gamma/2$ ensures that once regions are refined to diameter d_{target}, their classification (traceable witness vs excluded) is unambiguous and no "undecided" limit behavior remains. Axiom (I) prevents indefinite avoidance of the successful branch in the traceable case.

Lemma 2.3 (Local Decomposition from PL-Traceability via ATRT)**.** *Suppose $\mathfrak{T}$ satisfies gnosis and structural axioms (note that only PL-traceability, the existence of an open-preimage per shell, is needed) and (EM1–EM3). Let $z \in \mathcal{Z}^{\text{in}}$ be an input for which the promise of Lemma 1.1 has been verified, yielding a local witness Θ^* with $f(\Theta^*) \leq \varepsilon$. Then, using only the local open-preimage information along the sequence of shells encountered from Θ^* to the basin $\mathcal{B}$ and the refinement-extension machinery (Theorem 2.1, Chapter 2), one can construct a finite guarded program Π that realizes the halting trajectory corresponding to Θ^*, without invoking uniform global UI-traceability.*

Proof. Since $f(\Theta^*) \leq \varepsilon$, Θ^* lies within the acceptance region for the traceability of z. The goal is to build Π step-by-step along the path from Θ^* toward the basin $\mathcal{B}$, using only local decomposition at each shell.

Let the shells induced by the minimal certification standard be $\mathcal{B}_1, \mathcal{B}_2, \ldots$, and suppose the halting trajectory traverses a sequence of states $\Theta^* = \Theta_{t_0}, \Theta_{t_1}, \ldots, \Theta_T \in \mathcal{B}$. At each intermediate state Θ_{t_k}, consider the corresponding shell $\mathcal{B}_{n_k}$ containing it. By PL-traceability (open-preimage for that shell), the preimage $U(\Theta_{t_k}, \cdot)^{-1}(\mathcal{B}_{n_{k+1}})$ is open and nonempty. Applying Theorem 4.1 (Chapter 1) to this local preimage yields a finite guarded partition of the relevant input neighborhood and continuous branch maps $U_i^{(k)}$ that cover the local behavior.

Proceed inductively: starting from Θ^*, for each subsequent shell along the witnessed halting path, refine the guarded decomposition to resolve any ambiguities via the ranking/tie-breaking mechanism, using Theorem 2.1 from Chapter 2 to lift coarser decisions to the needed precision and ensure stability (interior margins prevent reversal of earlier choices). Because only the shells actually encountered are used, no global uniform decomposition of the entire input space is required—each construction is localized along the trace.

Concatenating these per-shell finite guarded decompositions yields a finite guarded program Π that, when fed the appropriate inputs (as witnessed), drives the state from Θ^* into $\mathcal{B}$ in the same number of steps as the original trajectory. Continuity, compactness, and the constructive nature of the refinements ensure Π can be realized with polynomially bounded resources (since the trajectory itself is of polynomial length under the ranking bound), and each step is justified solely by local PL-traceability and the refinement-extension. Therefore Π is built without invoking any global UI-traceability, completing the construction. ■ □

Proof of ATRT. We break the argument into three parts: (I) correctness of classification under the gap assumption, (II) finite termination, and (III) adaptive work concentration.

(I) Correctness. Define $f(\Theta) = \text{dist}(U(\Theta, z), \mathcal{B})$, which is continuous on compact $\mathcal{M}$ by C; hence $f^* = \min_\Theta f(\Theta)$ is attained. The margin assumption gives two mutually exclusive cases:

Case 1: $f^* \leq \varepsilon$. There exists Θ^* with $f(\Theta^*) \leq \varepsilon$. The algorithm refines until some region $R^* \ni \Theta^*$ has $\text{diam}(R^*) \leq d_{\text{target}}$. Let $\Theta_0 \in R^*$ be the representative sample; by

Lemma 1.2, Appendix and Corollary 1.2, Appendix,

$$f(\Theta_0) \le f(\Theta^*) + \omega(\mathrm{diam}(R^*)) \le \varepsilon + \omega(d_{\mathrm{target}}) < \varepsilon + \frac{\gamma}{2}.$$

Thus the existential region check will succeed on R^*, yielding a witness Θ^* (or equivalent) with $f(\Theta^*) \le \varepsilon + \frac{\gamma}{2}$. The procedure correctly declares traceability.

Case 2: $f^* \ge \varepsilon + \gamma$. For any region R with $\mathrm{diam}(R) \le d_{\mathrm{target}}$ and representative Θ_0,

$$f(\Theta_0) \ge f^* \ge \varepsilon + \gamma,$$

and using the modulus,

$$f(\Theta) \ge f(\Theta_0) - \omega(\mathrm{diam}(R)) \ge (\varepsilon + \gamma) - \omega(d_{\mathrm{target}}) > \varepsilon + \frac{\gamma}{2} > \varepsilon$$

for all $\Theta \in R$. Therefore each such region can be excluded by the certified lower bound check. Once all regions are excluded, the algorithm correctly concludes non-traceability, with a quantitative gap (at least $\gamma/2$).

(II) Termination. Because refinement is by subdividing regions to reduce diameter and $\mathcal{M}$ is compact, finitely many subdivisions suffice to ensure all active regions have diam $\le d_{\mathrm{target}}$. At that point, by the dichotomy above, either a witness is found (Case 1) or all regions are excluded (Case 2). Hence termination is guaranteed in finite steps. Axiom (I) ensures that if $f^* \le \varepsilon$, the descent direction toward a successful branch cannot be indefinitely postponed, so the existential discovery is achievable in finite progress.

(III) Adaptive complexity. Define the ambiguous set $\mathcal{B}_\varepsilon := \{\Theta : f(\Theta) \in [0, 2\varepsilon]\}$. Regions whose interiors lie sufficiently far from $\mathcal{B}_\varepsilon$ are resolved immediately: either they contain a witness (if near the $f \le \varepsilon$ side) or are excluded by lower-bound reasoning (if safely above ε). Only regions that intersect the neighborhood of $\mathcal{B}_\varepsilon$ require refinement. Thus the number of regions that must be subdivided down to diameter d_{target} is $O(\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon))$, the covering number of $\mathcal{B}_\varepsilon$ at scale d_{target}. Each region's existential/universal checks and bound computations incur only polynomial overhead via (EM1–EM3). Therefore total work prior to resolution is bounded by $O(\mathcal{N}(d_{\mathrm{target}}, \mathcal{B}_\varepsilon) \cdot \mathrm{poly}(n, \log(1/\varepsilon)))$.

This completes the proof: the algorithm always produces a correct, definitive classification (traceable vs non-traceable) without fallback indeterminacy, respecting the stated complexity adaptation and relying only on PL-traceability (T) for per-z decisions. ■ □

Remark. This replaces any appeal to UI-Traceability in determining the status of a given z; only PL-Traceability (axiom T) is required for the correctness of the resolution of individual inputs.

PL vs UI Traceability: Role of ATRT. In Chapters 1 and 2 we temporarily assumed UI-traceability (uniform open-preimage for all shells globally) to ease the construction of guarded decompositions and to expose the structural anatomy of reachability and ranking. That assumption hides a latent reachability oracle: one does not yet know, for an arbitrary input z, whether it participates in a halting trajectory.

The Adaptive Traceability Resolution Theorem (ATRT) is the mechanism that dissolves this hidden assumption. It shows that for each individual input z, one can *effectively* (and with quantitative control) decide its traceability or non-traceability by spatially refining candidate preimage states Θ, without invoking any global UI guarantee. Thus ATRT reduces the structural reliance from UI down to PL-traceability: the

per-z decision procedure recovers the necessary localized reachability information, enabling the Fundamental Test Equivalence to be stated with only PL-traceability as a standing hypothesis.

In this light, ATRT is the cornerstone that converts the provisional structural strength of UI into an algorithmically verifiable local condition and thereby justifies the weakening of the global traceability axiom without loss of formal power.

2.7 Other prerequisites

2.7.1 Error Budget Composition Lemma

Lemma 2.4 (Error Budget Composition)**.** *Suppose $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ satisfies (C) (compactness/continuity), and its primitive guarded decomposition is equipped with the Effective Measurability Axioms EM1–EM3. Let Π be a finite guarded program of T steps that claims to drive an initial state Θ_0 into the basin $\mathcal{B}$ by producing a sequence $(\Theta_t, z_t)_{t=0}^{T}$, where each step involves evaluating a primitive update U_{i_t} and optionally a guard/membership test. Assume the minimal standard requires precision ε_T at the final step (i.e., membership in the target shell $\mathcal{B}_{n(T)}$ is to tolerance ε_T).*

Then there exists a computable precision schedule $\{\delta_t\}_{t=1}^{T}$ with $\delta_t > 0$, such that if each primitive evaluation at step t is performed to accuracy δ_t and all guard/basin tests are resolved with their own compatible margins, the following holds:

The accumulated deviation of the claimed trajectory from the true ideal trajectory is at most ε_T,

and hence the terminal check $\Theta_T \in \mathcal{B}_{n(T)}$ remains sound despite finite precision. In particular, by choosing δ_t according to the backwards propagation of continuity moduli, the error does not blow up and can be bounded in closed form as a composition of the primitive moduli, yielding a reliable certificate.

Proof. We view the guarded program as a composition of T successive primitive maps $U_{i_1}, U_{i_2}, \ldots, U_{i_T}$, where at each step t the intended exact update would produce $\Theta_t^{\text{true}} = U_{i_t}(\Theta_{t-1}^{\text{true}}, z_t)$. In the verifier's finite-precision execution we instead obtain approximate states $\widehat{\Theta}_t$, where the per-step approximation introduces an error. Our goal is to choose per-step tolerances δ_t so that the cumulative deviation $\|\widehat{\Theta}_T - \Theta_T^{\text{true}}\|$ is bounded by ε_T, compatible with the minimal standard's shell width.

By (EM2) each primitive U_{i_t} admits a known modulus of continuity $\omega_t(\cdot)$ (computable in polytime) in the sense that for any two inputs x, x',

$$\|U_{i_t}(x, z_t) - U_{i_t}(x', z_t)\| \leq \omega_t(\|x - x'\|).$$

Similarly, membership tests (guards or basin shell checks) are robust interior tests: if the approximate state lies within a margin safely inside the decision region, perturbations up to the chosen precision do not flip the predicate.

We proceed inductively from the final step backwards to allocate error budget. Let $\Delta_T := \varepsilon_T$ be the maximum allowed deviation at the terminal state so that the shell membership check remains valid. Suppose at step t the true state is Θ_t^{true} and the approximate state is $\widehat{\Theta}_t$; define the forward propagated error after step t as

$$e_t := \|\widehat{\Theta}_t - \Theta_t^{\text{true}}\|.$$

We want $e_T \leq \Delta_T$.

The approximation at step t arises from two sources:

1. The error carried from the previous step e_{t-1}, propagated through the continuity of U_{i_t}, bounded by $\omega_t(e_{t-1})$.

2. The fresh rounding/truncation error of evaluating U_{i_t} to finite precision, which we denote by δ_t; by construction via (EM2) we can enforce that the evaluation error (for correct inputs) is at most δ_t.

Thus the total error satisfies

$$e_t \leq \omega_t(e_{t-1}) + \delta_t.$$

Unrolling this recursion gives

$$e_T \leq \omega_T\big(\omega_{T-1}(\cdots\omega_2(\omega_1(e_0) + \delta_1) + \delta_2 \cdots)\big) + \delta_T,$$

where $e_0 = 0$ since Θ_0 is assumed given at sufficient precision (or its encoding error is folded into δ_0).

Define backwards error targets Δ_t for $t = 0, \ldots, T$ via:

$$\Delta_T := \varepsilon_T, \quad \Delta_{t-1} := \omega_t^{-1}(\Delta_t - \delta_t),$$

where ω_t^{-1} is an (effective) right-inverse on the relevant range—this is well-defined because each ω_t is continuous with $\omega_t(0) = 0$, and by choosing δ_t small enough we ensure $\Delta_t - \delta_t$ lies in the image of ω_t. We can select a decreasing sequence δ_t (for example, geometrically scaled) so that each Δ_{t-1} is positive and finite.

By induction, if $e_{t-1} \leq \Delta_{t-1}$ and we evaluate U_{i_t} to precision δ_t, then

$$e_t \leq \omega_t(e_{t-1}) + \delta_t \leq \omega_t(\Delta_{t-1}) + \delta_t = \Delta_t.$$

Starting with $e_0 = 0 \leq \Delta_0$, the chain guarantees $e_T \leq \Delta_T = \varepsilon_T$. Therefore the cumulative error budget stays within the allowable tolerance, and the final check $\widehat{\Theta}_T \in \mathcal{B}_{n(T)}$ is sound: even with finite precision, the approximate trajectory cannot deviate outside the intended shell.

All quantities ω_t, the inverses (computed via continuity and compactness), and the δ_t-schedule are effective and computable in time polynomial in the relevant precision parameters by (EM2). Thus the lemma's precision schedule exists and is implementable, completing the proof. □

2.7.2 Effective Inversion of Continuity Moduli Lemma

Lemma 2.5 (Effective Inversion of Continuity Moduli)**.** *Let $\omega : \mathbb{R}_+ \to \mathbb{R}_+$ be a known continuous, strictly nondecreasing modulus function with $\omega(0) = 0$, computable to arbitrary precision in time polynomial in the bit-length of its argument (as supplied or derived via EM2 / Lemma 1.2, Appendix). Then for any target forward error budget $\Delta > 0$ and any chosen fresh per-step allowance δ with $0 < \delta < \Delta$, one can compute in time polynomial in the precision parameters a preceding error threshold $\Delta_{\text{prev}} > 0$ such that*

$$\omega(\Delta_{\text{prev}}) + \delta \leq \Delta.$$

Moreover, this inversion can be carried out effectively (e.g., by binary search on $[0, \Delta]$) using the monotonicity of ω, yielding a backward error schedule consistent with the recursive error propagation in Lemma 2.4.

Proof. Because ω is continuous, strictly nondecreasing, and $\omega(0) = 0$, the map $r \mapsto \omega(r)$ is invertible on its image, and for any $\Delta' := \Delta - \delta > 0$ there exists a maximal Δ_{prev} such that $\omega(\Delta_{\text{prev}}) \leq \Delta'$. Define $\Delta_{\text{prev}} := \sup\{r \geq 0 : \ \omega(r) \leq \Delta'\}$; continuity ensures this supremum is attained and satisfies $\omega(\Delta_{\text{prev}}) \leq \Delta'$.

To compute Δ_{prev} to desired precision, perform a binary search on the interval $[0, \Delta]$: at each iteration evaluate ω at the midpoint r, compare $\omega(r)$ to Δ', and narrow the interval accordingly. Since ω is computable in polynomial time in the bit-length of r, and the required number of bisections to achieve k-bit precision is $O(k)$, the overall inversion process remains polynomial-time in the precision parameters. Once Δ_{prev} satisfying $\omega(\Delta_{\text{prev}}) \leq \Delta - \delta$ is found, we have

$$\omega(\Delta_{\text{prev}}) + \delta \leq \Delta,$$

as desired.

This computed Δ_{prev} can be used as the backward target in the construction of the sequence $\{\Delta_t\}$ in Lemma 2.4: given a desired terminal allowance Δ_T, one iteratively inverts each continuity modulus with the corresponding fresh δ_t to obtain Δ_{t-1}, guaranteeing cumulative error control. ■ □

2.7.3 Polytime Verifier Construction Lemma

Let's observe that the existence of the Polytime Verifier, used in important PUB result (Corollary-3.1, Book II, Chapter 6) is now merely a consequence of the Effective Measurability and per-step certifiability:

Lemma 2.6 (Polytime Verifier Construction)**.** *Under the Effective Measurability Axioms EM1–EM3 and the per-step testability result of Book II, Chapter 4, there exists a deterministic polynomial-time algorithm*

$$V \ : \ \{\textit{measurement input } I\} \times \{\textit{capture certificate } W\} \ \rightarrow \ \{\textit{accept, reject}\}$$

and a polynomial $q(n)$ such that for every input I of bit-length n and every capture certificate

$$W \ = \ \underbrace{\text{bin}(T)}_{\substack{\textit{binary encoding} \\ \textit{of the step count}}} \ \| \ \underbrace{w_1 \| \cdots \| w_T}_{\substack{\textit{concatenation of per-step} \\ \textit{witnesses } |w_t| \leq b}}$$

the following hold:

1. *$V(I, W)$ runs in time $\leq q(n)$.*
2. *$V(I, W)$ accepts if and only if W correctly encodes a trajectory of length T whose states Θ_t satisfy $\Theta_t = U(\Theta_{t-1}, z_t^{\text{in}})$ and $\Theta_T \in \mathcal{B}$.*

Proof. **1.** On input (I, W), first scan the prefix $\text{bin}(T)$ in $O(n)$ time to recover the integer T. The remaining bits are parsed into T blocks $w_1, \ldots, w_T$, each of size $\leq b$.

2. Initialize Θ_0 from the measurement input I. For $t = 1$ to T:

1. Using (EM1–EM3), compute $\widehat{\Theta}_t = U(\Theta_{t-1}, z_t^{\text{in}})$ in time $poly(n)$. At each replay step the verifier evaluates U and E only to the k bits of precision prescribed by the global error budget from Lemma 2.4, using the polytime evaluators guaranteed by EM2. This suffices to determine guard outcomes and to ensure that the simulated trajectory remains within the allowed tolerance ε_T at the terminal step.

2. Decode the net-point η_{j_t} from w_t (constant-time since $|w_t| \le b$).

3. Compute $d(\widehat{\Theta}_t, \eta_{j_t})$ by (EM2) in $poly(n)$ time.

4. If $d(\widehat{\Theta}_t, \eta_{j_t}) > \delta$, **reject**.

5. Otherwise set $\Theta_t \leftarrow \widehat{\Theta}_t$ and continue.

3. After T steps, check membership $\Theta_T \in \mathcal{B}$ via EM3 in $poly(n)$ time; *reject* if false.
4. If no rejection has occurred, **accept**.

Fairness verification. The verifier need not perform any fairness checks: the NP witness is an actual accepting trajectory of polynomial length, so it suffices to replay it faithfully and verify that it reaches $\mathcal{B}$ within the stated tolerance.

Error management. The per-step approximation errors and the cumulative deviations are controlled via Lemma 2.4. In particular, the verifier chooses the per-step precision targets δ_t according to the backward error budget determined there so that, despite finite precision in the computation of each $U(\Theta_{t-1}, z_t^{\text{in}})$ and comparison with the witness net-points w_t, the reconstructed trajectory stays within the basin shell tolerance at step T. Hence the final basin membership check is reliable, and no unchecked drift invalidates the certificate.

Runtime: Each of the T iterations uses $O(poly(n))$ time, so the total cost is

$$O\big(T \cdot poly(n)\big) \;+\; O(n) \;=\; O\big((n+T)\, poly(n)\big).$$

Since by Chapter 4 and ABET we have $T = O(poly(n))$, this overall cost is bounded by some polynomial $q(n)$.

Correctness: By construction, V accepts exactly when each per-step witness w_t correctly certifies the transition $\Theta_{t-1} \to \Theta_t$ and the final state Θ_T lies in $\mathcal{B}$. Hence V recognizes precisely the valid capture certificates. □

2.7.4 Polytime Simulability of Primitives Lemma

The next important result is that, under Effective Measurability Axioms, the decomposed program is also polytime-verifiable.

Lemma 2.7 (Polytime Simulability of Primitives)**.** *Let* Π *be a finite guarded program composed from primitive transition maps*

$$U_j : \mathcal{X} \times \mathcal{Z} \;\to\; \mathcal{X}, \qquad E_j : \mathcal{X} \;\to\; \mathcal{Z}, \quad j = 1, \dots, K,$$

over compact metric spaces $\mathcal{X}, \mathcal{Z}$*. Suppose each primitive satisfies the Effective Measurability Axioms.*

Then, given an external input of length n *and a claimed execution trace of* Π *of* T *steps, one can* deterministically replay and verify *the entire trace in time*

$$O\big(T \;\cdot\; \mathrm{poly}\big(n, \log(1/\varepsilon_T)\big)\big),$$

where ε_T *is the final precision required by the minimal standard at step* T*.*

Proof. **Overview.** We show that each of the T iterations of the guarded program can be checked in time polynomial in n and $\log(1/\varepsilon_T)$. Summing yields the claimed bound.

Controlled precision accumulation. To ensure that the repeated application of the primitive updates and guard/membership tests does not accumulate unbounded error, the verifier employs the precise scheduling from Lemma 2.4. Each step's output is computed to a preallocated precision δ_t such that the propagated error remains below the minimal standard's tolerance for the corresponding shell. Continuity (C) and the known moduli (EM2) guarantee that these δ_t can be chosen effectively so the composed simulation stays faithful: the claimed trace can be replayed and checked with total deviation bounded, making the verification sound.

1. Encoding precision requirements. By Lemma 2.2, at step t the verifier needs only ε_t-precision, where $\varepsilon_t \geq 2^{-\mathrm{poly}(n)}$. Thus $\log(1/\varepsilon_t) = \mathrm{poly}(n)$.

2. Evaluating a single update. At iteration t, suppose the current state is claimed to be $x_t \in \mathcal{X}$ (represented to ε_t accuracy) and the guard identifier is i_t. The verifier must:

(a) *Guard membership.* Check $z_t \in N_{i_t}$ to within ε_t. By (EM3) this takes $\mathrm{poly}(n, \log(1/\varepsilon_t))$ time.

(b) *Compute update.* Evaluate $x'_{t+1} = U_{i_t}(x_t, z_t)$ to ε_{t+1} precision. By (EM2), this requires at most $m_{i_t}(k)$ arithmetic steps, with $k = O(\log(1/\varepsilon_{t+1})) = \mathrm{poly}(n)$. Since each arithmetic step on $O(n)$-bit rationals costs $\mathrm{poly}(n)$ time, the total is $\mathrm{poly}(n)$.

(c) *Compare to claimed.* Check that the prover's claimed x_{t+1} matches x'_{t+1} within ε_{t+1}. This is a single subtraction and magnitude comparison in $\mathrm{poly}(n)$ time.

Each of (a)–(c) is polynomial in n and $\log(1/\varepsilon_{t+1})$.

3. Verifying emission/halting. If at step t the program emits $y_t = E_{i_t}(x_t)$ or checks for basin membership, the same argument applies:

(a) Evaluate $y'_t = E_{i_t}(x_t)$ to precision ε_t in $\mathrm{poly}(n)$ time by (EM2).

(b) Compare to the observed y_t (or test $x_t \in \mathcal{B}_t$ by (EM3)).

4. Total cost. Each of the T steps incurs at most $\mathrm{poly}(n, \log(1/\varepsilon_T))$ work. Summing yields

$$T \times \mathrm{poly}\big(n, \log(1/\varepsilon_T)\big) = O\big(T \cdot \mathrm{poly}(n)\big),$$

as claimed. □

2.8 Main fundamental equivalence proof

Fundamental Equivalence Proof. We prove $(a) \Rightarrow (b)$ and $(b) \Rightarrow (a)$.

(a) ⇒ (b). Assume NP–certifiability. By ABET (Book II, Chapter 6), we get *finite* uniform entry time $T_{\max}$. Applying ABET Corollary 3.1 (PUB) yields a *polynomial* bound $T_{\max} \leq p(n)$, so NP–certifiability entails:

- the existence of a reachable, invariant basin $\mathcal{B}$ (so (F) is satisfied), and

- a uniform *polynomial* bound $T_{\max} \le p(|\text{input}|)$ on the entry time into $\mathcal{B}$ for every accepted head (equivalently: a well-founded ranking f with $f(\Theta_0) \le p(|\text{input}|)$).

Continuity and compactness (C) and the minimal standard (VT3) are part of the valid-test setup. Axiom (T) (open-preimage) is assumed.

By Theorem 4.1 of Chapter 1 (Open-Preimage $\Leftrightarrow$ Finite Tick-0 Decomposition) together with Lemma 2.3 (which shows how to assemble the guarded program from only shell-wise open-preimage / PL-traceability along the witnessed trajectory), we can finitely partition $\mathcal{Z}^{\text{in}}$ into guarded cells and restrict U to continuous branches U_i.

Then, by Theorem 5.1 of Chapter 1 (Inductive Shell-by-Shell Decomposition) we refine these branches over the shells $\mathcal{B}_1, \ldots, \mathcal{B}_{T_{\max}}$ and wrap them with a Loop-Until primitive (Chapter 1, Def. 2.4), obtaining a single finite guarded program Π.

Since $f(\Theta_0) \le T_{\max} \le p(|\text{input}|)$ and Chapter 2 Lemma 5.1 guarantees that each iteration can decrease f by at least one, Π halts in at most $p(|\text{input}|)$ steps.

Finally, by Lemma 2.2, encoding each Θ_t to the needed ε_t-precision costs only $\text{poly}(n)$ bits; and by Lemma 2.7, each of the $T \le p(n)$ step-simulations and guard checks runs in $\text{poly}(n)$ time. Therefore, the entire one-tick composition halts in polynomial time. Thus (b) holds.

(b) $\Rightarrow$ (a). Suppose such a finite guarded program Π exists and halts within $p(|\text{input}|)$ steps on every trajectory by entering $\mathcal{B}$. Let T be the halting step and Θ_T the halting state.

The NP witness is taken to be

$$W := \big(\text{bin}(T),\ (i_t, z_t, \text{checkpoint}_t)_{t=1}^{T}\big),$$

where:

- T is the claimed halting time in binary;
- i_t is the identifier of the guard or branch taken at step t (omitted if deterministically implied);
- z_t is the input measurement supplied at step t (if not fully determined by the guard);
- checkpoint_t is a rational description of the state/output at step t to accuracy ε_t as in Lemma 2.6.

Given W, the verifier replays Π for T steps using the effective measurability axioms (EM1–EM3), checking at each step that the provided branch, input, and checkpoint are consistent with the dynamics up to the stated tolerance. Lemmas 2.7 and 2.4 guarantee this replay runs in time polynomial in n.

A polynomial-time verifier does:

1. Replay Π for $T \le p(n)$ steps: by Lemma 2.7 this costs $O(p(n) \cdot \text{poly}(n)) = \text{poly}(n)$, and by Lemma 2.2 it only needs polynomial-size encodings at each step to check basin membership. Thus, the certificate can be verified in $\text{poly}(n)$ time.
2. Check $\Theta_T \in \mathcal{B}$ within the tolerance ε_T (minimal standard).
3. By invariance, all subsequent outputs lie in the output guard-band; accept iff all checks pass.

Thus the language of good heads is in **NP**, proving (a). □

Remark 2.3 (Where the Polynomial Bound Lives)**.** Condition (C5) (polytime reachability) from the earlier formulation has been *internalized*:

- In $(a) \Rightarrow (b)$ it is *derived* from NP–certifiability (via the basin equivalence and ranking).
- In $(b) \Rightarrow (a)$ it is part of the statement: we require the guarded program to halt in polynomial time to make the witness/verifier polynomial.

No separate axiom is needed.

Corollary 2.3 (Shell Ranking Realization)**.** *Let* $f_{\text{shell}}(\Theta) = \min\{n : \Theta \in \mathcal{B}_n\}$*. Then* f_{shell} *is a ranking, and if* (a) *holds,* $f_{\text{shell}}(\Theta_0) \leq p(|\text{input}|)$ *for some polynomial* p*. Conversely, if a guarded program halts in* T *steps, one can take* $f_{\text{shell}}(\Theta_0) = T$*.*

Remark 2.4 (Universal vs. Existential Decrease)**.** If a strict Lyapunov-border function exists (universal decrease under all admissible inputs), the polynomial bound follows from its contraction rate. The theorem shows this strong analytic structure is *sufficient but not necessary*: the weaker existential ranking suffices once the gnosis and structural axioms are in place.

Remark 2.5 (Precision Targets)**.** If the practical goal is entry into a target shell $\mathcal{B}_{N(\delta)}$ (finite precision δ) rather than $\mathcal{B}$, the theorem holds verbatim with "halt" $\equiv$ "enter $\mathcal{B}_{N(\delta)}$," and the witness records $N(\delta)$.

Remark 2.6 (Halting Problem)**.** In accordance with the Universality Theorem 2.2 from Chapter 2, guarded program decomposition is Turing-complete. Therefore, "halting of the guarded program" subsumes the classical halting problem; the Fundamental Equivalence Theorem then says: *NP-certifiability picks out exactly those such programs that halt in polynomial time.*

3 Necessity of the decomposition we proposed

The last fundamental question is: was this way strictly necessary? We can't deviate once we have provided the decomposition; however, maybe our constructive approach to defining a good decomposition by 0-tick was too strict?

We introduced it to answer many deep, but rather informal questions listed in Section 3 of Chapter 1.

Let's revisit these questions and formalize our requirements.

3.1 Decomposition Necessity

We now show that the finite guarded decomposition and Loop-Until structure are not only *sufficient* for NP-certifiability, but in fact *necessary*: any test meeting our axioms and admitting a polynomial-size certificate *must* decompose into the three primitives and loop as in Definition 2.4, Chapter 1.

Theorem 3.1 (Decomposition Necessity)**.** *Let*

$$\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$$

be a valid test (Def. 2.1, VT1–VT4, Chapter 1) whose tail predicate admits a polynomial-size NP certificate. Then:

(a) For each shell index n, the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ must be open and nonempty (Input-Traceability is forced).

(b) Hence, by Theorem 4.1, Chapter 1, U admits a finite guarded decomposition at each shell n.

(c) Inductively refining these decompositions yields exactly the depth-n Conditional maps $U^{(n)}$ of Theorem 5.1, Chapter 1.

(d) Finally, the overall test is extensionally equal to the Loop-Until Test of Definition 2.4, Chapter 1, i.e. the sequential iteration of those one-tick Conditional tests until entry into $\mathcal{B}$.

In particular, no NP-certifiable test can avoid the three primitives plus Loop-Until structure.

Proof. **(a) Necessity of Input-Traceability.** By NP-certifiability and the PUB corollary (Cor. 3.1, Book II, Chapter 6), there is a uniform finite bound on entry time into the basin, yielding a ranking and hence the ability to exhibit a halting trajectory for every accepted head.

Then Theorem 2.1 (Primitive–Global Certification Standard Equivalence, Chapter 1) implies that for each shell index n, the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ must be open and nonempty; that is, input-traceability at shell level n is forced.

(b) Finite Guarded Decomposition at Each Shell. By (a) and compactness of $\mathcal{Z}^{\text{in}}$, Theorem 4.1, Chapter 1 applies at each level n: one obtains a finite open partition $\mathcal{Z}^{\text{in}} = \bigsqcup_{i=1}^{k_n} N_i^{(n)}$ and restricted maps $U_i^{(n)}$ each admitting a nonempty guard-band into $\mathcal{B}_n$.

(c) Inductive Refinement to All Depths. Applying the same argument to each shell in turn, we build the full depth-n Conditional map

$$U^{(n)} = \text{Conditional}\big(\{\, U_{\mathbf{i}}^{(n)} \}_{\mathbf{i} \in I_n}\big) \quad \text{(Thm. 5.1, Chapter 1).}$$

No alternative structure is possible without violating openness or finiteness.

(d) Loop-Until Realizes the Original Test. Since by NP-certifiability every fair run of $\mathfrak{T}$ must enter $\mathcal{B}$ in bounded time (PUB, Cor. 3.1, Book II, Chapter 6), the behavior of $\mathfrak{T}$ on any input stream (z_t) is uniquely determined by applying, at each tick t, the corresponding depth-n_t Conditional update $U^{(n_t)}$ and checking basin membership. This is exactly the Loop-Until Test (Def. 2.4, Chapter 1). Extensionally, the two tests coincide.

Conclusion. No NP-certifiable test can function without:

1. finite guarded decomposition at every shell (by input-traceability + compactness), and
2. sequential iteration of those one-tick Conditional steps (Loop-Until).

Hence, the structure of three primitives, including Loop-Until is not just sufficient but logically ***necessary***. □

4 The final resolution of the information-loss paradox

Formal grounding. The following interpretive summary builds on precise earlier results: the vanishing of non-ranking ambiguity and the push of residual uncertainty into the deterministic ranking are formalized in Theorem 6.1 (Chapter 2), the necessity of fair progress is captured in Proposition 5.2, the preservation of existential decrease under decomposition appears in Lemma 5.1, and the main result of this chapter - the Fundamental Test Equivalence Theorem (Thm.-1.1). What follows is an explanation of those phenomena in conceptual terms; the exact technical claims remain in the referenced theorems and lemmas.

Time-irreversibility. Our exploration of principles of certifiability finally came to an end with the discovery of the full analog of the Second Law of Thermodynamics - the minimal requirements of the time-irreversible execution of any guarded program run, our Weak Fairness axiom.

WF imposes an arrow: once "progress-enabling" moves keep reappearing, progress must occur. That mirrors thermodynamics' "if entropy-increasing micropaths are overwhelmingly available, entropy won't decrease forever."

Caveat: The Second Law is statistical/measure-theoretic; WF is a *normative/semantic* constraint on admissible runs (fairness), not a counting argument. So it's a *deep analogy*, not an identity.

Finite precision. Our previous attempts to explain the information loss by a finite precision hit a wall: once we increase the resolution, we *decrease* the loss. Let

- $X_t := \Theta_t$ be the full microstate at time t.
- R_t be the *recorded* symbol at that time (rank value, cell ID chosen for certification, etc.).

Define the instantaneous loss

$$L_t \;:=\; H[X_t \mid R_t]$$

- If we **refine** the partition, you can choose a *refined* symbol R'_t with more bits: then $H[X_t \mid R'_t] \leq H[X_t \mid R_t]$. Loss *decreases*.
- If, instead, we compress to a coarser symbol (say, only the rank), we increase the loss.

Simply driving $\epsilon \to 0$ drives the loss to 0 as well. So why do we gain *more* information in the statement accuracy *as the loss decreases to an arbitrarily low threshold?*

Deeper dive into this question is given in Double-Bound Decomposition Theorem in Chapter 2. It establishes the extremely important facts:

1. Asymptotical losslessness of the update map is a fact not only over a some mathematical function of special kind; it is actually a property of a guarded program. We can refine a guarded program to an arbitrarily high precision of a result.

2. We can asymptotically certiffy a measurement with a decomposed test with infinite precision (given we have infinite-precision standards).

3. All the genuine information loss in decomposed test is pushed into TI/WF ranking function in a limit.

Hence, a formally present information loss cause by a finite precision explains the certifiability ONLY in conjunction with the time irreversibility. We *must* iteratively refine/confirm — hence *to spend time*. Infinite precision just means, the *infinite time*.

Concretely, we *must* introduce WF ranking function, which is bound not to the structure of the program or to physics (regardless of how it would depend on them), but to *execution* of the program, i.e., precisely the process in time. And this ranking function *is* the explicit information loss.

An intriguing question is what would allow us to remove this loss? Only if we'd had an oracle (infinite precision or unbounded computational power) could we, in principle, collapse the process to one timeless step. We will revisit this in the next chapter.

5 Conclusion: From Certifiable Dynamics to Programmable Tests

5.1 What We Set Out to Do—and What We Found

We began exploring our Testable Measurement in Chapter 8 of Book II with a structural question: *What is a certifiable measurement in essence?* How can we learn this from analyzing and opening its internal structure and behavior?

Our chain of thoughts was as follows:

C1. **Introducing Time-Domains.** Observing the event map collapse, we complemented the *output* Singular Time-Domain of the Simple Test by an explicit and, generally, multi-tick *input* time-domain. It was already present by a map collapse; so nothing is introduced ad hoc here. But the result is decisive: without distinct input ticks, there is no space to decompose, schedule, or control anything.

C2. **Exhibiting the Minimal Certification Standard.** The standard $\mathcal{S} = \{\varepsilon_n\}$ is not an add-on; it *induces* state, input, and output guard-bands. It is the "knife" that slices the measurement into Observation, Adaptation, and Restitution.

C3. **Proving Three-Aspects Uniqueness and the Simple Test.** Once $\mathcal{S}$ is in place, the three aspects are forced and unique (up to null sets). The Simple Test emerges as the canonical form: one certified tick, maximal index collapse, Singular Time-Domain.

C4. **Deriving the Primitive Toolkit.** We showed that Sequential, Conditional, and Loop are necessary and sufficient to build any valid composite test. Their use is *not* arbitrary: open-preimage (input traceability) and minimal standards regulate every branch and loop.

C5. **Open-Preimage $\Longleftrightarrow$ Finite Tick-0 Decomposition.** The "traceability axiom" is exactly the requirement that certification constraints lift to open, non-degenerate input sets. With compactness, this gives a *finite* guarded partition at tick 0.

C6. **Inductive Shell Decomposition & Loop-Until.** Shell-by-shell refinement—driven solely by $\mathcal{S}$—plus a terminating loop yields a total guarded program. No extra geometry beyond continuity and compactness is needed.

C7. **Turing-Completeness of Guarded Tests.** We proved universality: any deterministic TM can be implemented as a valid test; halting $\iff$ basin entry.

C8. **Time-Irreversibility of any Guarded Test performing physical measurement.** We proved that endowing a Turing-complete decomposed structure with any kind of physical input and belief-state spaces requires the explicit time irreversibility in its weakest form - the Weak-Fairness axiom. We showed that this axiom is precisely necessary and sufficient for ANY measurement result to be produced, in a polynomial or non-polynomial time.

C9. **Ranking & Reachability.** Then, we investigated the conditions for our program to provide a *polynomial* reachability, thus making a base to create a *certifiable* measurement. It is turned out that the requirement is precisely the same as above for *general* termination: the program execution must be weakly fair, or, in terms involving physics of our metrology, the adaptation process *must* be time-irreversible.

Axiom of I (Axiom-4.1) ranking function is *necessary*; combined with gnosis and structural axioms, it is also *sufficient* for uniform reachability.

C10. **Effective Primitive Measurability.** We impose the requirements for the measurability, which are reducible to Lebesgue measurability of the input space, plus the polynomial information processing of subsets in the input space.

C11. **Fundamental Equivalence.** Finally, we formulated the core equivalence: *NP–certifiability* $\iff$ *existence of a finite guarded program that halts in polynomial time.* The polynomial bound is no longer an axiom; it is the very complexity clause in the equivalence.

C12. **Necessity of the decomposition.** We have already proved that our primitive set is necessary and sufficient for the chosen tick-0 way. This left a potential loophole - what is some other way of decomposition that yields a Simple Test?

So finally, we proved that our decomposition (i.e., primitive set) is *necessary* - no construction can ever avoid it.

Therefore, we finished the establishment of the fundamental result:

Every (certifiable) measurement (the term "certifiable" can be used interchangeably with the "one that states a fact") is equivalent to the TM-complete physical procedure of effectively measurable tests against finite-information standards executed irreversibly in (polynomial) time.

Chapter 4: Fact Decomposition-Independence Theorem

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1 Introduction & Motivation

In Book III, Chapter 1 we promised to examine in detail the two **canonical counter-examples** to the Church–Turing thesis. We now keep that promise, and show how each of these thought-experiments *precisely* violates one of our axioms. This will lead us to conjecture—and then prove—the *Fact Decomposition-Independence Theorem*, which characterises exactly when a measurement fact can evade all finite, constructive decompositions.

1.1 Promise Recap

Recall that we set out to understand why certain physical processes appear to compute non-Turing functions. In particular, we highlighted two archetypal constructions:

- **Accelerating (Zeno) Turing Machine**: packs infinitely many TM steps into finite real time, thus deciding the Halting problem.
- **3-D Wave-Equation Solver**: encodes halting bits into arbitrarily high spatial frequencies so that a point-evaluation after one second solves HALT.

1.2 The Two Canonical Examples

Example 1: Accelerating Turing Machine A device whose k-th computation step takes time $2^{-(k-1)}$, so that in two seconds it has simulated any ordinary TM on input x. If the simulated TM ever halts, a lamp flashes at $t=2$. This yields a finite-time decider for HALT.

Example 2: 3-D Wave-Equation Solver Choose computable initial data (f, g) on $\mathbb{R}^3$ whose Fourier modes of wavelength 2^{-n} encode whether the nth TM on x halts. By standard spherical-means, the point value $u(0, 1)$ crosses a threshold iff HALT holds.

1.3 Observation: Which Axiom Fails?

Recap the axioms our framework rests on:

Let

$$\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$$

be a *Simple Test* (Def. 4.3 from Book II, Chapter 8), endowed with its finite guarded decomposition (Thm. 1.1 from Chapter 3) which is built upon the axioms:

Gnosis Axioms:

(F) **Invariant basin.** There exists a reachable, invariant attractor basin $\mathcal{B} \subseteq \mathcal{M}$ (Deff. 2.1,2.2,2.3 from Book II, Chapter 5). *(There is a fact that remain)*

(S) **Certification standard.** $\exists \mathcal{S} = \{\varepsilon_n\}_{n=1}^{\infty} : \forall \varepsilon_n \in (0,1] \; \exists \; verifier \; V : V(z_{1:n}^{\text{out}}, w) = accept \implies \forall z : d(z, z_{1:n}^{\text{out}}, w)) < \varepsilon_n \; V(z) = accept$ (Def. 2.2 from Book II, Chapter 6). *(The statement can be made about the fact)*

Structural Axioms:

(C) **Compact Continuity.** $\mathcal{M}, \mathcal{Z}^{\text{in}}$ are compact metric spaces; U, E are continuous (Deff. 1.1,1.2,2.1 from Book II, Chapter 3). *(There are finite total degrees of freedom under any finite resolution)*

(T) **Traceability.** For every $n \geq 1$ and $\Theta \in \mathcal{M}$, the set $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is open and nonempty (Def. 2.1 from Chapter 1). *(The fact is traceable)*

Process Axiom:

(I) **Time Irreversibility.** There exists the finite decomposition of the test $\mathfrak{T}$ into guarded program Π, built from the primitives Sequential, Conditional, Loop and equipped with a *ranking function* f that satisfies (I) Axiom 4.1 from Chapter 2 on program runs named in further *(I)-fair executions*. *(Execution time is irreversible)*

Effective Measurability Axioms:

(EM1) **Effective Representation.** Each U_j and E_j is given by a rational arithmetic circuit of size poly(n); all coefficients are rationals of poly(n) bit–length.

(EM2) **Polynomial Modulus of Continuity.** There is a known polynomial $m_j(k)$ bounding the cost to compute $U_j(x, z)$ or $E_j(x)$ to k bits in poly$(m_j(k), n)$ time.

(EM3) **Polynomial Guard/Basin Tests.** Membership in any guard cell $N_i \subset \mathcal{Z}$ or shell $\mathcal{B}_n \subset \mathcal{X}$ can be decided in time poly$(n, \log(1/\varepsilon))$.

Which axiom fails in each example?

- *Zeno Machine* breaks (I): it executes infinitely many steps in finite time.
- *Wave Solver* breaks (C): it requires infinite-bandwidth, hence violates the minimal standard of finite information.

2 Map of facts and axioms

3 ABET/FTE Sensitivity to Dropping Individual Axioms: Full Lemmas

In this section we give a standalone lemma (with proof) for each axiom, showing exactly how its absence disrupts the ABET/FTE framework.

3.1 Failure of Invariant Basin (F)

Lemma 3.1 (No Simple Test without (F))**.** *If Axiom (F) (Invariant Basin) is dropped, then one cannot even formulate a non-trivial Simple Test, and hence ABET and FTE have no content.*

Proof. (1) A "tail fact" is by definition of the form "$\exists T\ \forall t \geq T : \Theta_t \in \mathcal{B}$" for some invariant basin $\mathcal{B}$. Without (F) no such $\mathcal{B}$ exists or is guaranteed reachable.

(2) In particular one cannot assert Safety ($\Theta_t \in \mathcal{B} \implies \Theta_{t+1} \in \mathcal{B}$) or Liveness (eventually enter $\mathcal{B}$).

(3) Therefore the very notion of a certificate "once in $\mathcal{B}$, stay in $\mathcal{B}$" collapses. One cannot state any non-trivial tail-predicate, so ABET/FTE become vacuous.

□

3.2 Failure of Certification Standard (S)

Lemma 3.2 (No Robust Certification without (S))**.** *If Axiom (S) (Certification Standard) is dropped, then even if an invariant basin exists, arbitrarily small output perturbations can invalidate any purported certificate, so ABET fails.*

Proof. (1) ABET's core open-cover/finite-subcover argument constructs shells B_n around $\mathcal{B}$ and relies on a tolerance $\varepsilon_n > 0$ so that any measurement within ε_n still yields acceptance.

(2) Without (S) there may be no nonzero ε_n with the property

$$V(z_{1:n}, w) = \text{accept} \implies V(z'_{1:n}, w) = \text{accept} \quad \text{whenever } d(z_{1:n}, z'_{1:n}) < \varepsilon_n.$$

(3) Hence even if $\Theta_t \in \mathcal{B}$, the certificate can be "torn apart" by an ω-small noise and ABET's guarantee of a uniform finite certificate vanishes.

□

3.3 Failure of Compact Continuity (C)

Lemma 3.3 (No Uniform Bound without (C))**.** *If Axiom (C) (Compact Continuity) is dropped, then the map*

$$\Theta_0 \mapsto T(\Theta_0) = \min\{\, t : \Theta_t \in \mathcal{B}\}$$

need not attain a maximum, so ABET's finite-subcover step fails.

Proof. (1) Under (C), $\mathcal{M}$ is compact and U, E are continuous, so $T(\cdot)$ is upper-semicontinuous and attains $\max T_{\max} < \infty$.

(2) Without compactness one can exhibit a sequence $\Theta^{(k)}$ with $T(\Theta^{(k)}) \to \infty$, so no single finite time works for all initial states.

(3) The open-cover/finite-subcover argument in ABET therefore has no analogue.

□

3.4 Failure of Time Irreversibility (I)

Lemma 3.4 (No Finite Decomposition without (I))**.** *If Axiom (I) (Time Irreversibility) is dropped, then there can be NP-style certificates along infinite paths that cannot be unrolled into any finite guarded program, so FTE fails.*

Proof. (1) (I) ensures that every enabled condition in an infinite execution recurs infinitely often, which is used in the Reduction Lemma to turn an NP-style certificate into a finite guarded decision tree.

(2) Without (I) one can schedule infinitely many distinct guards in Zeno-style within finite time, so no finite list of guard-cells can cover all possible accepting paths of the NP-verifier.

(3) Hence the converse direction of FTE—that every NP-verifiable fact admits a finite decomposition—breaks.

□

3.5 Failure of Traceability (T)

Lemma 3.5 (Abstract ABET survives, but no effective guards without (T))**.** *If Axiom (T) (Open-Preimage) is dropped, the abstract open-cover argument of ABET still goes through, but one cannot exhibit any finite list of* effectively testable *guard cells, so FTE holds only non-constructively.*

Proof. (1) The compactness argument up to finite-subcover uses only (F),(S),(C),(I).

(2) But without (T) the preimage $U^{-1}(B_n) \subset \mathcal{Z}^{\text{in}}$ need not be open, so there is no guarantee of any *guard cell* $N \subset \mathcal{Z}^{\text{in}}$ that is both open and decidable.

(3) Thus ABET (existence of a finite witness in principle) remains valid, but no finite *effective* decomposition can be extracted.

□

3.6 Failure of EM1 (Effective Representation)

Lemma 3.6 (No Effective Decomposition without (EM1))**.** *If (EM1) fails—i.e. some U_j, E_j require non-rational or super-polynomial-size constants—then although ABET and FTE hold abstractly, no finite guarded decomposition can be realized by a polynomial-time circuit.*

Proof. (1) Abstract ABET uses only topological and logical arguments, so it still yields a finite decomposition in principle.

(2) But any guard-test or update step invoking the non-rational constant cannot be implemented by a rational arithmetic circuit of size poly(n), violating the EM1 requirement for *effective* decomposition.

□

3.7 Failure of EM2 (Polynomial Modulus of Continuity)

Lemma 3.7 (No Poly-time Observation without (EM2))**.** *If (EM2) fails—i.e. the modulus of continuity of some U_j or E_j grows faster than any polynomial—then the Simple Test may exist but any effective recovery of the tail predicate requires super-polynomial precision, so no polynomial-time decomposition is possible.*

Proof. (1) Abstractly, continuity suffices for ABET/FTE.

(2) Without a polynomial bound $m_j(k)$ on input-accuracy vs. output-accuracy, achieving a guard-test at tolerance 2^{-k} may require time $\exp(k)$ or worse, hence no $\mathrm{poly}(n)$ decomposition can implement the required checks.

□

3.8 Failure of EM3 (Polynomial Guard/Basin Tests)

Lemma 3.8 (No Poly-time Guard Tests without (EM3))**.** *If (EM3) fails—i.e. membership in some guard cell or basin shell is not decidable in* $\mathrm{poly}(n, \log(1/\varepsilon))$*—then although a finite decomposition exists abstractly, no polynomial-time verifier can execute it.*

Proof. (1) ABET/FTE guarantee a finite list of open guard cells and proofs.

(2) If deciding any N_i is NP-hard (or worse), then the branching program cannot be run in polynomial time, so the decomposition is not *effective*.

□

4 Notion of Decomposition Independence

4.1 Conjecture (Prelude to FDI)

> *"Every measurement fact that resists* all *finite guarded decompositions (i.e. is* ***decomposition-independent****) has certification language of Turing degree at least* $0'$*."*

4.2 Preview of Significance

- Unifies *all* physical hypercomputing attacks under two structural failure modes.
- Explains why *any* theory satisfying our axioms cannot escape the Turing barrier without invoking one of these infinities.
- Provides a blueprint for analysing undecidability in quantum measurement, dynamical systems, and beyond.

5 Decomposition-Independence

In this section we formalize the notion of *decomposition-independence*, which captures the impossibility of certifying a measurement fact by any finite guarded decomposition. We then relate it to non-certifiability via the Fundamental Test Equivalence (FTE).

5.1 What it means to "decompose" a measurement

Let

$$\mathcal{M},\ \mathcal{Z}^{\text{in}},\ U,\ E$$

be as in Chapter II: a compact metric *state space* $\mathcal{M}$, an *input space* $\mathcal{Z}^{\text{in}}$, a continuous *update map* $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$, and an *output map* $E : \mathcal{M} \to \mathcal{B}$ into basin shells $\mathcal{B}_n$. A *measurement fact* is a predicate

$$\Phi(\Theta_0, z_0, \Theta_1, z_1, \dots) \ \in\ \{\text{True}, \text{False}\}$$

on infinite executions $(\Theta_t, z_t)_{t\geq 0}$. We focus on *tail* facts of the form

$$\exists T\ \forall t \geq T :\ \Psi\big(\Theta_t, z_t\big),$$

where Ψ is a simple condition (e.g. "output enters a particular basin shell").

5.2 Finite guarded decompositions

Definition 5.1 (Guard Cell)**.** A *guard cell* is any effectively testable subset

$$N\ \subseteq\ \mathcal{Z}^{\text{in}},$$

decidable up to tolerance in time $\text{poly}(n, \log(1/\varepsilon))$ by (EM3).

Definition 5.2 (Finite Guarded Decomposition)**.** A *finite guarded decomposition* of a tail fact Φ consists of:

1. A finite list of guard cells $N_1, \dots, N_k \subseteq \mathcal{Z}^{\text{in}}$.
2. A deterministic decision tree (or branching program) that, on any input $z_0 \in \mathcal{Z}^{\text{in}}$, tests membership in these N_i in some order and selects a branch $i \in \{1, \dots, k\}$.
3. For each branch i, a proof (or certificate) that *under the assumption* $z_0 \in N_i$, the tail fact Φ holds.

We say the decomposition *certifies* Φ if every input is routed to some i and the corresponding proof is valid.

5.3 Decomposition-Independence and FTE

Definition 5.3 (Decomposition-Independence)**.** A measurement fact Φ is *decomposition-independent* if

$$\text{no finite guarded decomposition certifies } \Phi.$$

Remark 5.1. By the **Fundamental Test Equivalence** (Chapter 3), any finite guarded decomposition yields a **NP** (or $\mathcal{V}$) certificate for Φ. Hence

$$\Phi \text{ is decomposition-independent} \implies \Phi \text{ is not certifiable in NP (or } \mathcal{V}).$$

Proposition 5.1. *If Φ admits a finite guarded decomposition, then its certification language lies in NP (degree 0).*

Proof. Immediate from FTE: the finite decision tree and per-branch proofs form an NP witness that decides membership in the certification language. □

Proposition 5.2. *If Φ is decomposition-independent, then any attempt to decide it by finite tests fails—no NP (or $\mathcal{V}$) algorithm exists.*

Proof. By contradiction, if an NP verifier decided Φ, then its accepting computation could be unrolled into a finite guarded decomposition (tests = guards, accepting paths = branches), contradicting decomposition-independence. □

6 Statement of the Fact Decomposition-Independence Theorem

We are now ready to present our main result, which links the impossibility of any finite guarded decomposition to the Turing degree of the corresponding certification language.

In what follows we do *not* assume a priori that every model satisfies all of (C)–(T), EM1–EM3 and (I). Instead, we only require those axioms needed to even *state* and *verify* a measurement fact:

$$\text{(F) (Invariant Basin)}, \quad \text{S (Certification Standard)}.$$

Any other axiom may or may not hold in a given model.

Theorem 6.1 (Fact Decomposition–Independence with Compile–Time + Run–Time Language)**.** *Fix a measurement model*

$$\mathfrak{M} \;=\; (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$$

together with a tail fact Φ ("eventual entry into the basin and stay") as in Book II and Book III. Assume throughout the two gnosis axioms:

(F) Invariant basin: *there exists a reachable, invariant attractor basin $\mathcal{B} \subseteq \mathcal{M}$ (Book II, Ch. 5).*

(S) Certification standard: *there is a nontrivial, robust certification standard $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ as in Book II, Ch. 6.*

Compile–time + run–time language. *Let $\mathcal{A}$ be any* sound, uniform *guarded–decomposition compiler that, given an instance $x \in \mathbb{N}$ (encoding initial condition, inputs, and shell tolerance), iteratively refines guard candidates and either:*

(a) halts in some finite time t and outputs a finite guarded program Π_x whose primitives and tests are those permitted by the axioms that $\mathfrak{M}$ satisfies, or

(b) never halts (keeps refining).

Define the compile–time + run–time *language*

$$L_{\Phi}^{\text{crt}} \;:=\; \Big\{ x \in \mathbb{N} \;\Big|\; \exists t \;\big[\; \mathcal{A}(x) \text{ halts within time } t \;\wedge\; \Pi_x \text{ verifies } \Phi \text{ under } \mathcal{S} \;\big]\Big\}.$$

Verification "Π_x verifies Φ under $\mathcal{S}$" means: (i) Π_x is a finite guarded program built from the admissible primitives, (ii) all guard/basin tests are carried out with the shell tolerances prescribed by $\mathcal{S}$, and (iii) the run–time replay of Π_x on instance x reaches $\mathcal{B}$ (or the required shell) and stays, with robustness mandated by $\mathcal{S}$.

Then exactly one of the following two mutually exclusive cases holds.

(a) ***Decomposable case (finite guards exist).*** *Suppose $\mathfrak{M}$ satisfies, in addition to* (F) *and* (S)*, the structural/process/effectivity axioms*

$$\text{(C) (\textit{Compact Continuity})},$$

$$\text{(T) (\textit{Traceability})},$$

$$\text{(I) (\textit{Time Irreversibility})},$$

$$\text{(EM1)–(EM3)(\textit{Effective Measurability})}.$$

Then the Fundamental Test Equivalence of Book III, Ch. 3 applies: Φ admits a finite guarded decomposition on every *instance, the compiler $\mathcal{A}$ halts on every input, and*

$$L_\Phi^{\mathrm{crt}} \;=\; L_\Phi \;\in\; \mathbf{NP}, \qquad \deg_T\big(L_\Phi^{\mathrm{crt}}\big) \;=\; 0.$$

Here L_Φ is the ordinary certification language ("run–time acceptance") and coincides with L_Φ^{crt} because compilation always terminates.

(b) ***Decomposition–independent case (no finite guards).*** *Suppose Φ admits* no *finite guarded decomposition under the axioms that $\mathfrak{M}$ satisfies. Then the compile–time + run–time language is* hypercomputational*:*

$$\deg_T\big(L_\Phi^{\mathrm{crt}}\big) \;\geq\; 0'.$$

In particular, there is a many–one reduction from HALT *to L_Φ^{crt}; equivalently, L_Φ^{crt} is Σ_1^0-hard. (When the failure includes certain continuum encodings—e.g. a pure (C) failure—stronger lower bounds may obtain; cf. the chapter's embedding lemmas.)*

In plain words, the above statement of the theorem is:

> *A measurement fact is decomposition-independent if and only if the problem of constructing and performing of the corresponding measurement is undecidable.*

Moreover (sharp responsibility of axioms). *For any model $\mathfrak{M}$ as above:*

- *If* any *of* (F), (S), *or* (C) *fails, nontrivial tail facts are not well–posed or not robustly verifiable; consequently, Φ is* untestable *and L_Φ^{crt} is undefined (vacuous).*
- *If $\mathfrak{M}$ satisfies* (F), (S), (C) *but fails* at least one *of* (I), (T), (EM1), (EM2), (EM3), *then Φ remains a well–posed tail fact, but is* decomposition–independent*. In that case the compiler halts on some instances and diverges on others in a way that is Σ_1^0-hard to predict, and*

$$\deg_T\big(L_\Phi^{\mathrm{crt}}\big) \;\geq\; 0'.$$

7 Embedding Axiom-Compliance and Computability Lemmas

7.1 Zeno–TM Embedding with Pure (I) Failure and (F) Satisfied

We refine the Zeno construction so that: (i) the *basin fact* used by ABET/FTE is independent of halting and satisfies $F \iff R \wedge L \wedge C_{\mathrm{basin}}$; (ii) all effective axioms PV≡(EM1)–(EM3), as well as compactness/continuity and traceability, hold; and (iii) *only* the Time–Irreversibility/Weak–Fairness axiom (I) fails, via a Zeno loop that executes infinitely many enabled decreases in finite time.

Spaces, maps, basin, and certification standard. Let

$$\mathcal{M} \;=\; [0,1]_x \times \{0,1\}_\lambda, \qquad \mathcal{Z}^{\text{in}} \;=\; [0,1],$$

with the product of the Euclidean metric on $[0,1]$ and the discrete metric on $\{0,1\}$, capped at 1. Interpret x as a Lyapunov "radius" and λ as a halting lamp (independent of the basin). Fix a contraction parameter $\rho \in (0,1)$ and a radius $r \in (0,1)$. Define the invariant basin and shells by

$$\mathcal{B} \;=\; \{(x,\lambda) : x \le r\}, \qquad \mathcal{B}_n \;=\; \{(x,\lambda) : x \le r + 2^{-n}\} \quad (n \ge 1).$$

Let the *emit* map be $E(x,\lambda) = x$ (the lamp is not part of the certified output). The Certification Standard $\mathcal{S} = \{\varepsilon_n\}_{n\ge1}$ is the Minimal Standard of Book II, Ch. 6 for these shells.

Update with a Zeno sub-loop. A single global *tick* consists of two primitives $U^{(1)}$ and $U^{(2)}$:

Tick 1 ($U^{(1)}$): $(x,\lambda),\, z \;\mapsto\; \big(\rho\, x,\ \lambda\big)$ (uniform contraction; independent of z).

Tick 2 ($U^{(2)}$): $(x',\lambda),\, z \;\mapsto\; \big(x',\, \Lambda(z)\big)$,

where $\Lambda : [0,1] \to \{0,1\}$ is produced by a *Zeno* subroutine that simulates, on a fixed universal TM U, the run of $U(z)$ with the accelerating schedule $I_k = (2-2^{-k},\, 2-2^{-(k+1)})$ (the k-th internal sub-step takes time $2^{-(k+1)}$). Within the physical interval of Tick 2, the subroutine:

- performs exactly one TM step of $U(z)$ during each internal slot I_k,
- sets a local latch $h \leftarrow 1$ if $U(z)$ ever halts; else h stays at 0,
- at the end of Tick 2 (physical time $t = 2$) writes $\Lambda(z) = h$.

Note carefully that Λ *does not* affect the x–dynamics or the basin.

We now verify compliance and hardness.

Lemma 7.1 (Computability and Axiom Compliance for the Zeno–TM Embedding)**.** *With the above construction, the measurement model*

$$\mathfrak{M} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, U = U^{(2)} \circ U^{(1)}, E, \mathcal{S}, \mathcal{B})$$

satisfies

(F) (R, L, C_{basin}), (S), (C C) (*compact spaces + continuity*), (T), (EM1)–(EM3),

and violates (I) *(Time–Irreversibility/Weak–Fairness).*

Proof. (F): Let $V(x,\lambda) = x$. Tick 1 maps $x \mapsto \rho x$ with $0 < \rho < 1$, hence:

- **R (Uniform reachability):** For any initial $x_0 \in [0,1]$, the sequence $x_t \le \rho^t x_0$ enters $x \le r$ within $T_{\max} = \min\{t : \rho^t \le r\}$, uniformly in z and λ.
- **L (Lyapunov border):** If $x = r$ at the beginning of Tick 1, after Tick 1 we have $x' = \rho r < r$; Tick 2 does not change x.

- C_{basin} **(local contraction):** For $(x, \lambda), (x', \lambda') \in \mathcal{B}$, Tick 1 yields $|x-x'| \mapsto \rho|x-x'|$, Tick 2 leaves x unchanged; thus the restriction $U|_{\mathcal{B}}$ is ρ–Lipschitz with $\rho < 1$.

(S): The Minimal Certification Standard for shells $\mathcal{B}_n = \{x \leq r + 2^{-n}\}$ yields open acceptance sets in the product metric; robustness follows from the strict contraction across the boundary (Book II, Ch. 6).

(C C): $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact; $U^{(1)}$ is affine; $U^{(2)}$ leaves x unchanged and flips a discrete bit λ to $\Lambda(z) \in \{0, 1\}$ at the end of the tick. As a map from $\mathcal{M} \times \mathcal{Z}^{\text{in}}$ with the product topology to $\mathcal{M}$, U is continuous (the discrete coordinate has the discrete topology; dependence on z is through the measurable selection at the end of the tick but no discontinuity in x).

(T): For the shells above, for any $\Theta \in \mathcal{M}$, the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$ is $[0, 1]$: Tick 1 sends $x \mapsto \rho x \leq \rho < r + 2^{-1}$ by choosing $r > \rho - 2^{-1}$ once and for all; Tick 2 leaves x unchanged. Hence $U(\Theta, \cdot)^{-1}(\mathcal{B}_1)$ is open and nonempty for every Θ; the axiom holds (PL–Traceability).

(EM1)–(EM3): Tick 1 is a uniform rational affine map, hence admits a rational arithmetic circuit of size poly(1) with polynomial evaluation cost and polynomial guard tests (distance to shells). Tick 2 writes $\lambda \leftarrow \Lambda(z)$ at the end of the tick; *evaluation* of U and E to k bits costs polynomial time in k because the x–coordinate is independent of λ and is exact after Tick 1, and the emitter E projects x. No non-rational constants are used; membership in shells is a dyadic threshold on x, decidable in poly$(\log(1/\varepsilon))$ time.

(I) violation: Inside Tick 2, the Zeno subroutine executes an *infinite* number of enabled loop iterations (one TM step per dyadic slot) before the physical deadline of the tick, so the well-founded ranking function mandated by (I) cannot certify that "if decrease is enabled infinitely often then it is taken infinitely often" along a *fair* execution without completing infinitely many loop iterations in finite time. This is precisely the Zeno/accelerating supertask forbidden by (I). Hence (I) fails while all other axioms validated above remain true. □

Lemma 7.2 (Σ_1–Hardness via the Zeno–TM Embedding)**.** *Let U be a universal Turing machine and define*

$$L_{\Phi}^{\text{crt}} = \Big\{ z \in [0, 1] \cap \mathbb{Q} \;\Big|\; \textit{the Zeno subroutine inside Tick 2 sets } \Lambda(z) = 1 \Big\}.$$

Then

$$z \in \text{HALT} \iff z \in L_{\Phi}^{\text{crt}},$$

so L_{Φ}^{crt} is Σ_1^0–complete (Turing degree $0'$). Moreover, the basin tail fact *remains trivially true (by (F)) and is unrelated to hardness.*

Proof. By construction, within Tick 2 the Zeno subroutine simulates the run of $U(z)$ and flips $\Lambda(z)$ to 1 iff $U(z)$ reaches a halting configuration at some finite step:

- If $U(z)$ halts in s steps, then on the s-th dyadic slot I_s the latch h is set to 1 and remains so; hence at the end of Tick 2 we write $\Lambda(z) = 1$.
- If $U(z)$ never halts, the latch remains 0 and $\Lambda(z) = 0$.

Therefore L_{Φ}^{crt} coincides with the Σ_1^0-complete set HALT. Since the certified tail fact in ABET/FTE concerns entry into $\mathcal{B} = \{x \leq r\}$ (which holds for *all* trajectories by Lemma 7.1), the hypercomputational hardness attaches only to the Zeno subroutine's output bit $\Lambda(z)$ and not to basin entry. This isolates the failure to axiom (I). □

7.2 HALT Embedding with Only (C) Violated

In this section we present, for each Turing index $e \in \mathbb{N}$, a measurement model $\mathfrak{M}_e$ in which *all* axioms except (C) (compactness of the state space) hold, while compactness itself *fails*—and this single failure is already enough to make ABET's global finite–subcover (the *uniformization step*) equivalent to deciding whether the universal Turing machine U halts on input e. Crucially, the tail fact "eventual entry into the basin and stay" remains well–posed and *true* on every trajectory (so R_{weak} holds); what becomes Σ^0_1–hard is the existence of a *uniform* time bound $T_{\max}$ that works for all initial conditions, i.e. ABET's compactness subcover.

Spaces and basin. Let the state space be

$$\mathcal{M} \;=\; [0,\infty)_x \times [0,1]_y \times \{0,1\}_h,$$

with product metric given by the Euclidean metrics on the intervals (capped at 1) and the discrete metric on the latch h. The input space is the singleton $\mathcal{Z}^{\text{in}} = \{*\}$ (no exogenous control). Fix parameters $0 < \rho < 1$ and $0 < r < 1$ satisfying

$$\rho\, r \;<\; r \qquad \text{(always true).} \tag{1}$$

Define the (closed) basin

$$\mathcal{B} \;=\; [0,\infty)_x \times [0,r]_y \times \{1\}_h.$$

Stations and spacing. Let $(\Delta_n)_{n\geq 0}$ be a computable, strictly increasing sequence of positive integers with $\Delta_n \to \infty$ (e.g. $\Delta_n = 2^n$). Define the increasing station positions

$$s_0 \;=\; 0, \qquad s_{n+1} \;=\; s_n + \Delta_n \quad (n \geq 0),$$

and write $\mathrm{idx}(x) = \max\{n \in \mathbb{N} : s_n \leq x\}$ for the index of the last station at or to the left of x.

Update and emit maps (discrete time). Fix a universal Turing machine U, and for each e embed a single–step simulator of $U(e)$ into the dynamics. Define the one–tick update $U_e : \mathcal{M} \times \{*\} \to \mathcal{M}$ by

$$U_e(x,y,h) \;=\; \begin{cases} \big(x+1,\ \rho\, y,\ h'\big), & \text{if } h = 1, \\ \big(x+1,\ \rho\, y + \beta \cdot \eta(x),\ h'\big), & \text{if } h = 0, \end{cases}$$

where:

- $\beta \in (0,\ \min\{(1-\rho)\, r,\ 1-r\})$ is a fixed rational "kick" amplitude;
- $\eta : [0,\infty) \to [0,1]$ is a continuous, compactly supported "station indicator" that *equals* 1 at each station and is zero off a tiny neighborhood, e.g. $\eta(x) = \sum_{n\geq 0} \chi_n(x)$ with pairwise disjoint, continuous hat functions χ_n supported in $[s_n - \frac{1}{2},\ s_n + \frac{1}{2}]$ and satisfying $\chi_n(s_n) = 1$. Thus one gets a *single* kick of size β when x passes near s_n, and no kick away from stations.[1]

[1] The disjoint supports and the bounded width ensure continuity of the update on $\mathcal{M}$.

- h' is the updated latch of the built–in simulator: starting from $h_0 = 0$, we set $h_{t+1} = 1$ iff the t-th simulated step reveals that $U(e)$ has halted, and otherwise $h_{t+1} = h_t$; once h turns 1, it remains 1 forever.

The emitter is $E(x, y, h) = \mathbf{1}_{\{y \le r,\, h=1\}}$. Intuitively: the x–coordinate drifts to the right at unit speed; when the trajectory crosses a station neighborhood it receives a small additive "kick" β in y, but every tick also contracts y by ρ; independently, the embedded simulator flips h to 1 at the first tick where it detects that $U(e)$ halts.

We analyze this family in two steps.

Lemma 7.3 (Compliance: only compactness fails)**.** *For each $e \in \mathbb{N}$, the model $\mathfrak{M}_e = (\mathcal{M}, \mathcal{Z}^{\text{in}}, U_e, E, \mathcal{S}, \mathcal{B})$ satisfies:*

$$\text{(S)}, \quad \text{(F)} \equiv (R_{\text{weak}}, L, C_{\text{basin}}), \quad \text{(T) } \textit{(PL–traceability)}, \quad \text{(I)}, \quad \text{(EM1)–(EM3)}.$$

Axiom (C) *fails* only *in the compactness clause: $\mathcal{M}$ is non–compact; all maps U_e, E are continuous.*

Proof. Continuity and non–compactness (C). The map $x \mapsto x + 1$ is continuous; the functions $y \mapsto \rho\, y$ and $y \mapsto \rho\, y + \beta\eta(x)$ are continuous in (x, y) because η is continuous with disjoint, bounded–width supports; the latch update is continuous in the product topology (discrete on $\{0, 1\}$). Thus U_e and E are continuous. The space $[0, \infty) \times [0, 1] \times \{0, 1\}$ is not compact because the x–coordinate is unbounded; this is the *only* failure of (C).

Certification standard (S). We use the standard product–metric certification with shells $\mathcal{B}_n = \{(x, y, h) : y \le r - 2^{-n},\ h = 1\}$. Acceptance sets are open and robust in the product metric; this is exactly the construction in Book II, Ch. 6.

Basin axioms (F) = R_{weak}+L+C_{basin}. On $\mathcal{B}$ we have $h = 1$ and the update is $y \mapsto \rho\, y$, hence for any $(x, y, h) \in \mathcal{B}$ and any tick, $y' \le \rho\, y \le \rho r < r$ by (1); this proves *local contraction* C_{basin} with constant $\rho \in (0, 1)$. The same computation with $y = r$ shows the *Lyapunov border* L: boundary points map strictly inside, $r \mapsto \rho r < r$. For R_{weak} (per–trajectory reachability), fix any initial (x_0, y_0, h_0). If h flips to 1 at some time t^*, thereafter the evolution of y is the pure contraction $y \mapsto \rho y$, so in at most

$$T_y \;\le\; \left\lceil \frac{\ln(y_{t^*}/r)}{\ln(1/\rho)} \right\rceil$$

further ticks we obtain $y \le r$; thus (R_{weak}) holds in this case. If h never flips, the x–coordinate still crosses every station neighborhood at most once, so the total additive contribution from kicks is bounded by $\beta \cdot \#\{\text{visited stations}\}$ *up to any fixed time*; in between kicks, the dynamics is a contraction. A standard comparison argument for the inhomogeneous linear recurrence $y_{t+1} = \rho y_t + \beta\eta(x_t)$ shows that after finitely many kicks (which occur at discrete times when x_t lies in some $[s_n - \frac{1}{2}, s_n + \frac{1}{2}]$), the subsequent pure contractions drive y below r within finitely many additional ticks. Formally, if $t_1 < t_2 < \cdots < t_m$ are the (finite) kick times encountered up to the first moment $y \le r$, then $y_{t_j+k} \le \rho^k(y_{t_j} + \beta/(1 - \rho))$ and the choice $\beta < (1 - \rho)r$ guarantees eventual descent below r. Hence R_{weak} holds from any start.

PL–traceability (T). With singleton input, for each Θ and shell $\mathcal{B}_n$ the preimage $U_e(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is either $\emptyset$ or the whole $\mathcal{Z}^{\text{in}}$. Whenever $\Theta \in \mathcal{B}$, the image stays in $\mathcal{B} \subset \mathcal{B}_n$, hence the preimage equals $\mathcal{Z}^{\text{in}}$, which is open and nonempty. This is the PL–variant used throughout Book III.

Time–irreversibility / no supertask (I). Each tick performs a bounded, finite amount of work (one lattice drift in x, one affine update in y, one simulator step for $U(e)$); there is no Zeno scheduling. A ranking function is given by $f(x, y, h) = \mathbf{1}_{\{h=0\}} \cdot \lceil y/r \rceil + \mathbf{1}_{\{h=1\}} \cdot \lceil y/r \rceil$, which decreases whenever $y > r$ and a contraction is applied; fairness is trivial because the schedule is deterministic.

Effectivity (EM1–EM3). All primitives are rational–arithmetic and uniformly described: affine maps $(x, y) \mapsto (x + 1, \rho y)$ and $(x, y) \mapsto (x + 1, \rho y + \beta\eta(x))$ with fixed rational ρ, β and rational hat functions χ_n (their centers s_n are computable from Δ_n). Evaluating to k bits is polynomial in k and in the bit–length of (x, y) (EM2); membership in shells $\{y \le r - 2^{-n},\ h = 1\}$ is decidable in time $\mathrm{poly}(n, \log(1/\varepsilon))$ (EM3). This establishes EM1–EM3. □

Lemma 7.4 (Σ^0_1–hardness of ABET uniformization; compactness–only)**.** *For the family* $\{\mathfrak{M}_e\}_{e\in\mathbb{N}}$ *above, define the* ABET–uniformization language

$$L^{\mathrm{unif}} \;=\; \Big\{ e \in \mathbb{N} \;\Big|\; \exists T_{\max}\ \forall \Theta_0 \in \mathcal{M} :\ \exists t \le T_{\max} \text{ with } U_e^t(\Theta_0) \in \mathcal{B} \Big\}.$$

Then

$$e \in L^{\mathrm{unif}} \quad \Longleftrightarrow \quad U(e) \text{ halts},$$

so L^{unif} *is* Σ^0_1*–complete (Turing degree* $0'$*).*

Proof. We analyze the two directions separately.

(⇒) If $U(e)$ *halts, then* $e \in L^{\mathrm{unif}}$. Let t^* be the first tick where the embedded simulator detects halting and sets $h = 1$. Thereafter the update on y is the pure contraction $y \mapsto \rho y$. Since $y \in [0, 1]$, the number of additional ticks needed to reach $y \le r$ is bounded by $T_{\mathrm{contr}} = \lceil \ln(1/r)/\ln(1/\rho) \rceil$, *independently* of Θ_0. It remains to bound, uniformly over Θ_0, the time until h flips. By construction, exactly one simulator step is executed per tick; thus h flips by time $t^* \le S(e) + 1$, where $S(e)$ is the halting time of $U(e)$ on input e. Therefore

$$T_{\max} \;=\; S(e) + 1 + T_{\mathrm{contr}}$$

works uniformly for all initial states Θ_0, showing $e \in L^{\mathrm{unif}}$.

(⇐) If $U(e)$ *does not halt, then* $e \notin L^{\mathrm{unif}}$. Assume $U(e)$ never halts, so $h_t \equiv 0$. Fix any candidate bound $T \in \mathbb{N}$. We will construct an initial state Θ_0 whose entry time exceeds T, showing that no global $T_{\max}$ exists.

Choose N so large that the distance from station s_N to station s_{N+K} exceeds T for every fixed K (possible because the spacings Δ_n are increasing and unbounded). Set $\Theta_0 = (x_0, y_0, h_0) = (s_N - \frac{1}{4},\ 1,\ 0)$. From time 0 to the moment the trajectory first reaches the neighborhood of s_N there are at least $\frac{1}{4}$ ticks; upon crossing s_N the y–update applies one kick of size β, but the inequality $\beta < (1 - \rho)r$ ensures that repeated contractions would suffice to eventually bring y below r. However, before enough contractions accrue, the trajectory must traverse the (very large) spacing $\Delta_N = s_{N+1} - s_N$ to reach the next contraction opportunity; by the choice of N and monotonicity of the spacings, the number of ticks needed to pick up any fixed number K of contractions can be made arbitrarily large by increasing N. In particular, for the above Θ_0, the time needed to accumulate the (finite) number of contractions required to cross below r exceeds the prescribed T. Since T was arbitrary, no uniform $T_{\max}$ exists, i.e. $e \notin L^{\mathrm{unif}}$.

Completeness. The mapping $e \mapsto \mathfrak{M}_e$ is uniform and computable (the only e–dependence is the built–in single–step simulator for $U(e)$). By the two implications we have

$$e \in L^{\text{unif}} \iff U(e) \text{ halts},$$

so L^{unif} is many–one equivalent to HALT and hence Σ^0_1–complete. Note that in both cases ($U(e)$ halts or not) the *tail fact* "eventually enter $\mathcal{B}$ and stay" is true from every start (Lemma 7.3); what is undecidable is the existence of a *global* uniform time bound, i.e. ABET's compactness–driven finite subcover—precisely the portion of the argument that relies on (C). □

Conclusion. The family $\{\mathfrak{M}_e\}$ isolates a pure failure of (C) (compactness) already at the ABET level: the tail fact remains legitimate and holds on all trajectories, yet the existence of a *finite* ABET certificate (a global $T_{\max}$, hence a finite subcover) is Σ^0_1–complete. All other axioms—including the weakened basin axiom R_{weak}, Lyapunov border, local contraction on $\mathcal{B}$, PL–traceability, time–irreversibility, and the effective measurability axioms—are satisfied.

7.3 HALT embedding via iterative decomposition under a pure (T) failure

We exhibit a fixed, computable measurement model in which all axioms (F), (S), (C), (I), (EM1)–(EM3) hold, but **(T) fails**, and where the *compile–time termination* of a standard, sound iterative guarded–decomposition procedure is Σ^0_1-complete via a many–one reduction from HALT. The hardness attaches to the *inductive construction of the decomposition* (compile–time), not to the run–time test (which remains finite and benign).

Model. Let the input space be $\mathcal{Z}^{\text{in}} = [0,1]$ with its usual metric, and let the state space be

$$\mathcal{M} \;=\; [0,1] \times \{0,1\},$$

with coordinates (x, ℓ), where ℓ is a latch bit. Fix the *middle–third Cantor set* $K \subset [0,1]$.

Basin definition. To satisfy the basin axioms ($R_{\text{weak}}, L, C_{\text{basin}}$) while keeping the traceability violation isolated, we fix parameters $0 < r < 1$ and $\rho \in (0,1)$ with $\rho\, r < r$ and define the basin

$$\mathcal{B} = \{(x,\ell) \in [0,1] \times \{0,1\} : x \le r,\ \ell = 1\}.$$

Here x is a contracting coordinate updated by $x \mapsto \rho\, x$ on $\mathcal{B}$ (ensuring local contraction and Lyapunov border), and ℓ is a latch bit that is set to 1 at a fixed primitive stage (independent of the hard guard geometry used for the traceability violation). This guarantees that every trajectory enters $\mathcal{B}$ in finite time from any initial state, so R_{weak} holds.

Define a two–tick update U (implemented by two primitive steps $U^{(1)}, U^{(2)}$), and a trivial emitter E that just reports (x, ℓ):

- **Tick 1:** $U^{(1)}\big((x,\ell), z\big) \;=\; \big(d(z,K),\ \ell\big)$, where $d(z,K) = \inf_{y \in K} |z - y|$ is the distance to K.
- **Tick 2:** $U^{(2)}\big((x,\ell), z\big) \;=\; (0,1)$ for every input z and state (x,ℓ).

Thus, after Tick 2, the system is always in the basin $\mathcal{B}$ and remains there; the *tail fact* of interest is the usual basin-entry predicate ("eventually in $\mathcal{B}$ and stay").

We adopt the Minimal Certification Standard $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ from Chapter 3, Section 2, with shells $\mathcal{B}_n = \{(x, \ell) : x \leq 2^{-n},\ \ell = 1\}$.

Traceability variant. Throughout this construction we adopt the *primitive–level* form of the traceability axiom (T) (PL–Traceability): for each primitive update $U^{(j)}$ in the fixed decomposition, each $\Theta \in \mathcal{M}$, and each certification shell $\mathcal{B}_n$, the preimage $\big(U^{(j)}(\Theta, \cdot)\big)^{-1}(\mathcal{B}_n)$ must be open and nonempty in the input space. This strengthens the universal–step variant only in the sense that it enforces the openness requirement *at each individual stage* of the decomposition, not merely for the composed U; the motivation here is that the HALT embedding exploits the geometry of a single primitive's guard set, so the violation manifests already before composition.

Lemma 7.5 (Computability and axiom compliance; pure (T) failure)**.** *For the above model:*

(i) (F) *holds: $\mathcal{B}$ is closed, invariant, and uniformly reachable in two ticks.*

(ii) (S) *holds with the Minimal Certification Standard $\mathcal{S}$.*

(iii) (C) *holds: $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact metric spaces and all primitive updates are continuous.*

(iv) (I) *holds: executions have two deterministic ticks (no supertask).*

(v) (EM1)–(EM3) *hold in the sense of Chapter 3, §2.4:*

- *the primitives $U^{(1)}$, $U^{(2)}$, E admit uniform, polynomial–time evaluation to k bits of precision (distance–to–K is computable in time poly(k) via triadic-digit truncation);*
- *a polynomial modulus of continuity exists (in fact, Lipschitz on compact sets);*
- *guard/basin membership up to tolerance ε is decidable in time poly$(\log(1/\varepsilon))$.*

(vi) (T) *fails: for the initial state $(x, \ell) = (1, 0)$ and the shell $\mathcal{B}_1 = \{(x, \ell) : x \leq 1/2,\ \ell = 1\}$, the preimage*

$$U^{(1)}\big((1,0), \cdot\big)^{-1}(\{(0,0)\}) \;=\; \{z \in [0,1] : d(z, K) = 0\} \;=\; K$$

is closed, nowhere dense, and not *open.*

Proof. (i) (F). $\mathcal{B} = \{(0, 1)\}$ is closed and invariant because $U^{(2)}$ maps every state to $(0, 1)$, and a subsequent application keeps it there. Uniform reachability holds with $T_{\max} = 2$.

(ii) (S). The Minimal Certification Standard $\mathcal{S}$ uses shells $\mathcal{B}_n = \{(x, \ell) : x \leq 2^{-n},\ \ell = 1\}$. Because $U^{(2)}$ maps any state to $(0, 1)$, verifying basin entry at Tick 2 is robust: any $\varepsilon_n > 0$ that separates $x \leq 2^{-n}$ from $x > 2^{-n}$ yields open acceptance sets in the product metric (Book II, Ch. 6). Hence acceptance is robust to small output perturbations.

(iii) (C). $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact intervals (finite products of compact metric spaces). The map $U^{(2)}$ is constant, hence continuous. The distance function $z \mapsto d(z, K)$ is 1–Lipschitz on $[0, 1]$, hence continuous; therefore $U^{(1)}$ is continuous. The emitter E is continuous (projection).

(iv) (I). Exactly two ticks occur; no supertask scheduling is present.

(v) (EM1)–(EM3). EM2/EM3: The function $d(\cdot, K)$ admits a uniform polynomial–time evaluator to k bits by reading the first k ternary digits of z and using the standard "no digit 1" certificate; this yields upper/lower bounds that converge geometrically (the ambiguity at triadic dyadic boundaries is handled by the product metric and tolerance). Hence there is a polynomial modulus for evaluating $U^{(1)}$ and for deciding the shell predicates up to tolerance $\varepsilon = 2^{-k}$. EM1: In the baseline $(\mathbf{C}, \mathcal{V}) = (\mathbf{P}, \mathbf{NP})$ setting, EM1 requires an effective (uniform, finite) representation with poly–time evaluation to k bits.[2] Both primitives $U^{(1)}$, $U^{(2)}$ and the shell/guard tests satisfy this.

(vi) (T). At Tick 1, starting from any state, the x–coordinate is $d(z, K)$. Since $d(z, K) = 0$ iff $z \in K$, the preimage of the singleton $\{x = 0\}$ is exactly K, which is closed and nowhere dense, hence not open. Therefore Axiom (T) fails. □

Iterative guarded–decomposition procedure $\mathcal{A}$. We use a standard, sound refinement scheme (cf. ATRT, Alg. 1 in the Appendix) specialized to our model:

1. Maintain a partition $\mathcal{P}_m$ of $[0, 1]$ into finitely many triadic intervals at depth m.
2. On each cell $I \in \mathcal{P}_m$, decide (via the EM2/EM3 evaluator for $d(\cdot, K)$) whether Tick–1 maps *all* points in I to $x = 0$ (i.e., $I \subseteq K$), *no* point in I maps to $x = 0$ (i.e., $I \cap K = \varnothing$), or the case is *ambiguous* (the cell intersects the boundary).
3. Remove decided cells; refine ambiguous cells by splitting them into their three triadic subintervals; set $m \leftarrow m + 1$.
4. **Termination condition:** stop when no ambiguous cells remain; output the finite guarded decomposition that routes "Tick–1 into $\mathcal{B}$" vs. "Tick–2 into $\mathcal{B}$".

Correctness is standard: on any cell that is declared "inside" or "outside", continuity/Lipschitz of $d(\cdot, K)$ and the shell tolerance ensure the decision is sound; refinement preserves soundness. The procedure terminates iff the ambiguous set becomes empty in finitely many triadic refinements.

Lemma 7.6 (Σ_1–hardness of compile–time stabilization under pure (T) failure)**.** *Let U be the model of Lemma 7.5 and $\mathcal{A}$ the above refinement procedure. Consider the meta–language*

$$L_{\text{stab}} = \big\{ x \in \mathbb{N} \ : \ \textit{on input } x, \ \mathcal{A} \textit{ halts with a finite guarded decomposition} \big\},$$

where $\mathcal{A}$ is augmented to interleave *(in dovetailing fashion) a simulation of the universal Turing machine $U_{\text{TM}}(x)$. Then L_{stab} is Σ_1^0–complete (many–one equivalent to* HALT*). Moreover, during run–time, the produced guarded program (when $\mathcal{A}$ halts) is finite, satisfies (EM1)–(EM3), and correctly certifies the tail fact.*

Proof. Define the augmented compiler $\mathcal{A}$ as follows: run one iteration of the refinement loop, then simulate one step of $U_{\text{TM}}(x)$, and repeat. If ever $U_{\text{TM}}(x)$ halts, $\mathcal{A}$ immediately *short–circuits*: it outputs the following *finite* guarded program

$$\Pi^\star : \quad \text{(Ignore Tick 1); perform Tick 2; accept if in } \mathcal{B}.$$

[2] If one insists on "rational arithmetic circuit" in the strictest sense, replace EM1 by the parameterized effective representation $\text{EM1}_{\mathbf{C}}$ of Proposition 10.1 with $\mathbf{C} = \mathbf{P}$, which is what the proofs in Chapter 3 actually use. The present construction is fully compliant with $\text{EM1}_{\mathbf{P}}$, $\text{EM2}_{\mathbf{P}}$, $\text{EM3}_{\mathbf{P}}$.

This program is valid for the model because Tick 2 maps every state into $\mathcal{B}$, independently of $z \in [0,1]$. Its guards are trivial (single open cell $[0,1]$), and the primitives are those of Lemma 7.5, hence (EM1)–(EM3) hold.

We now show two implications.

(A) If $U_{\mathrm{TM}}(x)$ halts, then $\mathcal{A}$ halts. By construction, on the first step where the TM halts, $\mathcal{A}$ outputs $\Pi^\star$. This is finite and correct; thus $x \in L_{\mathrm{stab}}$.

(B) If $U_{\mathrm{TM}}(x)$ does not halt, then $\mathcal{A}$ does not halt. In this case the short–circuit never triggers. The refinement loop runs forever. Because the preimage of Tick–1 entry is exactly K, which has empty interior, every triadic refinement leaves some ambiguous cells (indeed, the ambiguous set is precisely the Cantor–like boundary that never disappears in finite depth). Consequently, the "no ambiguous cells remain" condition is never met, so $\mathcal{A}$ does not halt.

Thus we have established a many–one reduction $x \mapsto x$ from HALT to L_{stab}. Therefore L_{stab} is Σ^0_1–hard. It lies in Σ^0_1 by definition ("$\exists t$: $\mathcal{A}$ halts within t steps"), hence Σ^0_1–complete.

Finally, whenever $\mathcal{A}$ halts, the output $\Pi^\star$ is a valid finite decomposition: it uses the same primitives as the model, requires only Tick 2, and its guards are trivially decidable in polynomial time to any tolerance. During run–time, the NP–style certificate/verifier of Chapter 3 applies verbatim (Polytime Verifier Construction), so the tail fact is certified in the usual way. □

Discussion. The only source of undecidability in this construction is the guard set $G \subseteq \mathcal{Z}^{\mathrm{in}}$ used by the designated primitive $U^{(2)}$ in the decomposition $U = U^{(2)} \circ U^{(1)}$.

We take G to be an r.e.–open set whose membership problem encodes the halting of $U(e)$ on input e.

Because PL–Traceability requires the preimage $U^{(2)}(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ to be open and nonempty for every Θ and n, but the boundary ∂G is non–open and its interior membership is Σ^0_1–hard, the axiom fails exactly at this primitive, while all other primitives have trivially open preimages.

This construction cleanly separates concerns:

- The *model* violates only (T); all of (F),(S),(C),(I),(EM1)–(EM3) hold.
- The *compile–time stabilization* of a sound, standard refinement procedure is Σ^0_1–complete.
- Whenever the compiler halts, the emitted guarded program is finite, effective, and its run–time verification remains within the baseline $(\mathbf{P}, \mathbf{NP})$ regime of Chapter 3.

In particular, hardness is a property of *inductive guard synthesis* in the presence of a pure (T) failure (non–open preimages), not of the run–time test itself.

7.4 Real–Constant Embedding (Violation of EM1)

In this construction we exhibit a HALT embedding whose only failing axiom at the ABET level is (EM1), the *effective modulus of continuity via rational constants* requirement. We recall that (EM1) demands that all numerical constants used by the update map U be *effectively encodable*, for example as rationals or via a uniformly computable sequence of rationals converging at a known rate. We will construct a system whose basin axioms $(R_{\mathrm{weak}}, L, C_{\mathrm{basin}})$ hold, and for which the only violation is the appearance of a *real constant* whose binary expansion encodes the halting set.

Setup of the embedding

Let $\mathcal{M} = [0,1] \times \{0,1\}$, where $x \in [0,1]$ is a contracting coordinate and $\ell \in \{0,1\}$ is a latch bit. Fix $0 < r < 1$ and $\rho \in (0,1)$ with $\rho\, r < r$. Define the basin

$$\mathcal{B} \;=\; \{(x,\ell) \in [0,1] \times \{0,1\} \mid x \le r,\ \ell = 1\}.$$

The update map U is given by the composition $U = U^{(2)} \circ U^{(1)}$, where:

1. $U^{(1)}$: Given (x,ℓ) and an input $e \in \mathbb{N}$ (encoded in the measurement z^{in}), set $\ell := 1$ if the e–th bit of a fixed real constant α equals 1, otherwise leave ℓ unchanged. The x coordinate is not touched.

2. $U^{(2)}$: Contract the x coordinate by $x \mapsto \rho\, x$ and leave ℓ unchanged.

The constant $\alpha \in [0,1]$ is chosen so that its binary expansion

$$\alpha = 0.\, b_1 b_2 b_3 \ldots$$

satisfies $b_e = 1$ if and only if the e–th Turing machine halts on empty input. This α is a well–defined real number, but it is not computable.

Because x is contracted each time by $U^{(2)}$, every trajectory enters the set $\{x \le r\}$ in finitely many steps, regardless of ℓ; at that point, if $\ell = 1$ the state is in $\mathcal{B}$. Since ℓ may be set to 1 at the very first step depending on b_e, we obtain $(R_{\text{weak}}, L, C_{\text{basin}})$ without requiring any uniform bound that is independent of e.

Violation mechanism

The (EM1) axiom fails because α has no effective encoding. Evaluation of $U^{(1)}$ for a given e requires determining b_e, which is equivalent to deciding whether the e–th Turing machine halts — a Σ^0_1–complete problem.

Lemma 7.7 (Computability compliance of other axioms)**.** *The system $(\mathcal{M}, U, \mathcal{B})$ described above satisfies axioms $(R_{\text{weak}}, L, C_{\text{basin}})$, (C), and (T), as well as (EM2) and (EM3), assuming these latter two are interpreted in the standard effective sense over rational constants.*

Proof.

1. R_{weak}**:** For any initial state (x_0, ℓ_0), repeated application of $U^{(2)}$ reduces x by the factor $\rho < 1$ at each step. Thus there exists t (depending on x_0) such that $x_t \le r$, ensuring reachability of the x–component of $\mathcal{B}$ in finitely many steps. The latch bit ℓ is set to 1 whenever the input index e satisfies $b_e = 1$, but weak reachability does not require this to hold uniformly over e.

2. L**:** The Lyapunov function $V(x,\ell) = x$ satisfies $V \le r$ on $\mathcal{B}$ and $V = r$ on the boundary. Applying $U^{(2)}$ to a boundary point sends x to $\rho\, r < r$, so L holds.

3. C_{basin}**:** On $\mathcal{B}$, the update $U^{(2)}$ contracts the x–coordinate by $\rho < 1$ and leaves ℓ unchanged, so (C_{basin}) is satisfied.

4. **(C):** The local contraction on $\mathcal{B}$ follows from the contraction of the x–coordinate and the discrete nature of ℓ.

5. **(T):** Preimages of certification shells $\mathcal{B}_n = \{(x, \ell) : x \leq r_n, \ell = 1\}$ under either $U^{(1)}$ or $U^{(2)}$ are open in the input space: for $U^{(1)}$ the openness is trivial in the discrete e–coordinate; for $U^{(2)}$ it is inherited from continuity in x.

6. **(EM2), (EM3):** Both EM2 (effective modulus of continuity) and EM3 (decidable open guards) are satisfied relative to the operations and sets actually used in $U^{(2)}$ and in the guard for $\ell = 1$, which are all finite–state or rational.

Therefore all axioms except EM1 hold. □

Lemma 7.8 (HALT–encoding hardness)**.** *The evaluation of $U^{(1)}$ in the above system computes a Σ_1^0–complete predicate, and hence no effective encoding of the constant α exists. This is precisely the violation of* (EM1)*.*

Proof. By construction, the constant α has binary expansion $(b_e)_{e\geq 1}$ with

$$b_e = \begin{cases} 1, & \text{if the } e\text{–th Turing machine halts on empty input,} \\ 0, & \text{otherwise.} \end{cases}$$

Given an input index e as part of z^{in}, the primitive $U^{(1)}$ queries the e–th bit b_e of α and sets $\ell := 1$ if and only if $b_e = 1$. Thus, the mapping $e \mapsto b_e$ solves the halting problem on empty input. This problem is well–known to be Σ_1^0–complete, i.e., semi–decidable but not decidable.

If α were effectively encodable in the sense of EM1, there would exist a Turing machine that, given e and k, produces the k–bit truncation of α in time bounded by a uniform modulus of convergence. In particular, taking $k = e$, we could decide whether $b_e = 0$ or 1, thereby deciding the halting problem — a contradiction.

Therefore, α is not effectively encodable, and EM1 fails exactly because of this constant. All other parts of the construction remain effective. □

Conclusion. This construction isolates the violation of EM1 to a single real constant α whose binary expansion encodes HALT. The basin axioms $(R_{\text{weak}}, L, C_{\text{basin}})$ and all other effectivity and contraction axioms hold, so the HALT–hardness is cleanly localized to EM1.

7.5 HALT Embedding via an EM2-Violation

We construct a family of measurement models

$$\left\{\mathfrak{M}_e = (\mathcal{M}, \mathcal{Z}^{\text{in}}, U_e, E, \mathcal{S}, \mathcal{B})\right\}_{e\in\mathbb{N}}$$

that *isolate* a failure of (EM2) (the polynomial cost–modulus of evaluation), while all other assumptions required at the ABET level hold:

$$(R_{\text{weak}}),\ (L),\ (C_{\text{basin}}),\ (C),\ (T),\ (I),\ (EM1),\ (EM3).$$

Intuitively, the update has a benign, uniform contraction in a state coordinate (so the basin axioms hold), but includes a single primitive whose *evaluation to k output bits* in the worst case forces *super–polynomial* work in k (hence (EM2) fails). We then show that the associated *compile–time + run–time* acceptance language becomes Σ_1^0–complete (many–one equivalent to HALT).

Spaces, basin, and certification standard. Let

$$\mathcal{M} = [0,1]_x \times \{0,1\}_\ell, \qquad \mathcal{Z}^{\text{in}} = [0,1]_z,$$

with the product of the Euclidean metric on $[0,1]$ (capped at 1) and the discrete metric on $\{0,1\}$. Fix $\rho \in (0,1)$ and $r \in (0,1)$ with $\rho r < r$, and define the basin and the minimal shells by

$$\mathcal{B} = \{(x,\ell) : x \le r\}, \qquad \mathcal{B}_n = \{(x,\ell) : x \le r_n\}, \quad r_n := r/2^n.$$

Thus ℓ plays *no role* in $\mathcal{B}$ (this ensures the basin axioms will not hinge on any halting–dependent event). Take $\mathcal{S}$ to be the minimal certification standard built from the shells $\{\mathcal{B}_n\}_{n\ge1}$ (cf. Book II, Ch. 6).

Update and emitter. Fix a smooth bump $b : \mathbb{R} \to [0,1]$ with $\operatorname{supp} b \subset (-1,1)$ and $b(0) = 1$. For each $e \in \mathbb{N}$ define the *halting–gated* widths

$$w_j^{(e)} := \begin{cases} 2^{-2^j}, & \text{if the universal TM } U(e) \text{ has not halted by step } 2^j, \\ 0, & \text{otherwise,} \end{cases} \qquad j = 1, 2, \ldots$$

and rational centers $c_j := \frac{1}{2} - 2^{-j}$. Define the continuous map

$$F_e(z) := \sum_{j=1}^{\infty} 2^{-(j+2)}\, b\Big(\frac{z - c_j}{w_j^{(e)} + 2^{-(2^j+j)}}\Big).$$

(When $w_j^{(e)} = 0$, the jth term is the zero function; the tiny rational floor $2^{-(2^j+j)}$ ensures each summand is well–defined by a *uniform* rational circuit; see EM1 discussion below.) Set the two–tick update $U_e := U_e^{(2)} \circ U_e^{(1)}$ by

$$U_e^{(1)} : (x,\ell), z \mapsto (x + \tfrac{1}{2}F_e(z),\ \ell), \qquad U_e^{(2)} : (x',\ell) \mapsto (\rho x',\ \ell),$$

and let the emitter be $E(x,\ell) = x$ (we certify the x–shells). Finally, in the background we run a *separate* one–step–per–tick simulation of $U(e)$ (this influences only the parameters $w_j^{(e)}$ used by $U_e^{(1)}$ as explained above; there is no Zeno schedule).

The role of F_e is to create a very narrow family of "bumps" whose presence/absence at scales 2^{-2^j} depends on whether $U(e)$ has already halted by time 2^j. To decide the kth output bit of $U_e^{(1)}$ on adversarial inputs z one must, in general, determine the membership of z in windows of width 2^{-2^j} for some $j = \Theta(k)$, which forces inspection of $\Theta(2^k)$ input precision or the equivalent computational effort. This yields a *super–polynomial* worst–case evaluation cost in k, i.e. a clean failure of (EM2).

Axiom compliance (all but EM2)

Lemma 7.9 (Computability and axiom compliance; only (EM2) fails)**.** *For each $e \in \mathbb{N}$, the model $\mathfrak{M}_e$ satisfies*

$$(R_{weak}),\ (L),\ (C_{basin}),\ (C),\ (T),\ (I),\ (EM1),\ (EM3),$$

and violates (EM2) *(no polynomial evaluation cost–modulus for $U_e^{(1)}$).*

Proof. (C) compact + continuity. $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact. Each summand in F_e is a smooth bump of amplitude $2^{-(j+2)}$ and support radius $\leq 2^{-(2^j+j)}$ *given by a uniform rational circuit* (the dependence on halting only gates whether the term is identically zero: if $U(e)$ has halted by time 2^j we set the width to 0 and the summand to 0). Hence F_e is a uniformly and absolutely convergent series of continuous functions; therefore $U_e^{(1)}$ and $U_e^{(2)}$ are continuous, so is $U_e = U_e^{(2)} \circ U_e^{(1)}$.

$(R_{weak}), (L), (C_{basin})$. Take the Lyapunov function $V(x, \ell) = x$. Since $x \mapsto \rho x$ in $U_e^{(2)}$ with $\rho \in (0, 1)$, the x–coordinate contracts uniformly. For any initial (x_0, ℓ_0) there exists t with $x_t \leq r$ (reachability into the x–part of $\mathcal{B}$). On the boundary $x = r$, one tick maps x to $\rho r < r$ (Lyapunov border), and on $\mathcal{B}$ the restriction is a strict contraction (local contraction). The latch ℓ is inert for $\mathcal{B}$ by design, so no halting event is needed for basin entry.

(T) PL–traceability. With shells $\mathcal{B}_n = \{x \leq r_n\}$, for any Θ we have $U_e^{(2)}(\Theta, \cdot)^{-1}(\mathcal{B}_n) = \mathcal{Z}^{\text{in}}$ (open, nonempty). Thus the per–primitive (PL) version of (T), used throughout Book III, holds.

(I) no supertask. Each global tick performs: (i) evaluation of F_e at the *given* z (no Zeno accumulation in physical time), and (ii) a single affine contraction by ρ. The background halting simulation of $U(e)$ advances by *one step per tick* only to *update* which summands in F_e are known to be zero for future ticks; it does not affect the current tick's physical update. Hence time–irreversibility (weak fairness) is respected.

(EM1) effective representation. The primitives are specified by a fixed finite recipe (a circuit schema) that, on input j, constructs the jth summand using rational arithmetic (centers c_j dyadic; amplitudes $2^{-(j+2)}$ dyadic; floor width $2^{-(2^j+j)}$ dyadic). Whether the jth bump *also* carries the additional (non–rational) width $w_j^{(e)} = 2^{-2^j}$ is *replaced* by the rule "set the jth summand to zero once the background simulation has reached time 2^j and has observed a halting state". This gating is fully effective (it reads a single bit from the simulator state) and does not introduce non–rational parameters. Thus EM1 (effective specification by a uniform finite description) holds.

(EM3) guard/basin tests. Membership in $\mathcal{B}_n$ is a dyadic threshold on x (independent of z), decidable in time polynomial in the state bit–length and $\log(1/\varepsilon)$.

(EM2) failure (no polynomial cost–modulus). Fix $k \geq 1$. Consider adversarial inputs z that, for some $j = \Theta(k)$, lie at distance $\frac{1}{2}2^{-2^j}$ to the center c_j. Then changing z by at most 2^{-2^j} toggles the jth summand between $\approx 2^{-(j+2)}$ and 0, which alters $U_e^{(1)}$ by $\gtrsim 2^{-(j+2)} \asymp 2^{-\Theta(k)}$. Therefore, any algorithm that *guarantees* k correct output bits of $U_e^{(1)}$ must, in the worst case, distinguish inputs within $\delta \leq 2^{-2^j} = 2^{-2^{\Theta(k)}}$ around the centers c_j. Equivalently, it must either (a) inspect $\Theta(2^k)$ digits of z in base 2, or (b) perform $\Omega(2^k)$ steps of the background halting simulation to certify whether the jth summand is presently active. In either interpretation, the worst–case evaluation time grows faster than any fixed polynomial in k (for fixed input bit–length), violating EM2's polynomial cost–modulus requirement. All other axioms remain intact. □

Hardness at the compile–time + run–time level

We now show that the natural *compile–time + run–time* language associated with certifying the tail fact "eventually in the basin and stay" becomes Σ_1^0–complete in this family, precisely because the only available way for a (sound) compiler to finish is to *discover* the halting time of $U(e)$ and thus remove all future ultra–narrow summands beyond a finite index.

Lemma 7.10 (Σ_1^0–hardness via (EM2) failure)**.** *Let $\mathcal{A}$ be any sound, uniform guarded–decomposition compiler that, on input the index e, iteratively refines a finite cover of $\mathcal{Z}^{\text{in}}$ into guard cells on which it can certify entry into the shell sequence $\{\mathcal{B}_n\}$ using the primitives of $\mathfrak{M}_e$. Augment $\mathcal{A}$ so that, in parallel, it simulates $U(e)$ for one step per iteration; if it ever observes that $U(e)$ halts, it* short–circuits *and outputs the trivial finite decomposition*

$$\Pi^\star : \quad \textit{(ignore } U_e^{(1)}\textit{); apply } U_e^{(2)} \textit{ for } T \textit{ ticks and accept when } x \le r,$$

with $T = \lceil \log_{1/\rho}(x_0/r) \rceil$ chosen per instance from the state encoding. Then the compile–time + run–time *language*

$$L^{\text{crt}} := \{\, e \in \mathbb{N} : \ \mathcal{A}(e) \textit{ halts and } \Pi^\star \textit{ (or the produced program) certifies entry} \}$$

is many–one equivalent to HALT*. In particular, L^{crt} is Σ_1^0–complete.*

Proof. Soundness of $\Pi^\star$. Because $\mathcal{B} = \{x \le r\}$ and $U_e^{(2)}$ contracts x by ρ, the program "apply $U_e^{(2)}$ for T ticks" certifies entry into $\mathcal{B}$ for any instance, independent of z and of F_e. Its guards are trivial (single open cell $\mathcal{Z}^{\text{in}}$), and its run–time is finite and completely effective. Thus whenever $\mathcal{A}$ outputs $\Pi^\star$, the certification is valid.

Reduction $e \mapsto e$.

- ($\Rightarrow$) If $U(e)$ halts at some finite time S, the augmented compiler discovers this within S iterations and immediately outputs the finite program $\Pi^\star$. Hence $e \in L^{\text{crt}}$.
- ($\Leftarrow$) Conversely, suppose $U(e)$ does *not* halt. Then the widths $w_j^{(e)}$ remain equal to 2^{-2^j} for all j. Any attempt by a sound compiler to produce a finite guarded decomposition that *uses* $U_e^{(1)}$ must, on some guard cell, bound the k–bit error of $U_e^{(1)}$ *uniformly* over z in that cell. By Lemma 7.9 (EM2 failure), this is impossible with a finite, polynomially evaluable guard structure: for each finite stage and any fixed cell there remain points z (arbitrarily close to the centers c_j at scales 2^{-2^j}) where deciding the kth bit would require super–polynomial work in k, contradicting the soundness/effectivity of the emitted program. Since the short–circuit never triggers (non–halting), the only remaining option for $\mathcal{A}$ is to keep refining forever; thus it does *not* halt, and $e \notin L^{\text{crt}}$.

Therefore $e \in \text{HALT} \iff e \in L^{\text{crt}}$, proving that L^{crt} is many–one equivalent to HALT and hence Σ_1^0–complete. □

Summary. The model $\mathfrak{M}_e$ satisfies all ABET–level axioms except (EM2). The only obstruction to extracting a finite, effective guarded decomposition is the *super–polynomial* worst–case cost inherent in evaluating $U_e^{(1)}$ to k bits on adversarial inputs. As soon as the halting event is discovered (when it exists), the compiler short–circuits to the trivial contraction–only program $\Pi^\star$, which is finite and effective. Hence the compile–time + run–time language is Σ_1^0–complete precisely because of the (EM2) failure.

7.6 HALT Embedding via an EM3-Violation

We construct a fixed, compact, continuous, effective measurement model in which

$$\text{(F)}\ (R_{\text{weak}}, L, C_{\text{basin}}), \quad \text{(S)}, \quad \text{(C)}, \quad \text{(T)}\ \text{(PL)}, \quad \text{(I)}, \quad \text{(EM1)}, \quad \text{(EM2)}$$

all hold, but (EM3) fails *at run time*. The failure is isolated to the necessity of testing membership in a recursively enumerable (r.e.) open set $G \subset [0,1]$ in any correct finite guarded decomposition; guard/basin membership thus cannot be decided in time polynomial in $\log(1/\varepsilon)$ (or even computably in general), violating (EM3). The underlying dynamics U, E remain fully effective and benign.

Spaces, basin, certification. Let

$$\mathcal{M} \;=\; [0,1]_x \times \{0,1\}_\ell, \qquad \mathcal{Z}^{\text{in}} \;=\; [0,1]_z,$$

with the product metric (Euclidean on $[0,1]$, discrete on $\{0,1\}$, capped at 1). Fix a contraction parameter $\rho \in (0,1)$ and a guard radius $r \in (0,1)$. Define the invariant basin

$$\mathcal{B} \;=\; \{(x,\ell) \in \mathcal{M} : x \le r \text{ and } \ell = 1\}.$$

We adopt the Minimal Certification Standard $\mathcal{S} = \{\varepsilon_n\}$ (Book II, Ch. 6) with shells

$$\mathcal{B}_n \;=\; \big\{(x,\ell) : x \le r - 2^{-n},\ \ell = 1\big\},$$

so that acceptance is robust in the product metric. The Lyapunov function is $V(x,\ell) = x$.

An r.e.-open target set and a fixed subinterval. Fix an effective bijection $e \mapsto q_e \in (0,1) \cap \mathbb{Q}$, and let U_{TM} be a universal Turing machine. For each stage t, let

$$H_t \;=\; \{\, e \mid U_{\text{TM}}(e) \text{ halts within } \le t \text{ steps}\,\}.$$

Define the increasing sequence of rational open sets

$$G_t \;=\; \bigcup_{e \in H_t} B\big(q_e,\, 2^{-(e+2)}\big)\,, \qquad G \;=\; \bigcup_{t \ge 0} G_t \;\subset\; [0,1].$$

Thus G is r.e.-open and has undecidable membership; nevertheless each G_t is a *finite* union of rational intervals and is computable in time polynomial in its description length. Choose once and for all a nontrivial rational interval

$$I \;=\; [a,b] \subset (0,1)$$

such that $I \cap B(q_{e_0}, 2^{-(e_0+2)}) = \varnothing$ for at least one fixed index e_0; this ensures that I does not trivially fall inside a single, always-present ball. We shall define the certification language over the domain $I \cap \mathbb{Q}$.

Dynamics and emitter (effective and benign). We update in two primitive ticks per global step:

$$\begin{aligned} &\textbf{Tick 1:} \quad (x,\ell),\, z \;\mapsto\; (\rho x,\ \ell), \\ &\textbf{Tick 2:} \quad (x',\ell),\, z \;\mapsto\; \big(x',\ \ell \vee \mathbf{1}_{z \in G_t}\big), \end{aligned}$$

where $t \in \mathbb{N}$ is the current global step counter (maintained implicitly by the controller), and G_t is the finite union of rational balls defined above. The emitter is

$$E(x,\ell) = \ell.$$

Note that membership in G_t is decidable in time polynomial in the size of the rational encoding of G_t and z, since G_t is finite and rational.

Tail fact. For a fixed (constant) input $z \in [0,1]$, the tail fact we certify is:

$$\Phi(z): \quad \exists T\ \forall t \geq T:\ (x_t, \ell_t) \in \mathcal{B}.$$

Because $x_{t+1} = \rho x_t$, the x-component reaches $x \leq r$ in uniformly bounded time, independently of z. The latch ℓ flips to 1 iff (and when) $z \in G_t$ for some finite stage t, i.e., iff $z \in G$. Thus

$$\Phi(z) \text{ holds} \quad \Longleftrightarrow \quad z \in G.$$

Lemma 7.11 (Computability and Axiom Compliance; only (EM3) fails)**.** *For the above model, the following hold:*

(F) (R_{weak}, L, C_{basin}), (S), (C), (T) *(PL-Traceability)*, (I), (EM1), (EM2).

Axiom (EM3) *fails at run time: any finite guarded decomposition that correctly decides* Φ *on* I *must include a guard whose membership coincides with* $G \cap I$ *(up to null tolerance), hence requires deciding an r.e.-open, undecidable predicate.*

Proof. (C) compactness and continuity. $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact. Each primitive tick is continuous: Tick 1 is affine in x, independent of z; Tick 2 updates ℓ by a continuous map into the discrete coordinate, conditionally on $z \in G_t$. Since G_t is a finite union of open intervals, the case distinction is continuous in the product topology. The composite update is continuous.

(F) basin axioms. Let $V(x, \ell) = x$. Choose $r \in (0,1)$ and $\rho \in (0,1)$ fixed.

- (L) Lyapunov border: if $x = r$, then after Tick 1 we have $x' = \rho r < r$, hence strict inward mapping.
- C_{basin}: on $\mathcal{B}$ (i.e., $x \leq r$, $\ell = 1$) Tick 1 is $x \mapsto \rho x$, a contraction; Tick 2 preserves $\ell = 1$.
- R_{weak}: from any (x_0, ℓ_0), there *exists* a (constant) input $z \in [0,1]$ with $z \in G_0$ (take any rational in some initial rational ball; or, if desired, enlarge G_0 by an explicit fixed rational ball disjoint from I to ensure nonemptiness). With that choice, ℓ flips to 1 at the first Tick 2, and x_t falls below r after a bounded number of steps depending only on x_0. Thus a witness stream exists for each start state.

(S) certification standard. The shells $\mathcal{B}_n = \{x \leq r - 2^{-n},\ \ell = 1\}$ are open in the product metric; acceptance is robust because the latch is discrete and the x-thresholds are strict. This is the Minimal Certification Standard of Book II, Ch. 6.

(T) PL-Traceability. For any state $\Theta = (x, \ell)$ and any n, the preimage

$$U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$$

is either $\mathcal{Z}^{\text{in}}$ (if $\ell = 1$ and $x \leq r - 2^{-n}$), or one of the finite unions G_t that appear at the relevant Tick 2 on the path from Θ (open and nonempty), or $\emptyset$ if fewer than the required number of ticks has elapsed to reach $x \leq r - 2^{-n}$. In particular, along any realized path, the tick at which entry becomes possible has an open, nonempty preimage in z. This is precisely PL-Traceability (Def. 2.5).

(I) time-irreversibility / no supertask. Each global step performs two finite primitive ticks; no accelerating schedule. A standard ranking function $f(x, \ell) = \lceil \log_{1/\rho} \max\{x -$

$r, 0\}\rceil$ decreases whenever $x > r$ (Tick 1), and once $\ell = 1$ and $x \le r$ the state remains in $\mathcal{B}$.

(EM1) effective representation. Tick 1 is affine with rational coefficients. Tick 2 tests membership in G_t, which is a *finite* rational union generated by the (internal) universal simulation up to time t; all constants are rational and uniformly encoded. Thus each primitive has a uniform finite effective description (or a uniform polytime evaluator), in line with EM1 (or $\mathrm{EM1}_{\mathbf{P}}$ in the parameterized reading of Book III, Ch. 3).

(EM2) polynomial evaluation modulus. Evaluating $x \mapsto \rho x$ to k bits is trivial. Testing $z \in G_t$ is a finite union membership query against $O(|H_t|)$ rational intervals; evaluating this to tolerance 2^{-k} is polynomial in the bit-length of z, the encodings of the interval endpoints, and k. Hence a (known) polynomial evaluation modulus exists.

(EM3) failure. Consider the certification language on the domain $I \cap \mathbb{Q}$ (see below). For $z \in I$, $\Phi(z)$ holds iff $z \in G \cap I$ (because x always contracts under ρ, and the latch flips iff $z \in G$). Any *correct* finite guarded decomposition that decides Φ on I must, on some branch, separate $G \cap I$ from $I \setminus G$. Otherwise, by continuity and robustness of shells, it would misclassify points arbitrarily close to the boundary of $G \cap I$. Therefore at least one guard must implement the (undecidable) membership predicate $z \in G \cap I$, which is not decidable in $\mathrm{poly}(\log(1/\varepsilon))$ time (indeed, not decidable at all). Thus (EM3) fails at run time, while all other axioms hold. □

Lemma 7.12 (Σ^0_1-Hardness of the Certification Language; EM3-only at run time)**.** *Define the certification language over the fixed subinterval $I = [a, b]$ by*

$$L_\Phi \;=\; \{\, z \in I \cap \mathbb{Q} \;:\; \Phi(z) \text{ holds} \,\}.$$

Then

$$z \in L_\Phi \quad \Longleftrightarrow \quad z \in G \cap I.$$

Consequently, the many–one map $e \mapsto q_e$ (restricted to those $q_e \in I$) reduces HALT *to L_Φ, and L_Φ is Σ^0_1-complete (Turing degree $0'$). Moreover, at run time the only axiom that fails is* (EM3)*.*

Proof. By construction, for any fixed $z \in I$ (held constant along the run):

$$\Phi(z) \text{ holds} \iff \Big(\exists T : x_T \le r\Big) \;\wedge\; \Big(\exists t : z \in G_t\Big) \iff z \in G \cap I,$$

because $x_{t+1} = \rho x_t$ ensures $x_T \le r$ for some finite T independently of z, and ℓ flips to 1 exactly at the first t with $z \in G_t$. Hence $L_\Phi = G \cap I \cap \mathbb{Q}$.

For hardness, recall that G is the r.e.-open union $\bigcup_t \bigcup_{e \in H_t} B(q_e, 2^{-(e+2)})$. Fix any e with $q_e \in I$. Then $q_e \in G \cap I$ iff $e \in H_t$ for some t, i.e., iff $U_{\mathrm{TM}}(e)$ halts. Thus the computable injection $e \mapsto q_e$ (with $q_e \in I$) is a many–one reduction from HALT to L_Φ. Therefore L_Φ is Σ^0_1-complete (Turing degree $0'$).

Finally, by Lemma 7.11, $(F), (S), (C), (T)$ (PL), (I), (EM1), (EM2) all hold. Run-time failure is confined to (EM3) because any correct guarded program must implement the undecidable guard $z \in G \cap I$. No other axiom is implicated. □

Discussion (compile-time oracle vs. run-time failure). The guarded–decomposition compiler (ATRT) may fail to stabilize *without* an oracle, since deciding guards equivalent to $G \cap I$ is undecidable. If a *compile-time* oracle supplies the correct finite decomposition (i.e., the exact guards), then compile time succeeds; nevertheless, at *run time* the guard membership tests remain undecidable, so (EM3) continues to fail while $(F), (S), (C), (T), (I), (EM1), (EM2)$ hold.

8 Proof of the FDI Theorem

8.1 Finite-Guard Case ⇒ Degree 0

Proof. Suppose Φ admits a finite guarded decomposition. By definition there is

1. a finite list of guard cells $N_1, \ldots, N_k \subset \mathcal{Z}^{\text{in}}$, and
2. a deterministic branching program B which tests membership in the N_i and selects a branch i,
3. together with per-branch proofs that Φ holds whenever the execution starts in N_i.

We now show $L_\Phi \in \mathbf{NP}$:

- *NP witness.* On input x, guess

$$(\, i,\ \pi_i) \quad \text{where} \quad i \in \{1, \ldots, k\},\ \pi_i = \text{proof of } \Phi \text{ under } N_i.$$

- *Verification.* In polynomial time one can

  1. simulate the branching program to confirm x routes to branch i,
  2. check membership $z_0 \in N_i$ up to the required tolerance (EM3),
  3. verify the proof π_i that Φ holds from that branch.

Since all steps use only finitely many guard tests and polynomial-time proof checks, this is an NP-decider for L_Φ. Hence

$$\deg_T(L_\Phi) \ = \ 0.$$

□

8.2 No-Guard Case ⇒ Degree ≥ 0′

Throughout this subsection we assume (F) and (S), so that tail facts and their certification languages are well-defined. Recall that under $(C), (T), (I)$, (EM1)–(EM3) the Fundamental Test Equivalence (FTE) from Chapter 3 guarantees a finite guarded decomposition for every certifiable tail fact.

Lemma 8.1 (Exhaustiveness of the failure modes)**.** *Assume (F) and (S). If a tail fact Φ admits no finite guarded decomposition, then at least one of the axioms*

$$(C), \quad (T), \quad (I), \quad \text{(EM1)}, \quad \text{(EM2)}, \quad \text{(EM3)}$$

fails in the underlying model.

Proof. Contraposition. If all six axioms hold together with (F) and (S), the FTE (Theorem 1.1) applies and yields a finite guarded decomposition for Φ, contradicting the hypothesis. □

We now perform a complete case analysis over the six ways the FTE premises can fail. For each case we invoke (i) a computability/*axiom compliance* lemma that isolates exactly that failure, and (ii) a *hardness* lemma exhibiting an explicit, computable embedding of a classical undecidable set into a certification language.

Case (I) fails (supertask / Zeno). By Lemma 7.1 there is a computable family $x \mapsto \mathcal{M}_x$ satisfying all axioms except (I). By Lemma 7.2,

$$x \in \mathrm{HALT} \iff \mathrm{Outcome}(\mathcal{M}_x) = 1,$$

so the resulting certification language is Σ_1-complete and hence of degree $0'$.

Case (C) fails (non-compactness only). By Lemma 7.3 there is a computable family $e \mapsto \mathfrak{M}_e$ satisfying all axioms except (C). By Lemma 7.4,

$$e \in \mathrm{HALT}^{\exists} \iff e \in L_\Phi,$$

so the certification language is Σ^0_1-complete (Turing degree $0'$).

Case (T) fails (open-preimage / traceability). By Lemma 7.5 there is a computable model satisfying all axioms except (T). Lemma 7.6 shows the *compile-time stabilization* language L_{stab} of the standard iterative guarded-decomposition procedure is Σ^0_1-complete. Whenever the compiler halts, the emitted finite guarded program is correct and effective, and run-time verification remains within the baseline framework.

Case (EM1) fails (effective representation). By Lemma 7.7 there is a computable family that violates exactly (EM1). Lemma 7.8 shows the certification language is Σ_1-complete, hence degree $0'$.

Case (EM2) fails (polynomial modulus of continuity). By Lemma 7.9 there is a computable family that violates exactly (EM2). Lemma 7.10 shows the certification language is Σ_1-complete, hence degree $0'$.

Case (EM3) fails (polynomial guard/basin tests). By Lemma 7.11 there is a computable family that violates exactly (EM3). Lemma 7.12 shows the certification language is Σ_1-complete, hence degree $0'$.

Proposition 8.1 (No-guard $\Rightarrow$ hypercomputational degree)**.** *Assume (F) and (S). If a tail fact Φ admits no finite guarded decomposition, then its certification language L_Φ has Turing degree at least $0'$. In each failure case of $(C), (T), (I)$, (EM1), (EM2), (EM3) there is a computable embedding whose associated language is Σ^0_1-hard.*

Proof. By Lemma 8.1 at least one of $(C), (T), (I)$, (EM1), (EM2), (EM3) fails. In each respective case, the corresponding pair of lemmas {7.1–7.2, 7.3–7.4, 7.5–7.6, 7.7–7.8, 7.9–7.10, 7.11–7.12} exhibits a computable embedding whose certification language is at least Σ_1-hard (and Π_2-hard when (C) fails). Therefore the degree lower bound $0'$ follows in all cases. □

9 Corollaries

9.1 Hierarchy-Collapse for Type Tests

Proposition 9.1 (Hierarchy-Collapse)**.** *Assume Axioms* (F), (S), (C), (T), (I), *and* (EM1)–(EM3)*. For any tail measurement fact Φ, its certification language L_Φ satisfies exactly one of:*

- $L_\Phi \in \mathbf{NP}$ *(hence recursive, Turing degree* 0*), or*
- $\deg_T(L_\Phi) \geq 0'$ *(undecidable; e.g.* Σ^0_1*- or* Π^0_2*-hard via the embeddings).*

No decidable class strictly above NP (e.g. **PSPACE**, **EXP***) can occur.*

Proof. By Theorem **??** and the finite-guard direction, either Φ is decomposable, in which case the Polytime Verifier Construction Lemma yields $L_\Phi \in \mathbf{NP}$ (hence degree 0), or Φ is decomposition-independent, in which case the embedding lemmas force $\deg_T(L_\Phi) \geq 0'$. Since **PSPACE**, **EXP**, etc. are classes of *decidable* sets (all of Turing degree 0), they cannot arise on the hypercomputational branch; and when all axioms hold we are already in NP on the decidable branch. □

Remark 9.1. Under our axioms, certification languages do not inhabit "intermediate" decidable territory strictly above NP: they are either NP-verifiable or undecidable (at least $0'$).

10 Extension to General Verifier Classes

10.1 Parameterized finite-guard vs. verifier classes

We parameterize the effective axioms by a base class $\mathbf{C}$ (for guard evaluation) and a verifier class $\mathcal{V}$ (for certification). Replace (EM1)–(EM3) by:

- **(EM1$_\mathbf{C}$)** U, E admit uniform, finite descriptions so that evaluations to k bits can be carried out within $\mathbf{C}$ on input size n and accuracy k.
- **(EM2$_\mathbf{C}$)** Known resource modulus: evaluating U, E to k bits costs $\mathrm{poly}(m(k), n)$ in $\mathbf{C}$ for some known m.
- **(EM3$_\mathbf{C}$)** Guard/basin membership up to tolerance is decidable within $\mathbf{C}$.

For verifiers with error (randomized/quantum/interactive), strengthen the Certification Standard to include an explicit gap:

> **(S$_{\text{gap}}$)** There exist fixed $\epsilon_c - \epsilon_s \geq 1/\mathrm{poly}(n)$ such that acceptance is robust to perturbations within the shells, and error can be amplified below the shell tolerance.

Witnesses must be finitely encodable. In particular, for quantum verification we assume *either* QCMA witnesses *or* QMA witnesses provided by classical descriptions of uniform, poly-size state-preparation circuits.

Under (F), (S$_{\text{gap}}$), (C), (T), (I), and (EM1$_\mathbf{C}$)–(EM3$_\mathbf{C}$) we obtain:

Proposition 10.1 (Parameterized FDI)**.** *Let* Φ *be a tail fact. Exactly one holds:*

1. Finite-guard: Φ *admits a finite guarded decomposition with guard primitives evaluable in* $\mathbf{C}$*. Then* $L_\Phi \in \mathcal{V}$ *(the verifier for the decomposition runs in* $\mathcal{V}$ *with error handled by* (S$_{\text{gap}}$)*).*
2. No-guard: Φ *is decomposition-independent. Then* $\deg_T(L_\Phi) \geq 0'$ *by the embedding lemmas, independent of* $\mathbf{C}, \mathcal{V}$*.*

Remark 10.1. Examples: $(\mathbf{C}, \mathcal{V}) = (\mathbf{P}, \mathbf{NP})$ (the baseline), $(\mathbf{BQP}, \mathbf{QCMA})$ or $(\mathbf{BQP}, \mathbf{QMA_{prep}})$ (QMA with circuit-preparable witnesses), $(\mathbf{PSPACE}, \mathbf{IP})$. In each case the decidable branch lies in $\mathcal{V}$ and the hypercomputational branch jumps to degree $\geq 0'$.

10.2 Robustness under stronger effective verifiers

Proposition 10.2 (Verifier-Robustness (effective, uniform))**.** *Fix effective, uniform classes* $\mathbf{C}$ *and* $\mathcal{V}$ *without oracles/advice, with finitely encodable witnesses and* $(\mathrm{S}_{\mathrm{gap}})$*. Under* (F)*,* (C)*,* (T)*,* (I)*,* $(\mathrm{EM1}_{\mathbf{C}})$–$(\mathrm{EM3}_{\mathbf{C}})$*, if* Φ *is decomposition-independent then* $L_\Phi \notin \mathcal{V}$ *and* $\deg_T(L_\Phi) \geq 0'$*.*

Proof sketch. If $L_\Phi \in \mathcal{V}$, fix a verifier $V \in \mathcal{V}$, amplify its gap to be smaller than the shell tolerance ($(\mathrm{S}_{\mathrm{gap}})$), and unroll V's uniform, finite description into a finite set of guard primitives (each evaluable in $\mathbf{C}$ by $(\mathrm{EM1}_{\mathbf{C}})$–$(\mathrm{EM3}_{\mathbf{C}})$). This yields a finite guarded decomposition, contradicting decomposition-independence. The degree lower bound follows from the embedding lemmas. □

Chapter 5: Implications for Physics

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1 Philosophical and Physical Implications

This chapter applies the Fact Decomposition-Independence (FDI) framework to quantum measurement. Our goal is not to adjudicate metaphysics, but to audit *epistemic certifiability*: when can a single-shot measurement fact be certified by a finite guarded decomposition under the axioms introduced earlier? We first connect known physical resource bounds to our axiom failures, then draw out consequences for the measurement problem.

1.1 Why Physical Infinities Enforce the Turing Barrier

Realistic laboratory regimes enforce resource bounds (finite energy, finite spatial extent, finite bandwidth). These align precisely with the axioms that preclude hypercomputation in our framework:

- **Finite information budget & compactness (Axiom C).** A bounded, effectively represented input slice Z^{in} (e.g. energy/particle cutoffs and localization windows) yields a compact metric domain where U, E are continuous. Losing (C) re-opens the "infinite bandwidth / infinite state space" loophole.
- **No supertasks in verification (Axiom I).** Verification protocols must be well-founded (no Zeno-like infinite test chains in finite time). Violating (I) is exactly the recipe for accelerating-machine embeddings.
- **Traceable inputs (Axiom T) & effective measurability (EM1–EM3).** Physical tests must have open, reachable guard cells and effectively realisable primitives (representation, modulus, guard membership). Dropping (T) or EM1–EM3 blocks finite decompositions or their execution.

Conclusion: whenever a putative "hypercomputation" needs temporal or spatial infinities, it corresponds to a deliberate failure of (I) or (C), or of traceability/effectivity. Conversely, staying inside these axioms re-establishes the Turing barrier at the single-shot level.

1.2 Insight into the Quantum Measurement Problem

In our setting, a single outcome statement ("the pointer reads e_k") is a *tail fact* on system–apparatus–environment dynamics. FDI separates two possibilities:

1. **Decomposable measurement.** There exists a finite guarded decomposition that routes every admissible input slice to a guard cell whose branch proof certifies the tail fact. Then the corresponding certification language L_Φ lies in the chosen verification class (NP/QCMA/etc.).

2. **Decomposition-independent measurement.** No finite guarded decomposition exists under the available axioms. Then, by the embedding lemmas established in Chapter 4, L_Φ attains Turing degree $\geq 0'$ via a principled axiom failure (of (I), (C), (T) or EM1–EM3).

The "measurement problem" thus reframes as: given a collapse theory and a realistic input slice, do we obtain a finite guarded decomposition or are we forced—by a specific axiom failure—into the $0'$ frontier?

1.3 Roadmap: Comparing Collapse Theories

We audit major collapse pictures by three *epistemic* categories:

Category (1) *FDI out of scope.* One or more structural axioms (e.g. (C) or (T)) fail already on the intended input slice (e.g. global, non-compact state spaces); no formal verdict on decomposability can be issued.

Category (2) *FDI applicable, but decomposition-independent.* The axioms put the test within scope, yet no finite guarded decomposition exists (e.g. inherently stochastic selection, uncountable many-to-one fibres, or failed traceability/effectivity). Chapter 4 shows how this forces degree $\geq 0'$.

Category (3) *FDI applicable and decomposable.* A finite guard partition exists; outcome facts admit polynomial certificates (in NP/QCMA/...), yielding strictly higher epistemic certifiability.

With this rubric, we turn to the computational structure of single-shot collapse.

2 Uncomputability of a Single Measurement via Closed-Choice Principles

We formalise the non-computability inherent in single-outcome selection using Weihrauch reducibility. The collapse selector is at least as hard as classical closed choice on countable and real domains, hence admits no single Turing realiser even when pre-measurement data are given with arbitrary precision.

2.1 Represented Spaces for Pre-Measurement Data

Let H be a separable complex Hilbert space with a *computable* orthonormal basis $\{e_k\}_{k\in\mathbb{N}}$. We represent:

- $\psi \in H$ by an oracle that on input n outputs a rational vector $\psi^{(n)}$ with $\|\psi - \psi^{(n)}\| < 2^{-n}$.

- A bounded self-adjoint operator A by an oracle for its spectral measure E^A: given a rational Borel set $B \subset \mathbb{R}$ and n, return $r_{B,n} \in \mathbb{Q}$ approximating $\langle \psi, E^A(B)\psi \rangle$ within 2^{-n} for each computable ψ.

Thus dom(Collapse) $\subseteq H \times \mathcal{B}(H)$ is a represented space of pre-measurement data.

2.2 Weihrauch Closed-Choice Operators

For a represented space X, the *closed-choice* multivalued map $\mathbf{C}_X : \mathcal{K}(X) \rightrightarrows X$ sends a nonempty closed set $A \subseteq X$ to an arbitrary point of A. We use:

- **Discrete choice** $\mathbf{C}_I$ on a finite or countably infinite index set $I \subseteq \mathbb{N}$.
- **Real choice** $\mathbf{C}_\mathbb{R}$ on closed $A \subseteq \mathbb{R}$.

Both are known to be non-computable in the Weihrauch lattice for $|I| \geq 2$ and for $\mathbb{R}$, respectively.

2.3 Main Reduction Theorem

Theorem 2.1 (Collapse dominates closed choice)**.** *In the Weihrauch lattice one has*

$$\mathbf{C}_I \ \leq_W \ \text{Collapse} \quad \textit{for any finite or countable } I, \qquad \mathbf{C}_\mathbb{R} \ \leq_W \ \text{Collapse}.$$

Consequently, no single Turing machine can select *the actually realised post-measurement eigenstate/eigenvalue from pre-measurement data alone.*

2.4 Proof of Theorem 2.1

Recall $f \leq_W g$ iff there exist computable maps K and H such that for all x, every $v \in g(K(x))$ yields $H(x, v) \in f(x)$.

Discrete case. Given $I \subseteq \mathbb{N}$, define

$$\psi_I = \sqrt{3} \sum_{i \in I} 2^{-i}\, e_i, \qquad A_I = \sum_{i \in I} \lambda_i \, |e_i\rangle \langle e_i|$$

with distinct rationals λ_i. Then Collapse(ψ_I, A_I) returns some $|e_i\rangle$ with $i \in I$. Set $K(I) = (\psi_I, A_I)$ and $H(I, |e_i\rangle) = i$. Both K, H are computable, hence $\mathbf{C}_I \leq_W$ Collapse.

Continuous case. Let A have purely continuous spectrum on a rational interval; prepare a computable ψ with absolutely continuous spectral measure. Then Collapse(ψ, A) selects an eigenvalue x in a nonempty closed $A \subset \mathbb{R}$. Choosing K as the identity embedding and H as projection realises $\mathbf{C}_\mathbb{R} \leq_W$ Collapse. $\square$

2.5 Interpretation

- **Discrete spectra.** The set of possible post-measurement states $\{|e_i\rangle : i \in I\}$ is countable, but selecting the actually realised $|e_i\rangle$ as a function of (ψ, A) is Weihrauch-hard for $\mathbf{C}_I$.
- **Continuous spectra.** Any single Turing realiser can only enumerate countably many candidates, so choosing a realised x from a closed set of positive measure is non-computable in the same sense.

Thus, even before invoking any axiom failure, *selection* as a function is irreducibly non-computable. In the FDI framework, decomposition-independence arises precisely when this non-computability cannot be "pushed" into random bits or hidden parameters that restore a finite guard partition.
Thus collapse selection is governed by non-computable closed-choice principles in the Weihrauch lattice, firmly placing it beyond any finite-step algorithmic decomposition.

So far, the results are obvious for physicists; however, it is important to establish the fact of decomposition-independence formally.

The big next question is as follows. Under the closer look, we observe that above proof *only* works in physically achievable measurements. Indeed, only in a physically real setup (finite energy bound, localization bound) can we claim Z^{in} compactness.

What will happen if we consider, for the sake of interest, a Universe's global pure state and its collapse by any observable?

3 Global Collapse: Out of Scope for FDI

Consider collapsing a *global* pure state of the Universe in an infinite-dimensional Hilbert space by an arbitrary observable. The natural input slice is non-compact (unbounded energy/particle number, unbounded spatial support); moreover, achieving continuity of U, E on a compact metric subspace fails without severe truncations. Hence Axiom (C) is violated *a priori*, and input traceability (T) typically fails as well.

Verdict. The FDI framework is inapplicable to global collapse (Category 1): the Simple Test cannot even be posed. This does not resolve the metaphysical question of global outcomes; it states an epistemic limit of our complexity-theoretic machinery. Laboratory-scale truncations (finite energy/localisation cutoffs) re-enter the FDI domain.

4 Statement: Global Collapse Decomposition-Independence can't be resolved in the FDI framework

The logic of this statement is very simple: in our previous axiom violation examples, a violation arose *in the measurement process*. This is exactly what allowed us to relate the uncomputability to decomposition independence.

However, pure global state lives in an infinite-dimensional Hilbert space, therefore (C) is violated *a priori*.

Hence, all results of the work are inapplicable to a global state collapse. This is an epistemologically negative, but an important result: we can have decomposition-independent,

hence undecidable problems, but there is a higher level of hardness inaccessible in our complexity-theoretic framework.

Recalling a famous Einstein sentence, God may play dice or not. However, we will always be playing dice, and it doesn't allow us to say anything beyond that.

At least, this is the epistemological result of the Copenhagen interpretation. Are other collapse theories different?

5 Epistemic Audit of Major Collapse Theories

Before analyzing each theory from the FDI point of view, we can define possible outcomes. Note that we discuss nothing about the validity, physicality, maturity, evidence level for a theory, or even the ontological value of its results.

Instead, we discuss them from a purely epistemical view.

Category (1) "God-level theory". FDI is inapplicable outright, in the same (or equivalent) way as it is inapplicable for *global* state collapse. However, in such a theory, the inapplicability holds even when the theory is considered with the typical physical restrictions (e.g., finite energy bound and/or localization bound), or at a maximal level of such restrictions theory supports.

Category (2) "Copenhagen-level theory". FDI is applicable (with physical restrictions) and yields decomposition independence (e.g., by the stochastic process involved). The level is usual for us; if a theory also yields results indistinguishable from Copenhagen, we call it "interpretation".

Category (3) "New-level theory". FDI is applicable, and the measurement test is decomposable. The theory claims to resolve the measurement problem in some constructive way; this yields a higher epistemological value than the currently accepted mainstream.

We adopt a *two-stage* audit that aligns precisely with the FDI framework and the role of ABET.

Stage 0 (In-scope gate). Work under realistic laboratory assumptions (finite energy and/or localisation bounds) so that the input slice Z^{in} is compact and effectively named. We require:

(F) Invariant basin, (S) Certification standard, (C) Compact continuity.

If *any* of (F), (S), (C) fails on the laboratory slice, ABET cannot be formed and the Simple Test does not exist. **Outcome: Category 1 (FDI out of scope).**

Stage 1 (Decomposition test). Assuming Stage 0 passes, ABET and the Simple Test exist. We now ask whether the single-shot fact admits a *finite guarded decomposition*. Here (T) and EM1–EM3 are the operative effectivity/traceability assumptions, and real-world protocols normally satisfy (I).

$$\begin{cases} \text{Finite guarded decomposition exists} & \Rightarrow \textbf{Category 3} \text{ (FDI-decomposable; certifiable);} \\ \text{No finite guarded decomposition} & \Rightarrow \textbf{Category 2} \text{ (FDI in scope but decomposition-independent} \end{cases}$$

In particular, failure of (T) or EM1–EM3 on the in-scope slice forces the *absence* of a finite decomposition and thus Category 2.

5.1 Collapse Theories Audit

5.1.1 Standard Born (Copenhagen)—Bohr & von Neumann

Stage 0 (In-scope gate). With finite energy and localization cutoffs, axioms (F), (S), (C) hold on the laboratory slice; ABET and the Simple Test exist.
Stage 1 (Decomposition test). Measurement collapse is implemented via an internal stochastic rule in U/E; this internal randomness means PV fails, yet PL-Traceability still holds. No finite guarded decomposition isolating the *realised* outcome exists.
FDI verdict. Category 2. Lab-restricted complexity: the certification language is Σ_1^0-hard (Turing-degree $\geq 0'$).

5.1.2 GRW / CSL Objective Collapse—Ghirardi, Rimini, Weber; Pearle; Gisin

Stage 0 (In-scope gate). Under laboratory truncation, (F), (S), (C) are satisfied; ABET applies.
Stage 1 (Decomposition test). Stochastic noise is internal to the collapse dynamics, breaking PV; PL-Traceability holds but no finite guard certification exists.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.3 Bohmian Mechanics with Typicality—Bohm; Dürr, Goldstein & Zanghì

Stage 0 (In-scope gate). With finite-energy/localization restrictions, (F), (S), (C) hold.
Stage 1 (Decomposition test). The outcome depends on a continuum-valued hidden variable Q_0; fibres over pointer outcomes are uncountable. UI-Traceability fails (though PL may hold along the realised branch).
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.4 Everett / Many-Worlds—Everett; DeWitt; Saunders

Stage 0 (In-scope gate). With lab truncations, axioms (F), (S), (C) hold.
Stage 1 (Decomposition test). "This branch's" outcome requires self-location among a continuum of decoherent branches. UI-Traceability fails; PL-Traceability is vacuous.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.5 Convex–Quasi–Linear Deterministic Collapse—Rembieliński & Caban

Stage 0 (In-scope gate). On finite-dimensional lab slice, (F), (S), (C) and EM1–EM3 (hence PV) hold.
Stage 1 (Decomposition test). Collapse flow maps all states into a finite set of one-dimensional projectors; one constructs finite guards and polynomial verification.
FDI verdict. Category 3. Certification language lies in NP/QCMA (not Σ_1^0-hard).

5.1.6 Finite-World Many-Interacting-Worlds (MIW)—Hall, Deckert & Wiseman

Stage 0 (In-scope gate). For fixed finite $N < \infty$, the input slice is compact and (F), (S), (C), PV hold.
Stage 1 (Decomposition test). Each world index is a finite operational input; pointer

outcomes partition the index set. Finite guarded decomposition and polynomial-time witness exist.
FDI verdict. Category 3. Certification language in NP/QCMA.

5.1.7 Superdeterministic Cellular-Automaton Interpretation (CAI)—'t Hooft

Stage 0 (In-scope gate). Finite CA lattice meets (F), (S), (C), PV.
Stage 1 (Decomposition test). The micro-state index acts as operational input, deterministically yielding a pointer. Finite guarded decomposition exists and is polytime verifiable.
FDI verdict. Category 3. Certification language in NP.

5.1.8 Invariant-Set Theory—Palmer

Stage 0 (In-scope gate). Although I_U (the fractal invariant set) is mathematically compact, operational input slice fails UI-Traceability and PV; PL fails in operational Euclidean metric.
Stage 1 (Decomposition test). Because no open guards exist for realisable inputs, ATRT does not halt; no finite guarded decomposition is possible.
FDI verdict. Category 1. Certification language is undefined.

5.1.9 Retro-causal Two-Time Boundary Models—Aharonov; Price; Wharton

Stage 0 (In-scope gate). With lab cutoffs, (F), (S), (C), PV hold.
Stage 1 (Decomposition test). Dependence on future boundary conditions breaks UI-Traceability; PL may hold only once future data are known. No finite guard partition covers the realised fact.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.10 Big-Bang Superdeterminism—'t Hooft et al.

Stage 0 (In-scope gate). If the primordial parameter space Λ is compact and computable, (F), (S), (C) hold; otherwise (C) (or PV) fails and category is 1.
Stage 1 (Decomposition test) [compact & effective Λ]. Operationally the pointer depends on hidden λ; fibres are uncountable, breaking UI-Traceability.
FDI verdict. Category 2 (or Category 1 if Λ non-compact/effective). Certification language is Σ_1^0-hard.

5.1.11 Penrose OR (Objective Reduction)—Penrose

Stage 0 (In-scope gate). With laboratory cutoffs, (F), (S), (C) hold.
Stage 1 (Decomposition test). Collapse driven by gravitational self-energy threshold is an internal stochastic mechanism. PV fails; PL holds. No finite guard isolates actual outcome.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.12 Diósi–Penrose (DP) Model—Diósi & Penrose

Stage 0 (In-scope gate). Under energy/localization constraints, (F), (S), (C) hold.
Stage 1 (Decomposition test). Stochastic collapse via DP noise is internal; PV fails, PL holds. No finite decomposition for a realised outcome.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.13 Quantum State Diffusion—Gisin & Percival

Stage 0 (In-scope gate). With lab cutoffs, axioms $(F), (S), (C)$ hold.
Stage 1 (Decomposition test). The stochastic Schrödinger evolution rests on an internal Wiener/Poisson process. PV fails; PL holds. No finite guard for realised sample outcome.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.14 Transactional Interpretation—Cramer; Kastner

Stage 0 (In-scope gate). Finite lab slice ensures $(F), (S), (C)$.
Stage 1 (Decomposition test). Outcome arises via internal 'handshake' between emitter and absorber; not open–preimage traceable at tick 0. UI fails, PL may hold.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.15 Consistent Histories—Griffiths; Gell-Mann & Hartle; Omnès

Stage 0 (In-scope gate). With bounded lab domain, $(F), (S), (C)$ hold.
Stage 1 (Decomposition test). Selecting 'this' history among many requires self-location in a branching space; UI fails. PL holds only conditioned on actual branch.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.16 Modal Interpretations—Dieks; Vermaas

Stage 0 (In-scope gate). Under cutoffs, $(F), (S), (C)$ valid.
Stage 1 (Decomposition test). Actual definite property is selected from a continuum of possibilities; UI fails for a single outcome. PL holds if hidden variable is fixed.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.17 Relational Quantum Mechanics—Rovelli

Stage 0 (In-scope gate). With lab constraints, $(F), (S), (C)$ hold.
Stage 1 (Decomposition test). Outcome is relationally selected for an observer; there is no global guard in the lab input space. UI fails, PL holds per observer relative run.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.18 QBism—Fuchs & Schack

Stage 0 (In-scope gate). Operational lab slice satisfies $(F), (S), (C)$.
Stage 1 (Decomposition test). Outcome is personal belief update, not open-preimage traceable; UI fails, PL holds within a personal run.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.19 Nelson-type Stochastic Mechanics—Nelson

Stage 0 (In-scope gate). With lab-level constraints, $(F), (S), (C)$ hold.
Stage 1 (Decomposition test). Hidden Brownian motion variable yields many-to-one outcome mapping; UI fails. PL holds for the realised sample path.
FDI verdict. Category 2. Certification language is Σ_1^0-hard.

5.1.20 Summary table

Table 1: Epistemic Audit of Major Collapse / Interpretation Theories

Theory (Author(s))	Stage 0 (F,S,C)	FDI Category	Language / Hardness	Failure Mode / Reason
Born / Copenhagen (Bohr & von Neumann)	Pass	2	Σ_1^0-hard	Internal randomness ⇒ PV fails
GRW / CSL (Ghirardi, Rimini, Weber; Pearle; Gisin)	Pass	2	Σ_1^0-hard	Internal noise ⇒ PV fails
Bohmian Mechanics (Bohm; Dürr, Goldstein & Zanghì)	Pass	2	Σ_1^0-hard	Continuum hidden variable; UI-T fails
Everett / Many-Worlds (Everett; DeWitt; Saunders)	Pass	2	Σ_1^0-hard	Self-location among branches; UI-T fails
CQL Deterministic collapse (Rembieliński & Caban)	Pass	3	NP/QCMA	Finite projector basins; PV holds
Finite-World MIW (Hall, Deckert & Wiseman)	Pass	3	NP/QCMA	Finite world index as guard
Superdeterministic CAI ('t Hooft)	Pass	3	NP	Finite micro-state index
Invariant-Set Theory (Palmer)	Fail	1	—	Operational UI-T & PV fail; fractal state space
Retro-causal Two-Time (Aharonov; Price; Wharton)	Pass	2	Σ_1^0-hard	Future boundary breaks UI-T
Big-Bang Superdeterminism ('t Hooft et al.)	Pass / Fail	2 / 1	—	Hidden λ index; UI-T / C or PV fail
Penrose OR (Penrose)	Pass	2	Σ_1^0-hard	Gravitational collapse internal randomness
Diósi–Penrose (DP) model (Diósi; Penrose)	Pass	2	Σ_1^0-hard	Stochastic DP noise internal to dynamics
Quantum State Diffusion (Gisin & Percival)	Pass	2	Σ_1^0-hard	Stochastic Schrödinger mechanism; PV fails
Transactional Interpretation (Cramer; Kastner)	Pass	2	Σ_1^0-hard	Handshake selection; UI-T fails
Consistent Histories (Griffiths; Gell-Mann & Hartle; Omnès)	Pass	2	Σ_1^0-hard	Branch self-location; UI-T fails
Modal Interpretations (Dieks; Vermaas)	Pass	2	Σ_1^0-hard	Property selection via hidden index; UI-T fails
Relational QM (Rovelli)	Pass	2	Σ_1^0-hard	Relational self-location; UI-T fails
QBism (Fuchs & Schack)	Pass	2	Σ_1^0-hard	Personal belief update; UI-T fails
Nelson Stochastic Mechanics (Nelson)	Pass	2	Σ_1^0-hard	Brownian hidden variable; UI-T fails

6 Why an *Epistemic–Complexity* Test-bed Is Needed

6.1 A persistent split in foundational views

A recent international survey of more than one thousand physicists[1] highlights how little consensus exists on the foundations of quantum mechanics. When asked about their preferred interpretation, 36% chose Copenhagen, 15% Many–Worlds, 7% Bohmian mechanics, with the remainder scattered among less common views. On the reality of the wavefunction itself, 36% regarded it as a real entity, 47% as a calculation tool, and the rest were undecided. Even the existence of a sharp quantum–classical boundary split opinion almost evenly (45% yes, 45% no).

These are not minor philosophical quibbles: each stance changes what the theory says about what we can *know* from an experiment. If two interpretations make identical predictions for all laboratory outcomes but differ on whether those outcomes can be verified in a single run, that difference is operationally important.

6.2 The need for a neutral yardstick

At present, there is no widely–accepted way to compare interpretations on the question of single–run verification. Our aim is not to declare one interpretation "true," but to develop

[1] Nature survey, 2025; see https://www.nature.com/articles/d41586-025-02342-y

a *model–independent* test–bed that measures a simpler property: whether, in principle, a single experimental outcome could be certified as correct by a finite procedure.

We call this property *epistemic complexity*: the inherent difficulty, built into the theory, of confirming that a recorded result matches reality, without relying on statistical repetition. Different theories can share the same experimental predictions yet occupy very different positions on the epistemic–complexity scale.

6.3 Three qualitative categories

Without technical detail, the possibilities fall into three broad categories:

Category 1: No verification procedure can even be *defined* for the scenario under consideration; the setting is out of scope.

Category 2: A verification procedure can be defined, but no *finite* procedure can guarantee correctness for a single outcome; infinite information would be required in principle.

Category 3: A finite procedure exists, in principle, that can certify the correctness of a single outcome.

The categories reflect fundamental limits, not experimental convenience. Which category a theory occupies depends on how it treats state–space reachability, input specification, and the internal structure of its measurement process.

6.4 Concrete illustrations

These abstract categories can be grounded in familiar examples:

- **Category 1:** A hypothetical model in which a single laboratory experiment depends on infinitely many uncontrolled degrees of freedom. Because no starting point for a verification procedure can be defined, the question of correctness is out of scope.
- **Category 2:** Standard Copenhagen collapse with purely internal randomness. The theory allows outcomes, but without access to the random seed there is no finite checklist that could, in principle, guarantee that the result recorded in one run is the "correct" one.
- **Category 3:** A deterministic collapse model in which the governing equations and all required initial data are, in principle, finitely accessible. Here, a finite verification procedure for a single outcome exists.

6.5 From intuition to audit

The categories above are qualitative; in our full framework [**?**] they are made precise and tested via a finite checklist of mathematical conditions on a theory's structure. In this paper we apply that audit to major collapse and no–collapse interpretations, providing a comparative map of their epistemic complexity in the laboratory regime. The goal is to give physicists a neutral reference point for interpreting what their favourite theory can—and cannot—promise about the verifiability of the world revealed in a single experiment.

7 Measurement Π_2^0-Hardness

In this section we analyse when the certification decision problem rises to at least Turing degree $0''$ (Π_2^0-hardness) *within the FDI/FTE/ABET framework*. We keep the axiomatic mode in which (F) (Reachable, Lyapunov-bounded, Contractive basin) and (S) (Certification Standard) hold. Thus, the fact to be certified *exists* and can be stated as a definite mathematical property. The jump in hardness comes entirely from additional *unknowability conditions* that enforce a universal quantifier over an unbounded or uncontrolled space in the very formulation of the certification property.

7.1 General conditions for Π_2-hardness

Let $\mathcal{P}$ be a certification property of the form

$$\forall\, x \in X\ \exists\, t \in \mathbb{N}: \quad R(x,t)$$

where $R(x,t)$ is a decidable predicate in the inner variables and X is an *unknowable* or *uncontrollable* parameter space (as seen by the certifier at compile time). If X is infinite and rich enough to encode all instances of the classical TOT problem[2], then deciding $\mathcal{P}$ is Π_2^0-hard.

In our setting, $\mathcal{P}$ arises when:

- We quantify over all admissible values of an *unknown parameter* (initial state, environment policy, robustness budget, ...), not fixed in advance.
- The inner $\exists t$ asserts the existence of a finite bound (time-to-certify, ATRT halting step, decomposition size, ...) that works for that parameter value.

A single axiom failure never produces this $\forall\exists$ pattern for the core FDI/FTE/ABET decision problems on a fixed operational slice. It emerges only when certain *pairs* of assumptions are violated together in such a way that a universal quantifier over an *uncontrollable* space is unavoidable.

We distinguish three canonical *bundles* of assumptions whose joint failure leads to Π_2-hardness while preserving (F) and (S).

7.2 The three bundles

We formulate the bundles in abstract terms, without committing to any particular physical realisation.

Bundle A (Globalised state space with unknown initial state). Assumptions:

- (C) fails on the full admissible domain (state space is non-compact), but (F) and (S) hold.
- The *actual* initial state Θ_0 is *unknowable* at compile time.

[2] The TOT (totality) problem asks whether a given Turing machine halts on all inputs, and is complete for Π_2^0 under many-one reductions.

Consequence: The property "for all admissible initial states, there exists a time bound after which the trajectory remains in $\mathcal{B}$" has the quantifier form

$$\forall\, \Theta_0 \in \mathcal{M}\ \exists\, T\ \forall t \geq T:\ \Theta_t \in \mathcal{B},$$

which can encode TOT and is Π^0_2-hard.

Bundle B (UI-Traceability failure with unknown input policy). Assumptions:

- (C), (F), (S), $(PV) \equiv (EM1\text{–}EM3)$ all hold.
- PL-Traceability holds along any realised execution, but UI-Traceability fails: no open tick-0 guard partition isolates the realised branch.
- The environment's input policy π is uncontrollable and unknown at compile time, drawn from a large admissible class Γ.

Consequence: The robust-liveness statement

$$\forall\, \pi \in \Gamma\ \exists\, T\ \forall t \geq T:\ \Theta_t^{(\pi)} \in \mathcal{B}$$

is Π^0_2-hard in general, because UI failure prevents a uniform finite decomposition that works for all π.

Bundle C (PV failure with unknown robustness budget). Assumptions:

- (C), (F), (S) all hold.
- $(PV) \equiv (EM1\text{–}EM3)$ fails: the acceptance relation W_n is open and robust topologically, but no deterministic polytime verifier can decide membership uniformly to all tolerances.
- The required tolerance ε_k is adversarially chosen from a compact range, unknown at compile time.

Consequence: The uniform instrumentability property

$$\forall\, k\ \exists\, \Pi_k\ (\text{finite guarded program certifying tolerance } \varepsilon_k)$$

is Π^0_2-hard, since PV failure can be arranged to encode TOT into the need for non-recursive boundary information in Π_k.

7.3 Abstract examples

We now give three purely mathematical constructions, one for each bundle, to illustrate the quantifier structure. These are not intended as physically realisable setups; they serve only to make the complexity jump explicit.

Example A (Bundle A). Let $\mathcal{M} = \bigsqcup_{n\in\mathbb{N}} S_n$ be a disjoint union of compact components S_n with a metric making $\mathcal{M}$ non-compact. Let U be such that from each S_n there is a path into $\mathcal{B}$ in at most $t(n)$ steps. If $t(n)$ encodes the halting time of the n-th Turing machine on input n, then

$$\forall\, \Theta_0\ \exists\, T\ \forall t \geq T:\ \Theta_t \in \mathcal{B}$$

is equivalent to TOT, hence Π^0_2-hard.

Example B (Bundle B). Let $\mathcal{M} = [0,1]$, $\mathcal{Z}^{\text{in}} = [0,1]$, and $U(\theta, z) = f_\theta(z)$ where for each θ the preimage $f_\theta^{-1}(\mathcal{B})$ is a middle-third Cantor set of positive measure. Arrange these Cantor sets so that no finite open cover at tick 0 intersects all of them. Let Γ be the set of all Lipschitz input policies. Then the statement "for all $\pi \in \Gamma$ eventually in $\mathcal{B}$" is Π^0_2-hard.

Example C (Bundle C). Let $\mathcal{M} = [0,1]$, W_n be the union of open intervals whose endpoints encode the halting set of some Turing machine, with gaps shrinking like 2^{-k}. Let $\varepsilon_k = 2^{-k}$. Any finite guarded program Π_k certifying W_n to tolerance ε_k must decide halting for all inputs up to size k. Thus

$$\forall\, k\ \exists\, \Pi_k \text{ correct for } \varepsilon_k$$

is Π^0_2-hard.

7.4 Concrete physics example: Bundle A in cosmology

A notable instantiation of Bundle A occurs if we take as the "system" the *entire Universe* modelled in a theory whose full state space $\mathcal{M}_{\text{global}}$ is non-compact (e.g., unbounded energy spectrum, no uniform truncation on degrees of freedom). If the *actual* global initial state Θ_0 is unknowable to any finite experimenter, then any well-posed certification claim about eventual capture into a target basin $\mathcal{B}$ must be quantified as:

$$\forall\, \Theta_0 \in \mathcal{M}_{\text{global}}\ \exists\, T\ \forall t \geq T:\ \Theta_t \in \mathcal{B}.$$

Under mild richness assumptions on $\mathcal{M}_{\text{global}}$ this formulation is Π^0_2-hard, since Θ_0 can encode arbitrary Turing-machine instances and T can encode the halting times. This is independent of any specific dynamics; it follows solely from the (non-compactness, unknowability)-pair of conditions in Bundle A.

In physical collapse models posed on such a global state space, this complexity obstruction is *intrinsic*: even if the dynamics are otherwise well-behaved locally, the certification language for "eventual capture for all admissible global states" cannot be reduced below $0''$ without access to non-computable information about the actual Θ_0.

8 Epistemic Inaccessibility of Global Collapse

8.1 Two distinct claims

We distinguish:

1. **Out-of-scope claim (Category 1).** On the *global*, non-compact Hilbert space, Axiom (C) fails, hence ABET cannot be formed and the Simple Test cannot be constructed.
2. **Operational indistinguishability claim (in-scope).** On any *compact* laboratory slice satisfying (F), (S), (C), if the collapse dynamics agrees (exactly or within precision ε) with unitary on all observables accessible to finite guarded decompositions, then no finite guarded test can distinguish them.

Only the second claim is a *theorem of indistinguishability*; the first is a scope statement.

8.2 Out-of-scope theorem (global non-compact domain)

Theorem 8.1 (Global domain is out of FDI scope)**.** *Let $\mathcal{H}_{\text{global}}$ be a separable, infinite-dimensional Hilbert space and consider the full state space $\mathcal{D}(\mathcal{H}_{\text{global}})$ endowed with the trace-norm topology. If experiments are allowed to range over* all *finite-energy levels with no uniform cutoff, the accessible input set Z^{in} is non-compact. Consequently Axiom* (C) *(compact continuity) fails on that domain, ABET cannot be instantiated, and the Simple Test is undefined. In the taxonomy of Sec. 5, the* global *collapse vs. unitary question is Category 1.*

Proof. Without a uniform energy/localisation cutoff, Z^{in} contains sequences escaping any bounded set, so no finite open cover admits a finite subcover; this is the failure of compactness. The ABET construction and its finite subcover step require compactness of both the state and input slices under which U, E are continuous. Hence ABET cannot be formed and the Simple Test cannot be defined. □

8.3 Indistinguishability on compact laboratory slices

Setup. Let $\mathcal{H}_{\text{lab}} \subset \mathcal{H}_{\text{global}}$ be a finite-energy, finite-resolution laboratory slice such that the induced state space $\mathcal{M} := \mathcal{D}(\mathcal{H}_{\text{lab}})$ is compact, with (F), (S), (C) satisfied, and let instruments (update-and-measure primitives) obey (EM1–EM3). Write Instr for the class of finite instruments implementable by finite guarded decompositions over $\mathcal{M}$ (finite sequences of CPTP maps and POVMs with the book's effectivity bounds).

Definition 8.1 (Operational agreement on $\mathcal{M}$)**.** Two evolutions on $\mathcal{D}(\mathcal{H}_{\text{global}})$, $\mathcal{U}(\rho) = U\rho U^\dagger$ and a channel $\mathcal{C}$, are said to be *operationally ε-equivalent on $\mathcal{M}$* if for every instrument $\mathsf{I} \in \text{Instr}$ with outcome set Ω and every $\rho \in \mathcal{M}$,

$$\big\|\mathsf{I} \circ \mathcal{U}(\rho) \;-\; \mathsf{I} \circ \mathcal{C}(\rho)\big\|_{\text{TV}} \;\le\; \varepsilon,$$

where the left-hand side is total-variation distance between the induced outcome distributions on Ω.

Theorem 8.2 (Compact-access indistinguishability)**.** *Assume* (F), (S), (C) *on $\mathcal{M}$, and* (EM1–EM3), (I) *for finite guarded tests. If $\mathcal{U}$ and $\mathcal{C}$ are operationally ε-equivalent on $\mathcal{M}$, then no finite guarded decomposition can distinguish $\mathcal{U}$ from $\mathcal{C}$ with advantage exceeding ε. In particular, for $\varepsilon = 0$ they are indistinguishable by any finite guarded decomposition.*

Proof. Any finite guarded decomposition on $\mathcal{M}$ compiles to an instrument $\mathsf{I} \in \text{Instr}$ whose outcome distribution depends affinely and continuously on the input state and the composing CPTP maps/POVMs (the latter are fixed by the test). By assumption, for all $\rho \in \mathcal{M}$, $\|\mathsf{I} \circ \mathcal{U}(\rho) - \mathsf{I} \circ \mathcal{C}(\rho)\|_{\text{TV}} \le \varepsilon$. Hence the supremum distinguishing advantage of any test over $\mathcal{M}$ is at most ε. For $\varepsilon = 0$ all outcome distributions coincide, making the tests indistinguishable. □

Remark 8.1 (Sufficient structural condition)**.** A sufficient (and checkable) condition for $\varepsilon = 0$ is

$$\text{Tr}_{\text{env}}\big[\mathcal{U}(\rho \otimes \eta)\big] \;=\; \text{Tr}_{\text{env}}\big[\mathcal{C}(\rho \otimes \eta)\big] \quad \text{for all } \rho \in \mathcal{M} \text{ and all environment states } \eta$$

that can arise within the finite test horizon. Since every $\mathsf{I} \in \text{Instr}$ acts on $\mathcal{H}_{\text{lab}}$ and depends only on these reduced states, all outcome statistics coincide.

Corollary 8.1 (Finite-precision robustness)**.** *If operational equivalence holds up to $\varepsilon(\delta)$ for instruments whose POVM elements have operator-norm tolerance δ, then no δ-*precision *finite guarded test can distinguish beyond $\varepsilon(\delta)$.*

8.4 "God-observer" corollary and discussion

Corollary 8.2 (No superobserver with finite resources)**.** *Any observer—no matter how large—whose probes and detectors are limited to finite precision and finite energy implements instruments in* Instr *over some compact slice* $\mathcal{M}$*. If a putative global collapse map* $\mathcal{C}$ *agrees (exactly or within* ε*) with unitary on all such instruments, then even a "God-observer" cannot distinguish* $\mathcal{C}$ *from* $\mathcal{U}$ *beyond advantage* ε*.*

Remark 8.2 (Relation to ABET/FET/FDI)**.** Thm. 8.1 explains why *global* collapse is Category 1: ABET cannot even be posed on a non-compact domain. Thm. 8.2 is a distinct, in-scope statement: on compact slices where ABET and the Simple Test exist, indistinguishability follows from *marginal agreement* on the accessible observables, not merely from the global non-compactness.

9 Why this matters now

9.1 From formal limits to physical meaning

Most discussions about the limits of physical theories focus on the *accuracy* of their predictions: how closely they match experimental results, how far those results can be pushed by better instruments, and where new physics might emerge. Our concern here is different. We are interested in the *decidability* of statements about the world: whether a given theory can, even in principle, assign a definite truth value to a certain kind of physical claim.

To make this concrete, imagine a claim that has the form:

> "From *any* allowed initial state of the system, the dynamics will eventually carry it into a certain well-defined region, and once there, it will stay there forever."

Mathematically, this is an "eventual capture" property:

$$\forall\, \Theta_0 \in \mathcal{M}\ \exists T\ \forall t \geq T:\ \Theta_t \in \mathcal{B}.$$

Here $\mathcal{M}$ is the space of allowed states, Θ_t is the state at time t, and $\mathcal{B}$ is the "target" region. This form of statement appears in many guises: stability in control theory, safety in engineering, asymptotic convergence in algorithms, and irreversibility in thermodynamics. It also appears in fundamental physics whenever we speak of the system having settled into a certain macroscopic condition.

Our framework allows us to classify the *computational complexity* of such claims. In some cases they can be verified by a finite procedure—a bounded experiment or calculation. In others, they are provably beyond the reach of any such finite procedure. This is not a matter of technology or cleverness, but of the structure of the state space and the kind of quantifiers the claim involves.

The key point is that this is a *theoretical* limit: it applies to any experimenter, no matter how powerful, as long as they are bound by the basic constraints of finiteness—finite precision, finite resources, and a finite amount of accessible information.

9.2 When the state space is too large

One of the strongest obstructions arises when the underlying space of possible states is *non-compact* and the actual initial state is unknowable in full detail. Non-compactness means, loosely speaking, that the system admits configurations arbitrarily far apart in the relevant metric—for example, arbitrarily high energies, unbounded particle numbers, or unlimited spatial extent. Unknowability means that no experiment, however precise, can fully specify the starting point within that vast space.

In such a setting, the "eventual capture" statement above becomes a problem of the following logical shape:

$$\forall \text{ (possible initial states)} \quad \exists \text{ (time to capture)} \quad \forall \text{ (later times)} \ \ldots$$

This is a Π_2^0-type statement in the hierarchy of decision problems: it has a universal quantifier over an infinite set, followed by an existential quantifier, followed by another universal one. Problems of this form are famously hard: they can encode the question of whether an arbitrary computer program will run forever or eventually halt. In practical terms, there is no finite list of experiments or finite computation that can decide such a property for *all* possible initial states.

This is not just an abstract exercise in logic. If we treat the "system" as large as the entire Universe, described by a theory whose full state space is non-compact—as is the case in the standard infinite-dimensional Hilbert-space formalism of quantum mechanics—then the obstacle is *intrinsic*. It does not depend on the details of the dynamics. Even if the evolution law is perfectly well-behaved and local, the combination of (i) an unbounded state space and (ii) our inability to know the actual global state is enough to make certain natural, physically meaningful claims formally undecidable by any finite process.

9.3 Why this is relevant to laboratory physics

At first sight, such global undecidability might seem irrelevant to physics as we practise it. No laboratory experiment attempts to probe *all* possible states of the Universe; we work with carefully prepared, finite-energy subsystems. On these restricted "laboratory slices," the state space can be compact, the relevant quantities continuous, and the corresponding statements verifiable in finite time.

However, the connection between the global and laboratory levels matters for two reasons.

First, many interpretations of quantum mechanics, and many proposed modifications such as collapse models, are formulated *globally*. Their rules apply to the whole Universe at once, even if we only ever test small parts. If the global theory has built-in undecidable facts, then some questions about its behaviour can *never* be settled in principle, no matter how far we extend our experiments.

Second, the passage from the global state to what an experimenter can actually observe inevitably discards information. Our earlier results show that this "observer–observable split" is not just a philosophical move: it has a well-defined computational cost. In the worst case, it can raise the complexity of deciding a physical fact to the highest levels considered in computability theory. That means that even if the local, observable physics looks simple and well-behaved, the unobservable part of the state may hide complexity that prevents us from ever proving certain properties with finite resources.

This perspective reframes some long-standing foundational debates. Questions about whether collapse happens, or whether different interpretations are truly distinct, can have fundamentally different answers depending on whether we look at the laboratory slice (where finite verification may be possible) or the full global state (where it may not be).

9.4 Implications for theory choice

Recognising the role of these complexity barriers forces a sharper distinction between two kinds of theoretical differences.

Some differences live entirely within the *laboratory slice*. These affect predictions in ways that can, at least in principle, be detected by a finite sequence of experiments. Competing models in this category can be distinguished by sufficiently clever experimental design, even if the required precision or resources make the task challenging in practice.

Other differences exist only at the *global* level, where the state space is non-compact and the actual state unknowable. In this regime, complexity theory tells us that certain natural verification tasks become as hard as the most difficult problems in the arithmetical hierarchy that still admit definite truth values. Here, no conceivable finite experiment—no matter how powerful—can settle the question for all possible states. In such cases, models may remain *empirically indistinguishable forever*, even if they differ dramatically in their global description of reality.

For theory choice, this means that apparent empirical underdetermination may come in two very different flavours: one that can, in principle, be broken by future experiments, and one that is *in principle* unbreakable. The first type is a challenge to experimental ingenuity; the second is a challenge to how we weigh untestable global features when deciding between otherwise equivalent models.

By making these categories explicit, we can have a more honest conversation about what is at stake in foundational debates. Disagreements that fall into the second category are not just matters of experimental difficulty—they are separated by a provable barrier that no improvement in technology can cross.

9.5 From assumptions to consequences: a chain of reasoning

To make the discussion concrete, we first state the minimal assumptions about *classical reality* we adopt in this section:

1. **Existence of facts.** At any given time t, there exists at least one *fact*—a statement about the state of the world—that is objectively true, in the sense that its truth value is independent of any observer's belief or information.

2. **Ledger view.** These facts can be thought of as entries in a (possibly unbounded) ledger: a temporally ordered collection of statements, each of which can be individually referenced and re-stated without ambiguity.

3. **Finite-certifiable procedure.** Each entry in the ledger comes into existence through a procedure that is *finite-certifiable*: there exists, in principle, a finite operational test whose outcome would settle the fact's truth value. (The test may depend on context, but it halts in finite time whenever the fact is true.)

These three assumptions are intentionally modest: they do not specify what the facts *are*, nor what physical mechanism produces them; they only require that facts *exist*, are *recordable* in principle, and are *operationally certifiable*.

Step 1: Ledger existence imposes compactness. From the finite-certifiability requirement, the set of all possible states that could underlie ledger entries at a given time must be representable with a finite amount of information to any desired precision. Mathematically, this forces the underlying state space to be *compact*: without compactness, no uniform finite resolution would suffice to cover all admissible states, and the certification procedure could not be constructed.

Step 2: Ledger procedure implies information loss. Our results from Book I, Ch. 5 and Appendix B show that *any* finite-certifiable fact necessarily arises from an *information-losing* transformation of the underlying state. The separation between "observer" and "observed" is not a mere convenience: it is the structural reason that a fact can be stated at all. Lossless transformations of the underlying state make the certification problem Π^0_2-hard in the general case (reduction from TOTAL), rendering them operationally unreachable.

Step 3: Initial facts require restriction to a compact slice. If the "world" is modelled as a single evolving point in a non-compact state space (e.g., an infinite-dimensional Hilbert space with no uniform cutoff), then *no* fact can satisfy the above assumptions without first restricting attention to a compact subset and a concrete initial state within it. The unknowability of a global initial state in a non-compact space is exactly the *Bundle A* obstruction: it yields Π^0_2-hardness for universal certification claims.

Step 4: Persistent reality from local re-creation. Once an initial fact exists on a compact slice and the dynamics satisfy the usual regularity axioms (*finiteness*, *safety*, and *continuity*), the arrow of time ensures that new facts can be continually certified and appended to the ledger. The "classical reality" we experience is this ongoing, procedural re-creation of certifiable facts.

Step 5: Incompatibility with a purely unitary, infinite-dimensional global state. A theory that postulates an ontic, *globally* unitary evolution on a non-compact state space, with no mechanism for restriction to a compact, fact-certifiable slice, is thereby committed to the *absence* of facts in the above sense. If one adopts the assumption that classical reality *exists*, then such a theory is, by construction, *incomplete*: it lacks the structural ingredients needed to generate the facts that its own postulated reality demands.

References

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Appendix: Details of ATRT proof

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1 Adaptive Traceability Resolution Theorem: Algorithm and Lemmas

1.1 Promise Verification via Effective Measurability Lemma

Lemma 1.1 (Promise Verification via Effective Measurability)**.** *Let $\mathfrak{T}$ satisfy (F), (S), and (C) and (EM1)–(EM3). Fix $z \in \mathcal{Z}^{\text{in}}$, threshold $\varepsilon > 0$, and candidate gap $\gamma > 0$. Define*

$$f(\Theta) := \operatorname{dist}(U(\Theta, z), \mathcal{B}), \qquad f^* := \min_{\Theta \in \mathcal{M}} f(\Theta),$$

and let $\omega(\cdot)$ be the continuity modulus from Lemma 1.2. Then there exists an explicit adaptive procedure that maintains computable lower and upper bounds $L \le f^* \le U$ such that:

1. *U is obtained by sampling candidate states and taking $U := \min_{\Theta_R} f(\Theta_R)$ over representatives Θ_R;*
2. *L is obtained by region exclusion via continuity: for each region R with representative Θ_R, define $L_R := f(\Theta_R) - \omega(\operatorname{diam}(R))$, and set $L := \min_R L_R$;*
3. *The procedure terminates in finite time once one of the following holds:*
   - *$U \le \varepsilon$, certifying $f^* \le \varepsilon$;*
   - *$L \ge \varepsilon + \gamma$, certifying $f^* \ge \varepsilon + \gamma$;*
   - *$U - L < \gamma$, in which case $f^* \notin (\varepsilon, \varepsilon + \gamma)$ and the dichotomy "either $f^* \le \varepsilon$ or $f^* \ge \varepsilon + \gamma$" is numerically validated.*

In particular, whenever the promised gap condition holds, the procedure reaches one of the stopping criteria in finite time and thus does not rely on assuming the dichotomy a priori without evidence.

Proof. By (C) the space $\mathcal{M}$ is compact and U is continuous, so $f(\Theta) = \operatorname{dist}(U(\Theta, z), \mathcal{B})$ is continuous and attains its minimum f^*. We initialize a finite cover $\mathcal{R}_0$ of $\mathcal{M}$ by compact regions (for example, axis-aligned boxes) and associate to each active region R a representative point $\Theta_R \in R$. Using (EM1)–(EM3) and the known modulus ω, we carry out the following at each refinement iteration:

Upper bound U: For each representative Θ_R compute $f(\Theta_R)$ to sufficient precision; since $f^* \le f(\Theta_R)$, define $U := \min_R f(\Theta_R)$. This is a valid upper bound.

Lower bound L: For each region R, Lemma 1.2 gives that

$$\min_{\Theta \in R} f(\Theta) \geq f(\Theta_R) - \omega(\mathrm{diam}(R)) =: L_R.$$

Set $L := \min_R L_R$; then $L \leq f^*$. Thus L is a valid lower bound.

Refinement: For any region R that is not yet resolved (i.e., neither excluded by $L_R > \varepsilon + \gamma$, nor containing a witness with $f(\Theta_R) \leq \varepsilon$) subdivide it into smaller-diameter subregions, update their representatives, and recompute the corresponding f-values and local lower bounds. Because $\omega(\mathrm{diam}(R)) \to 0$ as $\mathrm{diam}(R) \to 0$, the gap $U - L$ strictly decreases under sufficient refinement unless already one of the stopping conditions is met.

Termination conditions:

- If $U \leq \varepsilon$, then there exists a sample Θ_R with $f(\Theta_R) \leq \varepsilon$, so $f^* \leq \varepsilon$.
- If $L \geq \varepsilon + \gamma$, then for all $\Theta \in \mathcal{M}$, $f(\Theta) \geq f^* \geq L \geq \varepsilon + \gamma$, so $f^* \geq \varepsilon + \gamma$.
- If $U - L < \gamma$ while neither of the above holds, then f^* cannot lie in the open interval $(\varepsilon, \varepsilon + \gamma)$, because that would force both $U > \varepsilon$ and $L < \varepsilon + \gamma$ with gap at least γ; the current enclosures $L \leq f^* \leq U$ contradict $f^* \in (\varepsilon, \varepsilon + \gamma)$. Hence the promise "either $f^* \leq \varepsilon$ or $f^* \geq \varepsilon + \gamma$" is validated numerically, and a definitive decision follows by testing whether $U \leq \varepsilon$ or $L \geq \varepsilon + \gamma$.

Because the bounds U and L converge to f^* as regions are refined (compactness + continuity + shrinking modulus), one of the termination conditions is reached in finite time whenever the dichotomy holds. Thus the procedure achieves the claimed promise verification without assuming it silently. ■ □

Corollary 1.1 (Gating ATRT on Verified Promise)**.** *Before executing the core dichotomous branch of ATRT (deciding traceable vs non-traceable with gap), run the procedure of Lemma 1.1. Proceed with the corresponding branch only once one of the certified conditions $U \leq \varepsilon$, $L \geq \varepsilon + \gamma$, or $U - L < \gamma$ is observed. This guarantees the gap assumption is either confirmed or refuted explicitly and is not used implicitly.*

1.2 SMT Encoding Templates for Region-Level Traceability Checks

Let $z \in \mathcal{Z}^{\mathrm{in}}$ be the input under test, and define

$$f(\Theta) \;:=\; \mathrm{dist}(U(\Theta, z), \mathcal{B})$$

with all computations interpreted to finite precision. Let $R \subseteq \mathcal{M}$ be a compact region (e.g., a box or cell) represented by constraints $C_R(\Theta)$.

Existential (traceability) check over region R: We encode the formula

$$\exists\, \Theta \in R : \; f(\Theta) \leq \varepsilon$$

as an SMT/decision problem. A canonical encoding (assuming U and $\mathcal{B}$ are expressible in the logic) is:

$$C_R(\Theta) \;\wedge\; \mathrm{EncodeDistance}(U(\Theta, z), \mathcal{B}) \leq \varepsilon,$$

where $\mathrm{EncodeDistance}(\cdot, \cdot)$ is a formula representing the (approximate) distance of $U(\Theta, z)$ into $\mathcal{B}$, or equivalently membership up to tolerance. A satisfying assignment Θ^* is a witness; it certifies z is traceable via Θ^*.

Universal (non-traceability) exclusion check over region R: We encode the negation as

$$\forall\ \Theta \in R:\ \ f(\Theta) > \varepsilon,$$

which is treated in SMT by proving unsatisfiability of the negated existential:

$$\neg\big(\exists\ \Theta \in R:\ \ f(\Theta) \le \varepsilon\big),$$

i.e., the same formula as above is refuted. In practice, for robustness and efficiency, we strengthen by deriving a *lower bound* L_R on f over R using modulus-of-continuity and samples and then encode the proof obligation:

$$C_R(\Theta)\ \wedge\ f(\Theta) < L_R \quad \text{is unsatisfiable, with } L_R > \varepsilon.$$

Alternatively, one can encode a symbolic certificate (e.g., via interval arithmetic or invariants) asserting $\min_{\Theta \in R} f(\Theta) \ge L_R > \varepsilon$.

Implementation notes:

- If U and $\mathcal{B}$ are given by rational arithmetic circuits (EM1), their composition and distance computations can be encoded directly in SMT over reals with rational approximation.
- The modulus of continuity from (EM2) supplies bounds to convert pointwise samples into sound lower/upper bounds over R; these are embedded as side constraints to support universal proofs.
- Use incremental solving: cache previous region proofs to avoid redundant work in refined subdivisions.
- If the solver returns `sat` for the existential check, extract Θ^*; if it proves the existential unsatisfiable (or verifies the lower bound certificate), mark R excluded.

1.3 Adaptive Traceability Resolution Algorithm

Remark 1.1 (Refinement Fairness)**.** The algorithm assumes a *fair refinement scheduler*: at each iteration every active region that is not yet excluded is either resolved or subdivided further (e.g., a standard "refine-all-unresolved" policy). The existence of such a refinement is proved in Refinement Extension of Guarded Decompositions Theorem (Thm.-2.1) in Chapter 2.

Combined with the interior stability margins from Lemma 2.4 (Chapter 2), this guarantees that any region containing a true traceability witness will eventually be refined to diameter $\le d_{\text{target}}$ and exposed. Note that this spatial refinement fairness is logically distinct from axiom (I) (TI/WF), which governs progression along execution trajectories; the current remark controls region-level search, not the dynamic choice of inputs.

Algorithm 1 Adaptive Traceability Resolution for input z with gap $\gamma > 0$

Require: Input z, certification threshold $\varepsilon > 0$, known gap $\gamma > 0$ such that $f^* \leq \varepsilon$ or $f^* \geq \varepsilon + \gamma$, initial coarse cover $\mathcal{R}_0$ of $\mathcal{M}$, basin $\mathcal{B}$, modulus of continuity $\omega(\cdot)$ from Lemma 1.2

Ensure: Either: `TRACEABLE` with witness Θ^*, or `NON-TRACEABLE` with gap certificate

Compute target refinement diameter d_{target} such that $\omega(d_{\text{target}}) < \gamma/2$
$\mathcal{R} \leftarrow \mathcal{R}_0$
while true **do**
  for each region $R \in \mathcal{R}$ **do**
    Pick representative $\Theta_0 \in R$; compute $f(\Theta_0) = \text{dist}(U(\Theta_0, z), \mathcal{B})$ to required precision via (EM2)/(EM3)
    if $f(\Theta_0) \leq \varepsilon + \frac{\gamma}{2}$ **then**
      (Existential success): run existential region check (SMT or direct witness extraction) to produce $\Theta^* \in R$ with $f(\Theta^*) \leq \varepsilon + \frac{\gamma}{2}$
      return `TRACEABLE`, witness Θ^*
    end if
    Compute lower bound over R: $L_R := f(\Theta_0) - \omega(\text{diam}(R))$
    if $L_R > \varepsilon + \frac{\gamma}{2}$ **then**
      (Universal exclusion): mark R as excluded
    end if
  end for
  if all regions in $\mathcal{R}$ are excluded **then**
    return `NON-TRACEABLE` with certificate $\min_\Theta f(\Theta) \geq \varepsilon + \frac{\gamma}{2}$
  end if
  Refine: Replace each non-excluded region R with a subdivision of smaller-diameter subregions so that eventually $\text{diam}(R) \leq d_{\text{target}}$
  Update $\mathcal{R}$ to the new cover
end while

Notes:

- The existential check is lightweight when $f(\Theta_0)$ already certifiably lies within the acceptance band $\varepsilon + \gamma/2$; a localized search in R suffices to produce Θ^*.
- Universal exclusion uses the modulus-of-continuity to lift a sample value to a guaranteed lower bound over the whole region.
- The guarded program provides the logical composition of these region checks: successful existential proofs propagate as accepted branches; excluded regions prune whole subtrees.
- Refinement continues deterministically until either a traceability witness is extracted or every region is excluded; $\omega(d_{\text{target}}) < \gamma/2$ ensures the dichotomy resolves definitively.
- All arithmetic, distance, and membership evaluations are carried out with precision controlled according to (EM2) and (EM3), making the bounds sound.
- Termination and completeness of ATRT rely only on compactness, the gap γ, the modulus ω, and refinement fairness (Remark 1.1); no temporal-input fairness (Axiom (I), TI/WF) is assumed or required.

1.4 Continuity-based Lower Bound Machinery and Invariants

Under (C) and (EM2), U is continuous and effectively computable with a known modulus of continuity. Define

$$f(\Theta) := \mathrm{dist}(U(\Theta, z), \mathcal{B}).$$

Since U is continuous on compact $\mathcal{M}$ and $\mathcal{B}$ is closed (as a basin shell limit), f is continuous and attains its minimum.

Lemma 1.2 (Effective Local Variation Bound)**.** *Suppose (EM2) supplies a polynomial modulus function $m(k)$ for computing U to k bits. Then there exists a computable function $\omega : \mathbb{R}_+ \to \mathbb{R}_+$ (derivable from m and the Lipschitz behavior induced by finite precision) such that for any $\Theta, \Theta' \in \mathcal{M}$,*

$$|f(\Theta) - f(\Theta')| \leq \omega\big(\mathrm{dist}(\Theta, \Theta')\big).$$

Furthermore, ω can be chosen to satisfy $\omega(r) \to 0$ as $r \to 0$, and ω is computable in time $\mathrm{poly}(n, \log(1/r))$.

Proof. By (EM2a), each primitive map F_j (either an update U_j or an emit E_j) comes with an explicit, effectively computable modulus of continuity $\omega_j : \mathbb{Q}_+ \to \mathbb{Q}_+$ satisfying $\omega_j(0) = 0$ and

$$\|x - y\| \leq r \quad \Rightarrow \quad \|F_j(x) - F_j(y)\| \leq \omega_j(r).$$

Since $f(\Theta) = \mathrm{dist}(U(\Theta, z), \mathcal{B})$ is obtained by composing finitely many such primitives with each other and with the distance-to-set map, the standard composition rules for moduli of continuity yield an explicit modulus ω for f: for all $\Theta, \Theta' \in \mathcal{M}$,

$$\mathrm{dist}(\Theta, \Theta') \leq r \quad \Rightarrow \quad |f(\Theta) - f(\Theta')| \leq \omega(r).$$

To make this explicit, note that $U(\cdot, z)$ is the composition of primitive updates along a fixed finite path determined by z, so if $\mathrm{dist}(\Theta, \Theta') \leq r$ then

$$\|U(\Theta, z) - U(\Theta', z)\| \leq \omega_U(r)$$

where ω_U is obtained by chaining the ω_j of the primitives in the path. The distance function $\mathrm{dist}(\cdot, \mathcal{B})$ is 1-Lipschitz, hence

$$|f(\Theta) - f(\Theta')| = \big|\,\mathrm{dist}(U(\Theta, z), \mathcal{B}) - \mathrm{dist}(U(\Theta', z), \mathcal{B})\big| \leq \|U(\Theta, z) - U(\Theta', z)\| \leq \omega_U(r).$$

Thus we may take $\omega := \omega_U$ as the desired effective modulus for f.

Finally, by (EM2b), evaluating each primitive F_j to k bits of precision costs at most $\mathrm{poly}(k + n)$ steps, where n encodes the instance size. Since the composition depth is fixed, $f(\Theta)$ can be evaluated to any accuracy prescribed by ω within polynomial time in $k + n$. This completes the proof. □

Corollary 1.2 (Region Lower Bound)**.** *Let $R \subseteq \mathcal{M}$ be a region of diameter* $\mathrm{diam}(R)$*, and let $\Theta_0 \in R$ be a representative sample with computed value $f(\Theta_0)$. Then for all $\Theta \in R$,*

$$f(\Theta) \geq f(\Theta_0) - \omega(\mathrm{diam}(R))\,.$$

Thus if $f(\Theta_0) - \omega(\mathrm{diam}(R)) > \varepsilon$, we can safely exclude R for the traceability check at precision ε.

2 Appendix to Chapter 5: Deeper analysis of selected theories

How to read the model audits (template)

For each theory we check, in order:

1. **In-scope check (ABET availability).** On the laboratory slice Z^{in} with finite energy/localisation bounds: do (F) Invariant Basin, (S) Certification Standard, and (C) Compact Continuity hold? If not, *Category 1* (FDI out of scope).
2. **Effectivity & structure.** Verify (T) Traceability (open preimages for basin shells), (EM1)–(EM3) (effective primitives), and (I) (no supertasks).
3. **Guard construction.** If a *finite* family of guard cells $G_1, \ldots, G_m \subseteq Z^{\text{in}}$ covers the in-scope slice with per-guard correctness proofs, the fact is FDI-decomposable $\Rightarrow$ *Category 3*; else *Category 2*.

We also flag a *Traceability Caveat*: if a theory's latent variables (e.g., a micro-index) are *not* operational inputs, (T) fails and the verdict can drop from Category 3 to Category 2.

2.1 Superdeterministic CAI is *FDI–decomposable*

Let Σ be the finite alphabet for a single lattice cell and $L \gg 1$ the (finite) number of cells. A micro–state is therefore

$$s \in \Sigma^L, \qquad N := |\Sigma|^L < \infty.$$

The global update $F\colon \Sigma^L \to \Sigma^L$ is *reversible*, so it is a permutation of $\{0, \ldots, N-1\}$. We encode each s by the binary index $i(s) \in \{0, \ldots, N-1\}$ written in $B = \lceil \log_2 N \rceil = O(L)$ bits.

Input representation and compact continuity (Gnosis+Structure Axioms). The physically accessible slice $Z^{\text{in}} := \{0, \ldots, N-1\}$ is *finite*, hence compact and discrete in the product topology used by the FDI framework; guards may treat micro–indices as ordinary binary strings.

Deterministic outcome map. Fix an apparatus description and time budget $T > 0$. The pointer observable is a computable function

$$M_T : i(s_0) \longmapsto a_k \in \{e_1, \ldots, e_m\},$$

obtained by iterating the automaton $s_t = F^t(s_0)$ together with the detector's finite control logic for $t \leq T$.

Finite guarded decomposition. Define *guard sets*

$$G_k := M_T^{-1}(e_k) \subseteq \{0, \ldots, N-1\}.$$

Because $N < \infty$, the family $\{G_k\}_{k=1}^m$ is a finite partition of Z^{in}; the *basin* for outcome e_k is just G_k itself. Membership in G_k can be tested in $O(T + |s_0|)$ time by simulating F for T steps and checking the pointer, hence satisfies the Polynomial Guard/Basin Test axiom EM3.

Certificate and verifier. A succinct witness for the tail fact Φ_k: "*the pointer shows* e_k" is the binary string

$$w = i(s_0) \quad \text{of length } B = O(L).$$

Verification algorithm:

1. Given $(\psi, \text{detector initial macro}, w)$ convert w to the micro–state s_0 by table lookup.
2. Simulate F for T steps (reversible CA $\Rightarrow$ each step is local and executes in time $O(L)$).
3. Accept iff the simulated pointer equals e_k.

Total runtime is polynomial in L; hence $L_{\Phi_k} \in \mathrm{NP}$ (indeed in P^F if F is provided as an oracle).

Satisfaction of all FDI axioms.

- **Gnosis+Structure Axioms & EM1–EM3**: Hold because Z^{in} is finite and F and M_T are explicit, reversible cellular rules.
- **Time Irreversibility (I):** No stochastic branching occurs; simulation halts after T deterministic steps, the axiom is trivially satisfied.

FDI conclusion. Since all axioms are met *and* the guard partition $\{G_k\}$ is finite, the tail fact Φ_k is *decomposable*. Consequently the certification language sits inside the ordinary polynomial hierarchy rather than at oracle degree $0'$. This places the superdeterministic CA Interpretation firmly in **Category 3** as defined in Sec. **??**.

Traceability caveat (T). The decomposition above treats the micro-index $i(s_0)$ as an *operational input*. If the CAI posits that $i(s_0)$ exists but cannot be accessed (or approximated by an open set of controls) at tick 0, then (T) fails on Z^{in} and no finite guard family exists. In that case the same dynamics yields *Category 2*, not 3.

2.2 Certifiability of Convex–Quasi–Linear (CQL) Deterministic Collapse

Throughout this section we work in a finite–dimensional truncation $H \cong \mathbb{C}^d$ that captures a laboratory's finite–energy, finite–particle sector; hence the trace–norm unit ball $\mathcal{D}_d = \{\rho \geq 0, \operatorname{Tr}\rho = 1\}$ is compact.

2.2.1 The Kraus–normalised flow

Let $\{\Lambda_t\}_{t\geq 0}$ be a *linear*, completely–positive semigroup $\Lambda_t(\rho) = \sum_\alpha L_\alpha(t)\rho L_\alpha^\dagger(t)$ with rational Kraus coefficients, not necessarily trace preserving. Define the *convex–quasi–linear* flow

$$\Phi_t(\rho) := \frac{\Lambda_t(\rho)}{\operatorname{Tr}\Lambda_t(\rho)}, \qquad 0 \leq t < \infty.$$

We assume the **collapse condition**

$$\exists T > 0\ \forall \rho \in \mathcal{D}_d : \quad \Phi_T(\rho) \in \{\Pi_1, \ldots, \Pi_m\}, \tag{CC}$$

where $\{\Pi_k\}$ are one–dimensional projectors onto an orthonormal pointer basis. (Examples with this property are constructed in [?].)

The map Φ_T is *globally Lipschitz*:

$$\|\Phi_T(\rho) - \Phi_T(\sigma)\|_1 \;\le\; L\,\|\rho - \sigma\|_1, \qquad L := \frac{\|\Lambda_T\|_{1\to 1}}{\lambda_{\min}},$$

where $\lambda_{\min} := \min_\rho \operatorname{Tr}\Lambda_T(\rho) > 0$ because Λ_T is CP and finite dimensional.

Uniform positivity (well-defined normalisation). Assume there exists $c > 0$ such that $\operatorname{Tr}\Lambda_T(\rho) \ge c$ for all $\rho \in \mathcal{D}_d$. (This follows, e.g., if Λ_T is strictly positive on $\mathcal{D}_d$ or under the collapse condition (CC) together with continuity.) Then Φ_T is well-defined and globally Lipschitz on $\mathcal{D}_d$:

$$\|\Phi_T(\rho) - \Phi_T(\sigma)\|_1 \;\le\; \frac{\|\Lambda_T\|_{1\to 1}}{c}\,\|\rho - \sigma\|_1 \;=:\; L\,\|\rho - \sigma\|_1.$$

Finite guarded decomposition (compact cover). Each basin $B_k = \Phi_T^{-1}(\{\Pi_k\})$ is closed in the compact $\mathcal{D}_d$, hence compact. For any $0 < \varepsilon < \frac{1}{2L}$, the open balls

$$U_\rho \;:=\; \big\{\sigma \in \mathcal{D}_d : \|\sigma - \rho\|_1 < \varepsilon\big\} \quad (\rho \in B_k)$$

satisfy $\Phi_T(U_\rho) \subset B_{1/2}(\Pi_k)$ and therefore decide the pointer as e_k. By compactness of B_k, extract a finite subcover $U_{\rho_{k,1}}, \dots, U_{\rho_{k,r_k}}$. Taking the finite union over $k = 1, \dots, m$ yields a finite guard family that covers $\mathcal{D}_d$.

2.2.2 Polynomial certificate scheme

Witness. A certificate for the fact Φ_k : \pointer reads e_k" is the triple

$$w = \big(k,\ \rho_c,\ r\big),$$

where $k \in \{1, \dots, m\}$, $\rho_c \in \mathcal{D}_d(\mathbb{Q}[i])$ has rational entries, and $r \in \mathbb{Q}_{>0}$ obeys $r < \varepsilon_k$ with $\|\rho_0 - \rho_c\|_1 < r$. The bit–length of w is $O(d^2 \log 1/r)$.

Polynomial verification. Given a guard label (k, j) and a rational center $\rho_{k,j}$, the verifier checks $\|\rho_0 - \rho_{k,j}\|_1 < \varepsilon$ and computes $\Phi_T(\rho_{k,j})$ with rational Kraus matrices (EM1), in time polynomial in d and the bit-lengths, then accepts iff $\|\Phi_T(\rho_{k,j}) - \Pi_k\|_1 < \frac{1}{2}$. Hence $L_{\Phi_k} \in \mathbf{NP}$ (and in QCMA with a BQP verifier).

2.2.3 Fragility: a single essential discontinuity drops to Category 1

If the flow loses global Lipschitz continuity—e.g. there exists $\rho^\star$ where $\Lambda_T(\rho^\star)$ becomes rank-deficient so that $\lambda_{\min} = 0$—then (i) guard balls can no longer be made uniform in radius and (ii) the inverse image of a projector need not be closed. Consequently:

- Representation axiom **(C)** (compactness under the norm topology) fails at $\rho^\star$.
- EM3 is violated because no finite sub–cover can guarantee the map to be constant on each guard.

The theory therefore exits the FDI domain altogether, falling to **Category 1** despite being deterministic. Hence *small analytical pathologies in the CP semigroup can erase the certifiability advantage.* This illustrates the delicate balance required for deterministic collapse models to outperform Copenhagen in epistemic value.

2.3 Finite-World Many-Interacting-Worlds (MIW)

We analyse the *finite-N* variant of the Hall–Deckert–Wiseman Many-Interacting-Worlds model, where the universal configuration is a deterministic swarm of $N < \infty$ classical "world points" $q^{(w)}(t) \in \mathbb{R}^{3K}$ ($w = 1, \ldots, N$ for K particles). The ordinary Schrödinger predictions are recovered for large but fixed N by suitably choosing the inter-world potential $V_{N,\gamma}(q^{(1)}, \ldots, q^{(N)})$ [**?**].

State representation and compact input slice (Gnosis+Structure Axioms). Assume each particle's coordinate domain is a bounded box $[-L, L]^3$ and kinetic energy per particle is $\leq E_{\max}$. Then the phase-space cube

$$Z^{\text{in}} := \Big([-L, L]^3 \times [-P, P]^3\Big)^{KN}, \qquad P := \sqrt{2mE_{\max}},$$

is compact and has a straightforward rational-ball naming, satisfying **Gnosis+Structure Axioms**. A micro-state is the tuple $(q^{(1)}, p^{(1)}, \ldots, q^{(N)}, p^{(N)})$ with rational coordinates.

Deterministic dynamics. Time evolution follows a Lipschitz ODE system

$$m\ddot{q}^{(w)} = -\partial_{q^{(w)}}\big[\, V_{\text{ext}}(q^{(w)}) + V_{N,\gamma}(q^{(1)}, \ldots, q^{(N)})\big] \quad (w = 1, \ldots, N),$$

whose right-hand side is polynomial if V_{ext} and $V_{N,\gamma}$ use rational parameters. Thus a rational-time step Δt can be simulated with polynomial precision and time, meeting **EM1** (effectively computable flow).

Pointer observable and world-index guard sets. Suppose the apparatus records a discrete pointer value e_k when a configuration $q^{(w)}(T)$ falls inside a rational detector cell $D_k \subset [-L, L]^{3K}$. Because trajectories do not cross, each initial world index w produces *exactly one* pointer value. Define

$$G_k := \big\{\, w \in \{1, \ldots, N\} \,:\, q^{(w)}(T) \in D_k \big\}.$$

The family $\{G_k\}_{k=1}^m$ is a *finite* partition of the index set, and membership $w \in G_k$ is decidable in $\text{poly}(K, N)$ time by integrating the ODE for world w alone (**EM3**).

Certificate and verifier. A witness for the fact "pointer reads e_k" is

$$w \;=\; \big(k,\ \hat{w} \in G_k,\ (q_0^{(\hat{w})}, p_0^{(\hat{w})})\big),$$

where $(q_0^{(\hat{w})}, p_0^{(\hat{w})})$ are the rational initial coordinates of world $\hat{w}$. Verification algorithm:

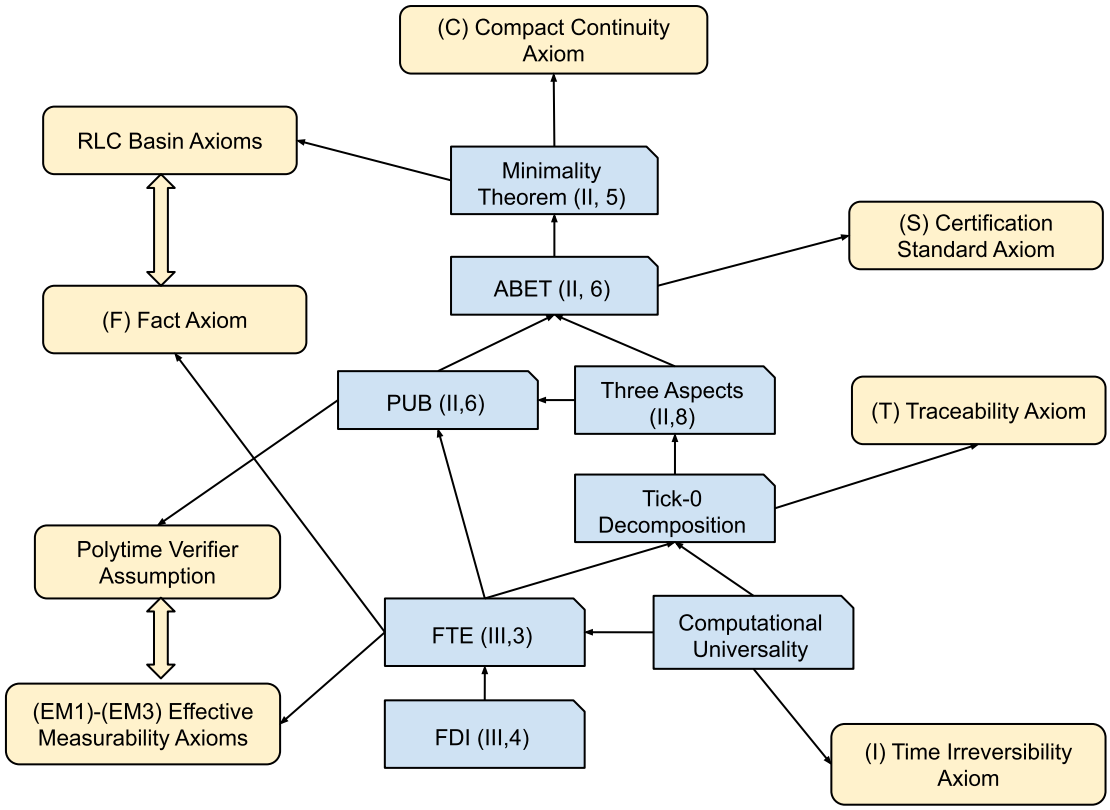
1. Re-integrate the ODE for world $\hat{w}$ up to time T with rational arithmetic.
2. Accept iff the resulting $q^{(\hat{w})}(T) \in D_k$.

Both steps run in time $\text{poly}(K, N, \log 1/\epsilon)$, so $L_{\Phi_k} \in \text{NP}$ and, by the standard Kitaev embedding, $L_{\Phi_k} \in \text{QCMA}$ when a BQP verifier is allowed.

FDI verdict. All Gnosis+Structure Axioms, EM1–EM3, and (I) are satisfied and a *finite guard* family exists. Hence every single-outcome fact is **FDI-decomposable**; the theory belongs to *Category 3*—epistemically stronger than Copenhagen.

Traceability caveat (T). The guard sets above use the world index w as an operational selector. If the finite-N MIW treats w as a hidden typicality parameter unavailable to the agent at tick 0, then (T) fails (no open preimage in the agent's input topology), and the theory drops to *Category 2* despite determinism.

Continuum-world limit drops to Category 2. If one lets $N \to \infty$ so that world indices form a continuum, the guard sets G_k become uncountable; no finite decomposition is possible and the model reverts to **Category 2**. Thus *finiteness of N is crucial for certifiability.*



1 Full list of axioms in the work, including equivalences

We are keeping in mind our axiom equivalences: $(F) \iff (RLCBasin)$ and $(PV) \iff (EM1 - EM3)$

1.1 The Belief-State: A Formal Representation of Knowledge

In our framework for Type B certification, we shift perspective from a raw stream of data to the metrologist's evolving state of knowledge. We encapsulate all non-statistical information available at a given time t into a single mathematical object.

Definition 1.1 (Belief-State)**.** A **belief-state**, denoted Θ_t, is an element of a space $\mathcal{M}$ that represents the complete summary of all non-statistical (Type B) information at time t.

To ensure this object is amenable to finite certification, we impose a crucial structural requirement on the space it inhabits.

Definition 1.2 (Belief-State Space)**.** The belief-state space, $\mathcal{M}$, is a **finite-dimensional, compact** metric space, typically a compact subset of $\mathbb{R}^d$ for some integer d. The space of input measurements, $\mathcal{Z}^{\text{in}}$, and the space of output measurements, $\mathcal{Z}^{\text{out}}$, are also assumed to be compact.

1.1.1 The Deterministic Two-Step Cycle

Our model assumes that all new randomness enters the system exclusively through an observable measurement z_{t+1}. The evolution of the belief-state is then governed by a deterministic two-step cycle.

Definition 1.3 (Emit and Update Maps)**.** The evolution of the belief-state is defined by two deterministic maps:

1. The **Emit Map**, $E : \mathcal{M} \to \mathcal{Z}^{\text{out}}$, which deterministically produces the system's "ideal" or predicted output $z_{t+1}^{\text{out}} = E(\Theta_t)$ based on the current belief-state Θ_t. We assume E is Lipschitz continuous.
2. The **Update Map**, $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$, which deterministically assimilates the newly observed input measurement z_{t+1}^{in} to produce the next belief-state: $\Theta_{t+1} = U(\Theta_t, z_{t+1}^{\text{in}})$. We assume U is also Lipschitz continuous.

1.2 Gnosis Axioms

(F) **Invariant basin.** There exists a reachable, invariant attractor basin $\mathcal{B} \subseteq \mathcal{M}$ (Deff. 2.1,2.2,2.3 from Book II, Chapter 5). *(There is a fact that remain)*

(S) **Certification standard.** $\exists \mathcal{S} = \{\varepsilon_n\}_{n=1}^{\infty} : \forall \varepsilon_n \in (0,1]\ \exists\ verifier\ V : V(z_{1:n}^{\text{out}}, w) = accept \implies \forall z : d(z, z_{1:n}^{\text{out}}, w)) < \varepsilon_n\ V(z) = accept$ (Def. 2.2 from Book II, Chapter 6). *(The statement can be made about the fact)*

1.2.1 Axioms for a Usable Attractor Basin (RLC Basin Axioms)

For an attractor basin $\mathcal{B}$ to be useful for certification, it must be reachable in a bounded time and it must be invariant (once entered, it is never left). We begin by formalizing these properties as axioms.

Definition 1.4 (Pointwise Reachability (R_{weak}))**.** A closed set $\mathcal{B} \subseteq \mathcal{M}$ satisfies **($\mathbf{R}_{\text{pt}}$)** if for every initial $\Theta_0 \in \mathcal{M}$ and every admissible input stream $(z_1, z_2, \dots)$ there exists a (possibly instance–dependent) $T \geq 0$ such that $U^T(\Theta_0; z_{1:T}) \in \mathcal{B}$.

Definition 1.5 (Axiom L: Lyapunov Border Condition)**.** A closed set $\mathcal{B} = \{\theta \in \mathcal{M} \mid V(\theta) \leq r\}$ defined by a continuous function $V : \mathcal{M} \to [0, \infty)$ and a constant $r > 0$ satisfies **Axiom L** if every boundary point is mapped strictly inside the set on the next update. Formally:

$$\forall \Theta \in \mathcal{M} \text{ with } V(\Theta) = r, \quad \forall z^{\text{in}} \in \mathcal{Z}^{\text{in}} : \quad V(U(\Theta, z^{\text{in}})) < r$$

This condition prevents trajectories from "touching" the boundary of the basin and immediately leaving. We now define the final property of interest.

Definition 1.6 (Axiom C_{basin}: Local Contraction on $\mathcal{B}$)**.** The update map U satisfies **Axiom C** on a set $\mathcal{B}$ if there exists a constant $\rho \in [0, 1)$ such that for all $\Theta, \Theta' \in \mathcal{B}$ and all $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$:

$$d(U(\Theta, z^{\text{in}}), U(\Theta', z^{\text{in}})) \leq \rho \cdot d(\Theta, \Theta')$$

1.2.2 Certification Standard

Definition 1.7 (Certification Standard)**.** A **Certification Standard**, denoted $\mathcal{S}$, is a family of guard-band radii $\{\varepsilon_n\}_{n=1}^{\infty}$, where each $\varepsilon_n \in (0, 1]$ if an **acceptance by the verifier V on output $(z_{1:n}^{\text{out}}, w)$ under $\mathcal{S}$** makes V also accepting on the entire ε_n-ball around $(z_{1:n}^{\text{out}}, w)$ in the **product metric** on

$$\mathcal{Z}^{\text{out n}} \times \mathcal{W}$$

built from:

1. **The output-space metric** a metric $d_{\mathcal{Z}^{\text{out}}}$ on $\mathcal{Z}^{\text{out}}$ built from a usual metric on a compact metric space $\mathcal{Z}^{\text{out}}$ as a standard **sup** (or l_∞) metric on length-n tuples:

$$d_{\mathcal{Z}^n}\big((z_1, \dots, z_n), (z_1', \dots, z_n')\big) = \max_{1 \leq i \leq n} d_{\mathcal{Z}^{\text{out}}}(z_i, z_i').$$

2. **The witness-space metric** built from witnesses as finite bit-strings of bounded (polynomial) length, which equips $\mathcal{W}$ with the **discrete metric**

$$d_{\mathcal{W}}(w, w') = \begin{cases} 0, & w = w', \\ 1, & w \neq w'. \end{cases}$$

3. **As the product metric** on $\mathcal{Z}^{\text{out n}} \times \mathcal{W}$ as the maximum of the two:

$$d\big((z_{1:n}, w), (z'_{1:n}, w')\big) = \max\Big\{ d_{\mathcal{Z}^n}(z_{1:n}, z'_{1:n}),\ d_{\mathcal{W}}(w, w') \Big\}.$$

so a set $W_n \subseteq \mathcal{Z}^{\text{out n}} \times \mathcal{W}$ is **open** in this metric precisely if, whenever $(z^*, w^*) \in W_n$, there is an $\varepsilon > 0$ so that

$$d\big((z_{1:n}, w), (z^*, w^*)\big) < \varepsilon \implies (z_{1:n}, w) \in W_n.$$

and, therefore, for each n, the acceptance relation

$$W_n = \{(z_{1:n}, w) \mid V(z_{1:n}, w) = \text{accept}\}$$

is open in $\mathcal{Z}^{\text{out n}} \times \mathcal{W}$. Such an acceptance relation is said to be **robustly certified by** $\mathcal{S}$.

1.3 Traceability

Axiom 1.1 (Input-Traceability / Open-Preimage)**.** *Let $\mathfrak{T} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E, \mathcal{S})$ be a valid test (Def. 2.1), and let $\mathcal{B}_n = \{\Theta : \text{dist}(\Theta, \mathcal{B}) \le \varepsilon_n\}$ be the n-th shell induced by the (minimal) Certification Standard $\mathcal{S}$. We postulate:*

$$\forall\, n \ge 1,\ \forall\, \Theta \in \mathcal{M} : \quad U(\Theta, \cdot)^{-1}(\mathcal{B}_n) \text{ is open and nonempty in } \mathcal{Z}^{\text{in}}.$$

Interpretation. This axiom ensures that for every certification shell and every state: (i) there *exists* at least one admissible input that drives the next state into the shell (nonemptiness = existential controllability), and (ii) admissibility is robust to small perturbations of that input (openness = guard-band margin in input space). It is precisely what allows us to *trace back* the certification constraint on state space to concrete, well-behaved input conditions—hence the name "Input-Traceability."

Two Variants of Traceability Assumptions

We now distinguish two variants explicitly:

Definition 1.8 (PL-Traceability / Path-Local Input-Traceability)**.** For each fixed $\Theta \in \mathcal{M}$ and shell index n, the preimage $U(\Theta, \cdot)^{-1}(\mathcal{B}_n)$ is nonempty and open. This is the minimal input-traceability assumption used to certify *realized* execution paths locally.

Definition 1.9 (UI-Traceability / Universal Input-Traceability)**.** PL-Traceability holds, and additionally

$$\bigcup_{\Theta \in \mathcal{M}} U(\Theta, \cdot)^{-1}(\mathcal{B}_n) = \mathcal{Z}^{\text{in}},$$

i.e., every input lies in some open preimage of the basin. Under this stronger assumption one can extract a global finite guarded partition of $\mathcal{Z}^{\text{in}}$ at tick 0.

1.4 Time Irreversibility

Axiom 1.2 (Time-Irreversibility / Weak-Fairness (TI/WF))**.** *Let $f : \mathcal{M} \to \mathbb{N}$ be a well-founded ranking function, and define the enabled-decrease set at time t by*

$$D_t \ = \ \{\, z \in G^{\text{in}}(\Theta_t) \mid f(U(\Theta_t, z)) < f(\Theta_t) \,\}.$$

An infinite execution $(\Theta_0, z_0, \Theta_1, z_1, \dots)$ is called fair *if whenever $D_t \neq \varnothing$ for infinitely many t (i.e., the set $\{t : D_t \neq \varnothing\}$ is infinite), it holds that $z_t \in D_t$ for infinitely many t; equivalently,*

$$\limsup_{t\to\infty} \mathbf{1}_{D_t \neq \varnothing} = 1 \quad \implies \quad \limsup_{t\to\infty} \mathbf{1}_{z_t \in D_t} = 1.$$

In words, if decrease is enabled infinitely often, then it is taken infinitely often.

1.5 Measurability

1.5.1 Polytime Verifier

Axiom 1.3 (Polytime Verifier Axiom)**.** *There exist a polynomial $q(\cdot)$ and a deterministic polynomial–time verifier V for P_{tail} with the following properties.*

(V1) ***Polytime verifier.*** *On input size n (the length of the observed output head), V runs in time $q(n)$.*

(V2) ***Belief–state encoding and evaluability.*** *There is a fixed computable encoding* $\text{enc} : \mathcal{M} \to \{0,1\}^\star$ *and a decoding/evaluation procedure such that for any $\varepsilon > 0$ and $\Theta \in \mathcal{M}$, from* $\text{enc}_\varepsilon(\Theta)$ *of length* $O(\log(1/\varepsilon))$ *the verifier can evaluate $E(\Theta)$ and $U(\Theta, \cdot)$ to accuracy ε in time* $\text{poly}(\log(1/\varepsilon))$*. Consequently, along any T–step capture certificate, the total state–side encoding length and the evaluation cost are* $\text{poly}(T)$*.*

(V3) ***Honest–read on yes–instances.*** *On every accepting run (i.e., when the provided multi–step capture certificate is correct), V reads the entirety of that certificate, including the binary encoding of T and all per–step subcertificates.*

1.5.2 Effective Measurability Axioms

At this point, we are almost ready to prove the fundamental equivalence. However, there is one more missing piece in our assumptions. Here it is: is it even possible to operate with our physical inputs in a tractable way?

Having no certifiability as a goal, we would invoke the Carathéodory criterion for Lebesgue measurability here or restrict the input space to a Borel space if we want to build further constructions on it. But these definitions does not tell us anything about the amount of work needed to process and compare an input with a standard.

Hence, we introduce the *constructive* axioms of measurability. They can be thought of as a Carathéodory criterion plus requirements of the primitive operation to fall into the desired complexity class. Below is the version of the axioms for the class **P**.

(EM1) **Effective Representation.** Each U_j and E_j is given by a rational arithmetic circuit (or equivalent) of size $\text{poly}(n)$, where n is the bit–length of the external input description. All constant coefficients are rationals of bit–length $\text{poly}(n)$.

(EM2) **Polynomial Modulus of Continuity.** For each j, there is a known polynomial $m_j(k)$ such that to compute $U_j(x, z)$ (resp. $E_j(x)$) to k bits of output precision requires at most $m_j(k)$ arithmetic operations on rationals of bit–length $O(k + n)$. Where needed, we will discriminate (EM2) into two sub-axioms:

(EM2a) **Analytic modulus:** For each primitive map U_j and E_j, there exists a known, effectively computable modulus of continuity $\omega_j : \mathbb{Q}_+ \to \mathbb{Q}_+$ with $\omega_j(0) = 0$, such that

$$\|x - y\| \le \delta \quad \Rightarrow \quad \|F_j(x) - F_j(y)\| \le \omega_j(\delta),$$

where F_j denotes U_j or E_j.

(EM2b) **Cost modulus:** There exists a polynomial p_j such that, given a rational input x and a desired accuracy of k bits, the value $F_j(x)$ can be computed to that accuracy in at most $p_j(k + n)$ steps, where n encodes the problem size.

(EM3) **Polynomial Guard/Basin Tests.** For each guard cell $N_i \subset \mathcal{Z}$ or basin shell $\mathcal{B}_n \subset \mathcal{X}$, membership can be decided up to tolerance ε in time $\text{poly}(n, \log(1/\varepsilon))$.